PHILIPPS-UNIVERSITÄT MARBURG
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
DISTRIBUTED SYSTEMS GROUP

Master's Thesis

Deep Learning based Insect Detection for Biodiversity Monitoring

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September 2026

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Marburg, 25 September 2026

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Contents

1 Introduction 1
1.1 Motivation 1
1.1.1 Insect decline and the limits of current monitoring 1
1.1.2 Policy drivers of monitoring demand 1
1.1.3 Limits of manual sample processing 2
1.1.4 Automated traps and the annotation bottleneck 3
1.1.5 The case for synthetic training data 4
1.2 Problem Statement and Objectives 4
1.3 Contributions 5
1.4 Outline 6

2 Background and Foundations 7
2.1 Biological and Taxonomic Background 7
2.1.1 The taxonomic hierarchy and the choice of rank 7
2.1.2 DNA barcoding and the BIOSCAN collections 8
2.2 Traps, Samples, and the Current Workflow 9
2.2.1 Trap types 9
2.2.2 Sample processing from trap to tray 9
2.2.3 Automated imaging 10
2.2.4 The current end-to-end workflow 10
2.3 Deep Learning Foundations 11
2.3.1 From convolutional networks to vision transformers 11
2.3.2 Self-supervised backbones 11
2.3.3 Object detection 11
2.3.4 Instance segmentation 13
2.3.5 Training a detector 13
2.3.6 Transfer learning and fine-tuning 14
2.3.7 Data augmentation 14
2.3.8 Data partitioning and leakage 14
2.4 Small Objects in Large Images 15
2.5 Learning from Synthetic Data 15
2.5.1 Cut-and-paste composition 15
2.5.2 Domain randomisation and the reality gap 15
2.5.3 Segmenting the source specimens 16

3 Related Work 17
3.1 General-Purpose Detectors as Transfer Baselines 17

3.2 Arthropod-Specific Detection and Segmentation . . . 18
3.3 Datasets for Insect Monitoring . . . 20
3.4 Multimodal Identification . . . 21
3.5 Field and Edge Deployment . . . 21
3.6 Synthetic Training Data in Ecological Vision . . . 22
3.7 Tiled Inference . . . 22
3.8 Positioning of This Work . . . 23

4 Synthetic Corpus Construction **24**
4.1 Pipeline Overview . . . 24
4.2 Source Material and Class Selection . . . 25
  4.2.1 The label set . . . 25
  4.2.2 Sampling, cutout cap and quality filter . . . 26
4.3 Specimen Segmentation . . . 27
  4.3.1 Cutout quality, fragment specimens and the blocklist . . . 29
4.4 Tray Backgrounds . . . 30
4.5 Placement and Physical Scale Calibration . . . 30
  4.5.1 Motivation for scale calibration . . . 30
  4.5.2 The size computation . . . 30
  4.5.3 Composition . . . 31
  4.5.4 Class-balanced generation schedule . . . 32
4.6 Modelling Tray Contamination . . . 35
  4.6.1 Characteristics of real soiling . . . 35
  4.6.2 The soil model . . . 35
  4.6.3 Occlusion by debris . . . 36
4.7 Specimen-Disjoint Splitting . . . 36
  4.7.1 Leakage through shared specimens . . . 36
  4.7.2 Splitting protocol . . . 37
  4.7.3 Measured effect of leakage . . . 37
4.8 Annotation Geometry . . . 38
  4.8.1 The box as the extent of the specimen . . . 38
  4.8.2 Verification against real annotation . . . 39
4.9 Real Annotated Trap Trays . . . 39
  4.9.1 The annotated set . . . 40
  4.9.2 Annotation recovery . . . 41
  4.9.3 Group-disjoint splitting . . . 43
4.10 Mixing Real and Synthetic Data . . . 44
4.11 Residual Scale Mismatch . . . 45
4.12 The Corpus . . . 48
4.13 Verification Built into the Pipeline . . . 49

5 Detection Models and Training **51**
5.1 Detection Models . . . 51
  5.1.1 RF-DETR . . . 51
  5.1.2 flat-bug . . . 51
5.2 Training . . . 53
  5.2.1 RF-DETR . . . 53
  5.2.2 flat-bug . . . 53
  5.2.3 Tiled training data . . . 53

5.3 Implementation and Reproducibility 54

6 Experiments 55

6.1 Experimental Setup 55
6.1.1 Protocol 55
6.1.2 Hardware and environment 56
6.2 Evaluation Metrics 56
6.2.1 Intersection over Union 56
6.2.2 Precision, recall, and Average Precision 57
6.3 Whole-Tray versus Tiled Inference 58
6.3.1 The two regimes 58
6.3.2 Expected benefit and cost of tiling 59
6.3.3 Confidence floors 59
6.4 Pretraining Domain versus Architecture 60
6.5 Results 61
6.5.1 Two evaluation paths 61
6.5.2 Synthetic test split, all five models 61
6.5.3 Whole-tray results under the framework-native evaluation 62
6.5.4 Tiled models 63
6.5.5 Per-class accuracy and specimen count 67
6.5.6 Real photographs 68
6.5.7 Training-time versus inference-time tiling 72
6.5.8 Inference cost 76
6.6 Discussion 77
6.6.1 Effect of tiling by detector 77
6.6.2 Duplicate detections at tile seams 80
6.6.3 Cost against accuracy 81
6.6.4 Naming errors versus detection errors 81
6.6.5 Agreement between detectors on weak classes 82
6.6.6 Disagreement between the detectors on Zoraptera 82
6.6.7 The residual scale offset on real data 82
6.6.8 Size of the domain gap 83

7 Conclusion and Future Work 84

7.1 Summary 84
7.2 Limitations 85
7.3 Future Work 86
7.3.1 Immediate experiments 86
7.3.2 Corpus extensions 87
7.3.3 Evaluation 87
7.3.4 Deployment 88

Acronyms 89

Bibliography 90

Abstract

A single manual step constrains insect monitoring programmes: each trap catch must be spread out, then counted and identified under a microscope. Traps that photograph their own catch already exist, yet an object detector is still required to convert those photographs into abundance data, and such a detector has to be trained on boxed images of bulk samples, which nobody has the time to produce.

This thesis asks whether that labelling bottleneck can be removed by generating the training data. A generator takes individual specimens segmented from the BIOSCAN museum-imaging collections and composites them onto photographs of empty trap trays, so that the box and the order-level label of every insect are known by construction. Each specimen is sized from a size proxy held in the BIOSCAN crop metadata, calibrated once against real trays, and rendered at the pixel density of the deployment camera. Soiling of the trays follows measurements taken from real soiled trays, and the partition is kept disjoint at the level of the physical specimen. A small hand-annotated real set (nineteen trays, most of them photographed repeatedly) is split by tray and mixed in at five percent of the training signal, with fourteen photographs of six trays held out. Two detectors, a set-prediction transformer (RF-DETR) and a convolutional model pretrained on arthropods (flat-bug), are compared under matched conditions, each with whole-tray and with tiled inference.

Across the fourteen held-out photographs (about 196 distinct animals), insect detection exceeds mAP@50:95 of 0.86 for every model. When scoring is restricted to the five annotated orders, the best model, flat-bug trained and run on native-resolution tiles, reaches 0.777 at order level. On the synthetic split the same five orders score 0.084 higher; because the synthetic trays are composed on photographs of trays from the same station, the domain shift is narrow. The single largest lever is tiling at native scale, which is worth 0.05 to 0.15, and training and inference share its benefit: running a model in a regime other than the one it was trained in costs 0.16 to 0.44. Among the orders with few specimens, the remaining errors are insects that are found but misnamed, not insects that are missed. Isolating the contribution of the real share against a real-only counterpart is identified as future work.

1. Introduction

1.1 Motivation

1.1.1 Insect decline and the limits of current monitoring

No group of animals on the planet has more species than the insects, and they do a disproportionate share of the work in terrestrial ecosystems. They pollinate crops and wild plants, break down organic matter, cycle nutrients, keep pest populations in check, and sit at the base of the food web that supports birds, bats, amphibians and reptiles. During the last decade, a series of long-term studies has reported that this group is declining, sharply in some places.

Among these, the study by Hallmann et al. [1] is cited most often: it analysed 27 years of standardised Malaise-trap catches from 63 protected areas in Germany and found that total flying-insect biomass had fallen by more than 75 %. Comparable losses in arthropod biomass, abundance and species richness were reported by Seibold et al. [2] for German grasslands and forests between 2008 and 2017, and these could be linked to drivers operating at the landscape level rather than purely local ones. The headline figure was moderated by a global meta-analysis from van Klink et al. [3], which estimated a loss of roughly 9 % per decade for terrestrial insects against an *increase* for freshwater insects, yet the direction of the terrestrial trend was left standing. Broader reviews have since set these results in context and listed the drivers, chief among them habitat loss, intensive agriculture, agrochemicals, climate change, invasive species and light pollution [4, 5].

A striking feature of this literature is that the disagreement between studies is largely a matter of method, not of ecology. Hallmann et al. measured *biomass* for a practical reason: a jar of ethanol-preserved insects is cheap to weigh, while its contents are expensive to count and identify. Those who criticise the “insect apocalypse” framing have noted that the time series on hand are short, clustered geographically in Europe and North America, inconsistent in both trap type and protocol, and seldom taken down to species or even to family level [6, 7]. Ecologists do not disagree about whether insects matter. The difficulty is that the measuring instrument itself, standardised trapping followed by sorting by hand, cannot be scaled up to the spatial and temporal coverage a continental trend estimate would demand.

1.1.2 Policy drivers of monitoring demand

The measurement gap has ceased to be a purely scientific matter. The European Union has bound itself to targets expressed as population trends, and such targets cannot be checked without population data. Under the EU Biodiversity Strategy for 2030 [8], 30 % of EU land and sea is to be protected and the decline of pollinators reversed; the revised EU Pollinators

Initiative, *A New Deal for Pollinators* [9], states the reversal of pollinator decline by 2030 as an explicit objective and asks for an EU-wide pollinator monitoring scheme (EU-PoMS) against which progress can be measured. Several of these commitments become binding obligations on Member States through Regulation (EU) 2024/1991, the Nature Restoration Regulation [10], among them the obligation to reverse the decline of pollinator populations and to monitor them at least once every six years. Periodic reporting on the conservation status of listed species, many of which are insects, is already required by Article 17 of the Habitats Directive [11].

Every one of these instruments presupposes that abundance and diversity can be measured again and again, in a comparable way, and on a national scale. For now they cannot be, or at least not at a price Member States are prepared to pay, and the sorting bench is the reason.

1.1.3 Limits of manual sample processing

In a century, the standard survey workflow for terrestrial arthropods (Figure 1.1) has barely changed. A trap is set out for a fixed interval (a pitfall trap sunk level with the soil surface, a Malaise tent that intercepts flying insects, a pan trap, a light trap), usually charged with an agent such as ethanol or propylene glycol that kills and preserves the catch. On a fixed schedule the trap is emptied and its contents go into a sample jar. Back in the laboratory the sample is washed, spread across a sorting tray and worked through under a stereomicroscope; specimens are picked out from soil, plant litter and debris, grouped by morphological similarity, counted, and identified as far as the expertise on hand permits.

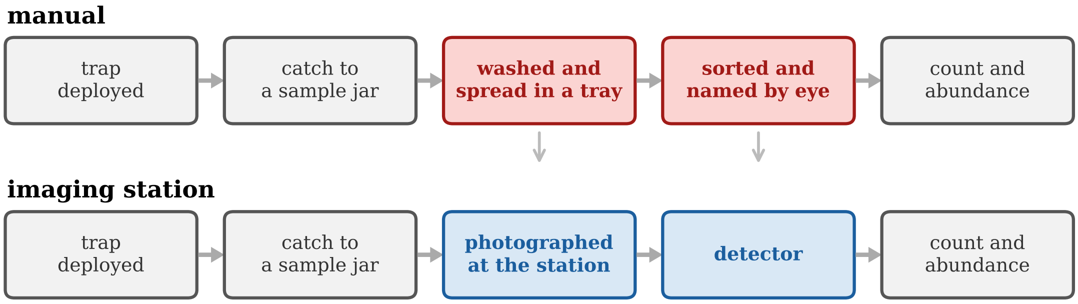


Figure 1.1: The monitoring workflow. Top: the manual chain from trap to count. The two highlighted steps (washing and spreading the catch in a sorting tray, and sorting and naming it under a stereomicroscope) are the ones that dominate cost and that cannot be parallelised without trained people. Bottom: the imaging-station variant, in which a photograph of the tray replaces the sorting bench and a detector replaces the technician's count. Everything before and after those two steps is unchanged.

This workflow is the binding constraint on monitoring capacity because of three of its properties.

Throughput. A single Malaise sample from a productive site may contain thousands of specimens; a pitfall sample from a temperate grassland typically contains tens to hundreds. Sorting time grows with the number of animals, so the samples that carry the most ecological information are precisely the ones that are most expensive to process. It is commonly reported that unsorted material in a monitoring programme piles up faster than the programme can work through it [12].

Dependence on taxonomic expertise. Identification below the level of order usually requires a trained taxonomist, and for many groups the number of people able to make such determinations is small and falling; the systematics literature has discussed this *taxonomic impediment* for decades [12]. Part of this pressure has been lifted by Deoxyribonucleic Acid (DNA) barcoding of the Cytochrome c Oxidase subunit I (COI) gene [13], which turns identification from a morphological judgement into a database lookup; reference libraries such as BOLD, and the BIOSCAN programme built on them [14, 15], now hold millions of specimens. Barcoding nevertheless destroys tissue from the specimen, has a cost per specimen, and still requires the animal to be physically picked out of the sample first. It lifts the bottleneck of identification but not the bottleneck of sorting.

Reproducibility. Sorting and identification are judgement tasks done under time pressure; two technicians working through the same tray will, in general, reach different counts, as will one technician on two different days [12]. Because manual sorting is expensive, a sample is almost never processed twice, so the variance stays hidden and is not open to correction.

1.1.4 Automated traps and the annotation bottleneck

These constraints have prompted the automation of the imaging step. A growing family of camera traps and imaging stations photographs insects in the field, or photographs the catch in a tray, and keeps images in place of, or in addition to, the physical material [16, 17, 12, 18]. This thesis is situated within one such system, the 3DTRAB project, whose imaging station is the source of every real tray photograph and every real annotation used here. In the configuration that produced this imagery, the catch from a pitfall trap is moved into a standardised tray and photographed from above under controlled illumination, at several camera heights and exposure settings, which yields high-resolution overview images of the whole tray.

The economics of monitoring genuinely change as a result. A trap can be emptied and imaged with no taxonomist present, archiving images is cheap, and the same tray can be analysed again later with a better model. On its own, though, this yields no single number an ecologist can use; a photograph of a tray is not an estimate of abundance. Getting from the one to the other means finding every insect in the image, delimiting it and assigning it to a taxonomic group. That is an object detection problem, and an unusually hard one at that (a representative image is shown in Figure 4.7):

- **The objects are small relative to the image.** An overview image of a tray spans several thousand pixels, while a single insect may occupy only a few tens of pixels. Most detectors rescale their input to a fixed resolution of a few hundred pixels, and in doing so discard precisely the evidence that tells one small arthropod from another.
- **The objects are numerous and touch each other.** In a drained tray, specimens clump together, overlap and pile up on one another.
- **The background is adversarial.** The catch from a pitfall trap comes out accompanied by soil, plant fragments, twigs and grit, and at tray scale a dark, irregular clump of soil imitates a small beetle very well.
- **The label space is fine-grained and extremely imbalanced.** How often an insect order is caught varies enormously: a few orders make up most of the specimens, and the rest are rare.

1.1.5 The case for synthetic training data

Annotated images are a prerequisite for supervised object detection, and annotating tray images costs nearly as much as the manual sorting it is supposed to replace. Boxing every insect in a photograph that contains several hundred of them, and giving each a taxonomic label, is a job for the same scarce experts, and the job must be done again each time the trap design, the tray, the camera or the illumination changes.

The position taken in this thesis is that, for this particular problem, training data ought to be *generated rather than annotated*. The central observation is that the two things which make annotation costly, knowing exactly where each insect is and knowing exactly what it is, come for free when the image is composed programmatically. Large collections of specimens photographed and identified one by one already exist. The BIOSCAN datasets [14, 15] supply millions of standardised specimen images, each already tied to a taxonomic determination backed by a DNA barcode. Once those specimens are segmented from their background and composited onto photographs of empty trap trays, the position, extent and label of every animal in the resulting image are known by construction. Such a label is exact rather than approximate, it costs nothing per image, and compute time is the only limit on how many images can be made.

Composition of this kind is nothing new in computer vision, where cut-and-paste augmentation and domain randomisation are established techniques [19, 20, 21, 22, 23]. Nor is it new to insect detection: one leading moth-monitoring benchmark trains its own localiser on a small copy-paste corpus of segmented crops, although it never reports how well that localiser performs [24]. Applying the construction to this problem raises questions that the generic treatments leave open. How realistic do the tray background, and the contamination on it, need to be? At what physical scale should specimens be pasted so that the model learns a size prior that transfers? How should the data be partitioned so that the reported accuracy measures generalisation to new animals rather than memorisation of the handful of specimen photographs available for the rarest orders? And finally, once a small real share is mixed in, how far does a model trained in this way transfer to a real trap image? Each of these is treated in the chapters that follow as a design decision whose consequence is measured.

1.2 Problem Statement and Objectives

The problem addressed by the thesis can be stated as follows. Starting from a collection of insect specimens that have been imaged and identified individually, together with photographs of empty trap trays, build a synthetic training corpus and a detection model such that the model localises insects in real trap-tray photographs and classifies them to the order level, well enough for abundance estimation, without needing a hand-annotated corpus of real tray images at training scale.

The approach rests on the premise that *synthetic training data supplies the volume and the label precision, while a small quantity of hand-annotated real imagery supplies the domain realism.* That combination is what the thesis builds and evaluates, and its objectives follow from it. The first objective is generation: constructing synthetic tray images from which a detector learns evidence that transfers. This calls for physically calibrated specimen scaling, a realistic model of tray soiling and debris, and a controlled relation between the number of generated images and the number of distinct source specimens. The second is evaluation design: partitioning the corpus and defining the annotations so that the reported accuracy measures generalisation instead of reflecting the generator. This turns on splitting at the level of the specimen rather than the image, and on the geometric convention used to record a bounding box. The third is a matched comparison, under identical training and evaluation conditions, of two detector families: a set-prediction transformer (RF-DETR, built on a self-supervised DINOv2 backbone) and an arthropod-specific convolutional detector (flat-bug, built on the eighth version of the You Only Look Once detector, YOLOv8). The fourth is the inference regime: measuring what is gained and what is lost when a tray, whose resolution far exceeds any detector's input, is processed as overlapping tiles at native scale rather than downscaled whole. The fifth is the synthetic-to-real gap: measuring, on real trap photographs held out of training, how much accuracy the combination retains.

1.3 Contributions

1. **A calibrated synthetic tray generator.** A pipeline in which a high-resolution dichotomous segmentation model cuts specimens out of BIOSCAN museum imagery. Each specimen is scaled by a size proxy derived from the BIOSCAN crop metadata and calibrated once against real annotated trays. The specimens are then pasted onto real tray backgrounds at the pixel density of the deployment imagery under a class-balancing schedule, and a soiling model whose parameters were measured on real soiled trays is applied. The generator writes exact Common Objects in Context (COCO)-format annotations for 22 arthropod orders, more than a million of them in the corpus used here, and verifies its own output (Chapter 4).
2. **Entity-level splitting applied to a generated corpus.** Grouped or blocked partitioning is established practice for structured data [25, 26]. The contribution here is to apply it to a synthetic corpus, where the entity is the source specimen, and to measure how far an image-level split inflates the reported accuracy. The inflation is largest for precisely the rare classes whose accuracy deserves the least trust (Section 4.7). The real data are treated by the same principle: they consist of nineteen physical trays, eleven of them photographed repeatedly, rather than ninety-one independent images (Section 4.9.3).
3. **An annotation geometry verified against independent real annotation.** Every generated instance is boxed by the extent of the rotated specimen matte, not by the rotated image canvas. The canvas would inflate the box by $4/\pi$ per side for a round body (about 1.29 when measured over the corpus). A synthetic test split shares the convention of its training split and so cannot expose an error in it; the boxes are therefore verified on the real photographs, where a box-scale sweep peaks at unity for every model (Sections 4.8 and 6.5.6).
4. **A mixed synthetic-and-real corpus**, built by splitting a small hand-annotated real set by physical tray, oversampling it to a fixed share of the training signal and merging it with the synthetic corpus, with a real test split held out (Sections 4.9 to 4.10).
5. **A matched comparison of two detector families and two inference regimes**, designed so that the choice of detector is kept separate from input scale. It shows that the gain from tiling is shared by training and inference and does not transfer from one regime to the other (Chapter 6).
6. **A per-class analysis of the trained detectors**, which shows three things. On the specimen-poor orders the residual errors are naming errors, not detection errors (missed 2 to 5 %, mislabelled 63 to 97 %). Over the range present in the corpus, the number of distinct source specimens does not predict per-class accuracy. Native-scale tiling recovers

the smallest order (Thysanoptera) while lowering the missed rate of the second (Polyxenida) (Chapter 6).

1.4 Outline

The background for the remainder is laid out in Chapter 2: the ecological and taxonomic setting, the way trap samples are produced and processed at present, and the deep learning foundations. The latter comprise object detection architectures and the particular difficulties of detecting small objects and of learning from synthetic data. Chapter 3 reviews related work on general-purpose and insect-specific detectors, on the datasets available for insect monitoring, and on field and edge deployments; it is a revised and reorganised version of the independent study the author completed beforehand [27]. The construction of the synthetic corpus, the principal contribution of this thesis, is described in Chapter 4. The two detection models and their training are introduced in Chapter 5. Chapter 6 sets out the evaluation protocol and metrics, the whole-tray and tiled inference regimes, and the experimental results together with a discussion of them. Chapter 7 gives the conclusions and the work that remains to be done.

2. Background and Foundations

This chapter sets out the vocabulary that the rest of the thesis depends on. The biological side comes first (which animals are counted, how they are classified, and the physical route by which a sample comes into being). The machine learning side follows: how object detection is formulated, why small objects in large images constitute a special case, and the present state of knowledge on training a detector with generated instead of photographed images.

2.1 Biological and Taxonomic Background

2.1.1 The taxonomic hierarchy and the choice of rank

Biological classification places organisms in a nested hierarchy. For the animals considered here the relevant ranks are, from the coarsest to the finest, *phylum* (Arthropoda), *class* (Insecta, Arachnida, Diplopoda, Chilopoda, ...), *order* (Coleoptera, Diptera, ...), *family*, *genus* and *species*. Each step down the hierarchy multiplies the number of categories and narrows the visual gap between neighbouring ones. Gross morphology (the number and texture of the wings, the type of mouthparts, the segmentation of the body) is enough to distinguish the roughly thirty insect orders. By contrast, two species within one genus may be separable only by genitalic structure or by the arrangement of a few setae, and features of that kind are not resolved by any overview photograph of a tray at a realistic camera resolution.

The rank adopted throughout this thesis is accordingly the *order*. This is a considered choice, resting on three grounds. To begin with, order is the finest rank that can be determined reliably from the dorsal view of an intact specimen at the resolution an overview image offers. Next, it is the rank at which the specimen imagery that is available carries its densest and most dependable labels. Finally, it carries ecological meaning: monitoring programmes routinely use order-level composition and abundance as indicators, and a change in, for instance, the ratio of Coleoptera to Collembola across a pitfall series can be interpreted without descending to species.

The label set of this work consists of twenty-two groups:

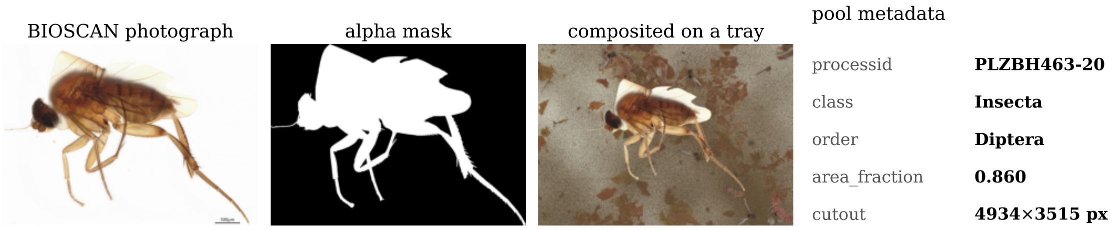
Lepidoptera
Phasmatodea
Archaeognatha
Hymenoptera
Neuroptera
Lithobiomorpha
Psocodea
Zygentoma
Dermaptera
Hemiptera
Araneae
Scutigeromorpha
Odonata
Thysanoptera
Diptera
Orthoptera
Opiliones
Zoraptera
Strepsiptera
Coleoptera
Trichoptera
Polyxenida

Rank consistency. Each entry in the list is an order, but only because the list was made uniform on purpose. Araneae (spiders) and Opiliones (harvestmen) sit in the class Arachnida rather than in Insecta; Polyxenida (bristly millipedes), Lithobiomorpha (stone centipedes) and Scutigeromorpha (house centipedes) are orders within the myriapod classes Diplopoda and Chilopoda. If the *class* names Diplopoda and Chilopoda had served as labels, each would have merged several genuinely distinct orders: a bristly Polyxenid, a flat-backed Polydesmid and a smooth cylindrical Julid would share one label, and a house centipede and a thread-like Geophilomorph the other. This is precisely what a field ecologist does when noting down "millipede", and as a reporting category it is defensible, but it requires a single classifier output to learn several unrelated appearances at once; for that reason the label set stays at order level (Section 4.2.1).

2.1.2 DNA barcoding and the BIOSCAN collections

Barcoding of DNA has gone some way towards relieving the identification bottleneck of Section 1.1.3. As Hebert et al. [13] demonstrated, a short standardised fragment of the mitochondrial COI gene differs sufficiently between species, and sufficiently little within them, to act as a species-level identifier across most animal groups. Once that fragment is sequenced and matched against a reference library, identification ceases to be an expert morphological judgement and becomes a database lookup. When no named species exists for a sequence, sequences that cluster closely enough receive a Barcode Index Number (BIN), which serves as an operational stand-in for a species.

At scale, this is what the BIOSCAN programme has done, and, of central importance to this thesis, it has published the specimen *photographs* together with the sequences. In BIOSCAN-1M [14], approximately one million RGB microscope images are paired with DNA barcodes and taxonomic annotations; the specimens came mostly from Malaise trapping in Costa Rica, Canada and South Africa and were imaged at 2880 × 2160 pixels on a Keyence VHX-7000 microscope. BIOSCAN-5M [15] enlarges this to 5,150,850 specimens from 47 countries, 10 classes and 55 orders, imaged at 1024 × 768 pixels, with geographic and specimen-size metadata added.



processid	**PLZBH463-20**
class	**Insecta**
order	**Diptera**
area_fraction	**0.860**
cutout	**4934×3515 px**

Figure 2.1: A BIOSCAN specimen record and the parts of it the pipeline retains. From left to right: the distributed microscope photograph; the alpha channel that segmentation produces; the cutout composited onto a tray background; and the metadata rows that travel through the pool. One of the two size references used by the generator is the radius of the circular imaging field (Section 4.5.2); it does not appear in the table, and for reasons of space the panel also leaves out the `scale_factor` row on which the size computation depends; `area_fraction` is carried but not used (Section 4.3). The photograph appears at the resolution of the display copy, which is larger than the 1024 × 768 distribution copy from which the pool is built.

These collections are suitable raw material for the generator of Chapter 4 for two reasons. First, the imaging is standardised (Figure 2.1): every specimen is photographed on its own, dorsally, against a uniform background and under controlled lighting, so that automatic figure-ground separation becomes tractable. Second, every record carries size metadata [15]: an `image_measurement_value`, the number of pixels occupied by the organism; an `area_fraction`, the share of the original photograph that the specimen crop covers; and a `scale_factor`, the ratio of the specimen crop to its resized version. None of these is a physical measurement, and neither the vessel nor the magnification is recorded in the collection. Since all specimens are nonetheless imaged at one fixed magnification, with the crop fitted to the animal, the crop ratio is proportional to the extent of the specimen. Once a single constant has been calibrated against real imagery, it can act as a size proxy (Section 4.5.2). In the absence of such a proxy, the size at which a specimen is pasted into a tray would have to be chosen arbitrarily.

Set against these advantages is a severe class imbalance. Because the collections mirror what Malaise traps really catch, Diptera and Hymenoptera outnumber everything else by orders of magnitude, groups such as Zoraptera appear as a handful of individuals, and a few orders are so sparse that they cannot be used at all. Every pipeline built on this material inherits the imbalance; how it is dealt with, and where it cannot be dealt with, recurs as a theme throughout this thesis.

2.2 Traps, Samples, and the Current Workflow

2.2.1 Trap types

Every trap type captures a different fraction of the arthropod community, and the imagery it produces differs accordingly.

Pitfall traps. A pitfall trap is a container sunk into the ground with its rim level with the soil surface, so that ground-active arthropods fall in; it is part-filled with a killing and preserving fluid. Its catch is the epigeic fauna: ground beetles (Carabidae), rove beetles (Staphylinidae), spiders, harvestmen, millipedes, centipedes, ants and Collembola. Because the opening lies at soil level, soil, grit, plant litter, seeds and small twigs enter as well, and it is this debris that makes pitfall imagery visually difficult. Pitfall traps are the source of the imagery used in this thesis.

Malaise traps. A Malaise trap is a tent-shaped interception trap in which flying insects are guided upward into a collecting head filled with ethanol. It samples the flying fauna, chiefly Diptera and Hymenoptera, and its catch is very large, frequently reaching thousands of specimens within one collection interval. Most of the BIOSCAN specimen images used as raw material here come from Malaise trapping.

Pan traps and light traps. These two complete the standard toolkit: pan traps sample flower-visiting insects and light traps the nocturnal fauna.

2.2.2 Sample processing from trap to tray

What a detector finally looks at is a *tray*, a shallow, flat container of uniform colour into which the drained catch of a trap is spread for examination. The detection problem is defined by how that tray looks.

Once the trap has been emptied, the preserving fluid is washed off the sample and the sample is tipped into the tray; with a pitfall trap, the soil and litter arrive along with it. The technician then spreads the specimens as evenly as possible, yet the spread is never perfect: animals form clumps, lie over one another, curl up (beetles and spiders preserved in ethanol take on a characteristic contracted posture) and come to rest at arbitrary orientations. Legs and antennae break off. Rather than a scatter of discrete blobs on a clean background, a soiled tray presents a continuous bed of fine tan-to-brown sediment, gathered organically into patches with cleaner zones in between; when that appearance has to be simulated, this distinction turns out to matter considerably (Section 4.6).

2.2.3 Automated imaging

In an automated imaging station, a camera mounted above the tray stands in for the technician's eye. The configuration used here photographs the tray from above under controlled illumination, at several camera heights and with several exposure settings, and the resulting overview images measure several thousand pixels on a side. Because a physical scale reference sits in the field of view, pixel measurements can be converted into millimetres.

The change this brings is to the order of the workflow. Under the traditional arrangement no data existed until sorting was done; with imaging, the data exist the moment the tray is photographed, and the analysis can be postponed, rerun, audited, or upgraded as models get better. What imaging does *not* alter is that a decision about where each animal lies and what it is must still be made somewhere. The rest of this chapter concerns the automation of that decision.

2.2.4 The current end-to-end workflow

Assembled from these pieces, a present-day monitoring cycle runs as follows:

1. **Deployment.** Traps are set out following a sampling design and left in place for a fixed interval.
2. **Collection.** On schedule, the traps are emptied; each sample is labelled with site, trap identifier and interval and kept in ethanol.
3. **Imaging** (in automated systems). The catch is spread across a tray and photographed.
4. **Sorting.** Under a stereomicroscope, a technician picks the specimens out of the debris and groups them by morphology.
5. **Identification.** Each specimen is determined to the finest rank that the expertise on hand permits; some subset may be barcoded.
6. **Data entry.** For every sample, the count per taxon is recorded.
7. **Analysis.** The counts are combined into indices of abundance, richness and diversity and compared between sites and years.

Most of the cost and the latency sit in steps 4 and 5, and these are the steps an automated detector is meant to take over. Taking them over calls for *detection* rather than classification alone: the model has to determine how many animals are present and where they are, not simply what a pre-cropped animal is.

2.3 Deep Learning Foundations

2.3.1 From convolutional networks to vision transformers

Convolutional Neural Networks (CNNs) learn a hierarchy of spatially local filters; locality and translation equivariance are built into the architecture itself. With residual connections [28], very deep convolutional stacks became trainable, yielding the backbone family that dominated visual recognition through most of the past decade.

The Vision Transformer (ViT) [29] swapped the convolutional prior for a sequence model. The image is divided into patches of fixed size, each patch is linearly embedded as a token, positional information is added, and a standard transformer encoder runs global self-attention across the resulting sequence. Under self-attention, every patch can attend directly to every other patch at every layer, a useful property whenever the evidence behind a decision is non-local: telling a beetle from a soil clump of similar size may hinge on context elsewhere in the tray. The price is complexity quadratic in the number of tokens. That is why input resolution is a first-class constraint for transformer detectors, and also why transformers tend to demand more data than CNNs: the inductive biases a convolution provides for free have to be learned instead.

2.3.2 Self-supervised backbones

Self-Supervised Learning (SSL) removes the need for labels when a backbone is pretrained. DINOv2 [30] trains a ViT on a large curated unlabelled corpus with a self-distillation objective, and the general-purpose features it produces transfer to dense prediction tasks with no fine-tuning of the backbone. In a task such as this one, where the target domain lies far from the natural images of COCO and labelled in-domain data is exactly the scarce resource, a backbone whose features were learned without labels makes an attractive starting point. One of the two detectors evaluated here, RF-DETR, rests on a DINOv2 backbone.

2.3.3 Object detection

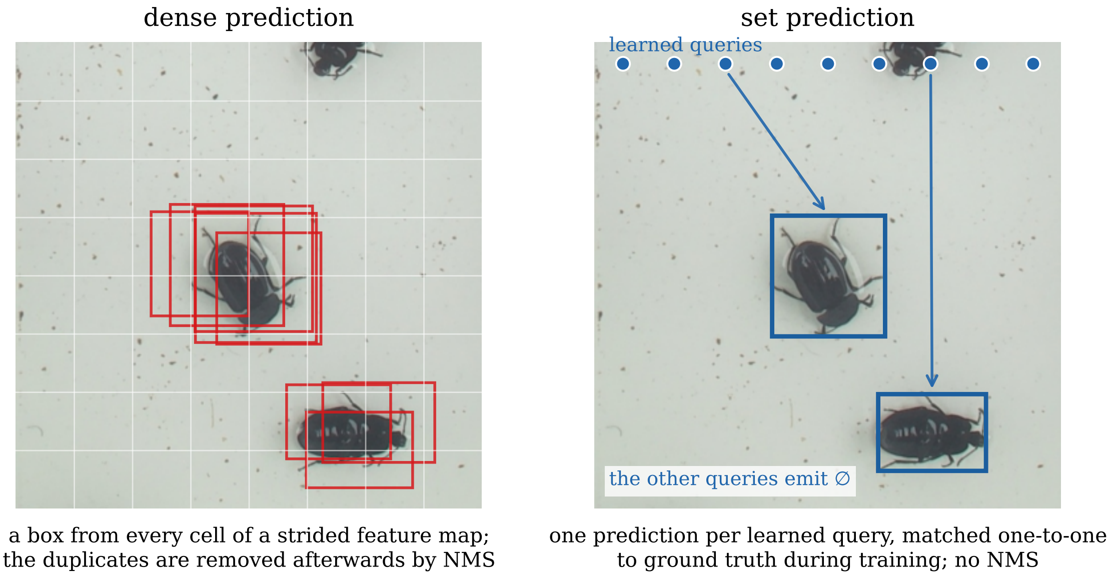
In object detection, the model must emit a set of records of variable length, each made up of a bounding box together with a class label and a confidence score. It is this unordered, variable-length output that makes the problem structurally harder than classification, and the handling of it is the principal point on which the architectural families below diverge.

Anchor-based and one-stage detectors. For many years the prevailing approach laid a dense grid of candidate boxes (*anchors*) over the image and had the network predict, per anchor, an objectness score, a distribution over classes and a correction to the anchor's geometry. Two-stage designs such as Faster R-CNN propose regions first and classify them afterwards, whereas one-stage designs such as the YOLO family [31] and RetinaNet [32] predict straight from the feature map, giving up some accuracy in exchange for a large gain in speed. Since many anchors respond to one and the same object, such detectors need Non-Maximum Suppression (NMS) as a post-processing step: the predictions are ordered by confidence, and any box whose overlap with a higher-scoring box of the same class exceeds an Intersection over Union (IoU) threshold is dropped. Cheap as it is, NMS adds two hyperparameters and misbehaves when genuine objects overlap heavily, which in a tray of clumped insects they do.

Recent YOLO releases [33] dispense with anchors (box geometry is predicted directly from grid cells), split classification and regression into decoupled heads, and are trained under a task-aligned assignment strategy; the dense-prediction-plus-NMS structure, however, remains. YOLOv8 also provides an instance-segmentation variant in which mask coefficients are predicted next to each box, and the flat-bug model used in this thesis is built on that variant.

Set prediction. With DEtection TRansformer (DETR) [34], detection was recast as direct set prediction. In a transformer decoder, a fixed number of learned *object queries* attend over the encoded image features, and every query yields one box and one class, where the classes include "no object". Bipartite matching drives the training: a Hungarian algorithm finds the optimal one-to-one assignment between predictions and ground-truth objects, and the loss is computed only along that assignment. As the assignment is one-to-one, duplicate predictions are penalised in training, so no NMS is needed at inference.

Convergence of the original DETR was very slow. Deformable DETR [35] tackled this by substituting deformable attention for dense attention: each query attends to a small learned set of sampling points around a reference location, which yields multi-scale features and far quicker convergence. Later work pushed the trade-off both ways. DETA [36] brought back one-to-many IoU-based assignment and NMS to converge faster; RT-DETR [37] preserved the end-to-end NMS-free property but redesigned the encoder for real-time use; and LW-DETR [38] demonstrated that a plain ViT encoder paired with a light decoder can reach YOLO-class throughput. RF-DETR [39] bundles this line of work into a deployable pipeline that combines a DINOv2 backbone with deformable attention.



a box from every cell of a strided feature map; the duplicates are removed afterwards by NMS

one prediction per learned query, matched one-to-one to ground truth during training; no NMS

Figure 2.2: Dense prediction against set prediction, illustrated over a real tray crop. Left: every cell of a strided feature map in a one-stage detector emits a box, so a cluster of near-duplicates gathers around each object and has to be removed by NMS afterwards. Right: a DETR-style detector produces exactly one prediction for each learned query, and during training these are matched one-to-one to the ground truth; queries with no match emit ∅, and NMS is not applied. The distinction is relevant here because duplicates that neither architecture removes by itself are reintroduced by tiled inference; see Section 6.3.1 and Figure 6.2.

The difference between the two families (Figure 2.2) goes beyond implementation, and it colours how the results of Chapter 6 are read. A dense one-stage detector predicts from a feature map at a fixed stride, so its localisation precision is bound to that stride; a set-prediction detector regresses boxes from a small number of queries equipped with global context, which alters both what it finds and how tightly it fits the box around it.

2.3.4 Instance segmentation

Instance segmentation goes beyond detection in demanding a per-pixel mask for each object instead of a box alone. Mask R-CNN [40] bolted a mask branch onto Faster R-CNN; YOLOv8-seg predicts a set of prototype masks for each image along with per-instance coefficients that combine them. In this domain the added cost of masks pays off for two reasons. An insect with its legs splayed fills only a small fraction of its bounding box, which makes box-IoU a poor gauge of the overlap between two neighbouring specimens; and deduplication on masks keeps apart touching animals that box-based NMS would fuse. This is exactly what flat-bug exploits when it deduplicates on polygon IoU instead of box IoU [41]. Because the synthetic corpus built here contains boxes only, the segmentation capability goes unused at training time; Section 5.1 discusses what follows from that.

2.3.5 Training a detector

One and the same supervised procedure fits every architecture described above: a loss is summed over the labelled objects in a batch of images, and the weights are updated by its gradient. The loss consists of a classification term and a localisation term; the two families differ in what is summed. A dense detector assigns cells or anchors to objects by a rule and then sums over every assigned cell, while a set-prediction detector sums only along the one-to-one Hungarian assignment of Section 2.3.3. This is why the latter learns during training to suppress duplicates rather than removing them afterwards. The localisation term matters for this thesis because object sizes span a wide range: over the 22 orders, the median box of the corpus runs from 25 pixels (Thysanoptera) to 157 pixels (Phasmatodea), a span of 6.3× (Table 4.7). A plain L_1 error on box coordinates charges the same penalty for a five-pixel miss on either, whereas an overlap-based term scales the penalty with the object. Every detector used here includes such an overlap term: for flat-bug it is the complete-IoU loss of the Ultralytics implementation [33], and for RF-DETR the generalised IoU term of the DETR family [34], in the latter case combined with an L_1 term on normalised coordinates.

Optimisation proceeds in epochs, each a single pass over the training set, and the learning rate follows a schedule rather than staying constant; the values used in this work are given in Section 5.2. During the opening epochs a *warm-up* holds the rate low, since the freshly initialised head yields large and poorly directed gradients. A *decay* then lowers the rate, either continuously (a linear decay) or in a single step (a tenfold drop at a set epoch). At a 1536-pixel input only a few images fit on one Graphics Processing Unit (GPU), so the gradient is *accumulated* across several small batches before a step is taken, which yields a larger effective batch. *Weight decay* regularises the model by penalising large weights. A run ends either at an epoch cap or by *early stopping*, which terminates it once the validation metric has failed to improve for a set patience and returns the best checkpoint rather than the last. The cap and the length of training therefore need not be the same. An *exponential moving average* of the weights is a widely used gain that is free at training time, and RF-DETR supports one; using it or not is a recipe choice (Section 5.2).

2.3.6 Transfer learning and fine-tuning

Neither detector is trained from scratch. Both start from weights pretrained on a large generic corpus: natural images for the DINOv2 backbone of RF-DETR (Section 2.3.2), real arthropod imagery for the YOLOv8 of flat-bug (Chapter 3). Each is then *fine-tuned* on the target data with every weight free to update. Since the pretrained backbone features deserve protection, fine-tuning frameworks expose a separate learning rate for the backbone, and by convention that rate is set lower. The reference recipe of RF-DETR sets it slightly higher instead (1.5×10^{-4} for the encoder, which is the backbone, against a base rate of 10^{-4}). The consequence for the thesis deserves a plain statement: "training on synthetic trays" always denotes fine-tuning a model that has already seen millions of natural images and, for flat-bug, more than a hundred thousand annotated real arthropods (Chapter 3). This is part of the background to any reading of the five percent real share, whose separate contribution Section 7.3 assigns to a synthetic-only counterpart. It also amounts to a confound between architecture and pretraining domain, which Section 6.4 names instead of hiding.

2.3.7 Data augmentation

Augmentation expands a training set by transforming its images on the fly. The transformations are geometric (flips, translation, rescaling), photometric (jitter in hue, saturation and value) or compositional (*mosaic*, which tiles four images into one, and *mixup*, which blends two). The flat-bug runs keep the Ultralytics defaults for every augmentation key; the values are listed alongside the recipe in Section 5.2. RF-DETR is trained with multi-scale input, which is the set-prediction counterpart of the rescale. Two consequences run through the rest of the thesis. The $\pm 50\,\%$ rescale absorbs some, though not all, of the residual scale offsets of Section 4.11, where three orders are rendered 1.75 to 2.58× smaller than they appear in real trays. The synthetic corpus is also augmentation carried to its limit: the same 148,285 cutouts are recombined into 27,000 trays, so additional trays supply new arrangements, backgrounds and co-occurrences of the same animals, but no new animals. This is precisely the distinction between specimen diversity and image count made in Section 4.2.2.

2.3.8 Data partitioning and leakage

A held-out set measures generalisation only if its rows are independent of the training rows; where they are not, a random split at the level of the stored file leaks. The failure is frequent enough to have its own literature. Kapoor and Narayanan [26] survey how widely it has produced irreproducible results across machine-learning-based science, and Roberts et al. [25] give the ecological treatment for data with temporal, spatial, hierarchical or phylogenetic structure, where the remedy is blocked cross-validation at the level of that structure. Both partition rules in this thesis apply that argument. In the synthetic corpus the same BIOSCAN individual is pasted repeatedly into many trays, so splitting by tray would put one animal on both sides; the split is therefore by *specimen* (Section 4.7). In the real set the frames of one tray are near-duplicates, the same animals at three camera heights, so the unit becomes the tray *group* (Section 4.9.3). The same distinction governs how much real data the training set contains: its real share consists of a small number of distinct photographs, each repeated several times, so its share of the training signal overstates its share of the variety (Section 4.10). It likewise governs how finely the real results can be read: the held-out frames come from few trays, and a bootstrap that resamples trays gives intervals wide enough that pairwise differences between models on the five-order metric are not resolved (Section 6.5.6). Split rule and sample size are a single fact seen twice.

2.4 Small Objects in Large Images

A detector takes input at a fixed resolution, usually somewhere between 640 and 1024 pixels on a side. Any larger image has to be reduced to fit, and the standard way of doing so, resizing the whole image, shrinks every object along with it. Resize a tray photograph 3,500 pixels wide to 800 pixels and a 40-pixel insect becomes 9 pixels, less than a single patch of a ViT with patch size 16, and indistinguishable from a speck of grit of the same size. No modelling deficiency is at fault here: the information has been destroyed before the network ever sees it.

Two responses are possible. The first is to raise the input resolution; memory limits this, quadratically under global attention, and returns diminish because the network was pretrained at a much lower resolution. The second is *tiled inference*. The image is cut into overlapping crops at or near native scale, the detector is run on each crop, and the predictions are mapped back into image coordinates and merged. Slicing Aided Hyper Inference (SAHI) [42] formalises the procedure for inference and fine-tuning alike, and for high-resolution bulk-sample imagery it is standard practice [43]. flat-bug goes one step further with a multi-scale *pyramid* of tilings, so that objects of widely differing sizes are each seen at a suitable scale [41].

The procedure is illustrated on a real tray in Figure 6.2 of Chapter 6. Tiling comes at a price, and three costs recur there. Objects lying on tile boundaries are cut off and may be detected twice, partially in each tile, so a merging rule is needed, usually an edge-proximity penalty on the confidence of boxes that touch a tile border followed by NMS over the union. The cost of inference is multiplied by the number of tiles. And, least obvious yet most important, tiling alters the *apparent size* of objects: a model trained on whole images downscaled by 0.5× has acquired a size prior at that scale, and when it is run on native-scale tiles it faces objects twice the size of anything it was trained on. Tiling therefore helps only when it moves the object scale at inference *closer* to the object scale at training; it does not help automatically.

2.5 Learning from Synthetic Data

2.5.1 Cut-and-paste composition

Building training images by pasting segmented object instances onto backgrounds is a long-established idea. Dwibedi et al. [19] showed that instance detectors trained on naive cut-and-paste composites do transfer to real images and, importantly, that local boundary artefacts count for more than global scene realism, because a detector may seize on paste seams as a shortcut. Ghiasi et al. [20] demonstrated that plain copy-paste is a strong augmentation for instance segmentation even when no attempt at realistic placement is made.

For this technique the tray domain is unusually accommodating. Pasting an object into a general scene calls for reasoning about support surfaces, occlusion order, perspective and the direction of the light. A tray, however, is a flat and uniformly lit plane seen orthogonally from above, and the specimens really do rest on it at arbitrary positions and orientations. The composition model (a dorsal view pasted at a random position and rotation) is close to the way the real image is actually formed, something that cannot usually be said of cut-and-paste pipelines.

2.5.2 Domain randomisation and the reality gap

Domain randomisation [21, 22, 23] takes the course opposite to photorealism. Rather than matching the real domain, it varies the nuisance parameters of the simulator so widely that the real domain becomes merely another draw from the training distribution, and the model is thereby forced onto features that are invariant to those parameters. The two strategies are combined in practice, so the design question is which factors to randomise and which to match. Position, orientation, instance density, class mixture and soil level are randomised by the generator of Chapter 4, while physical specimen scale and the statistical appearance of tray soiling are matched.

Three components of the gap between synthetic and real data are worth telling apart:

Appearance gap. The low-level statistics differ, in illumination, colour balance, noise, focus and compositing seams.

Content gap. What is depicted follows a different distribution: which taxa are present, in what proportions, at what density and in which postures. Under a microscope a dorsally photographed specimen is flat and splayed; in a tray the same animal is curled and may lie on its side.

Label gap. The annotation *convention*, meaning what is taken as the extent of an object, differs. Among the three it receives the least discussion, and Section 4.8 argues that it is the one that a synthetic benchmark is unable to detect by itself.

2.5.3 Segmenting the source specimens

For compositing, every source specimen is needed as an Red-Green-Blue-Alpha (RGBA) cutout with an accurate alpha matte. That is a dichotomous image segmentation problem, one salient foreground against a background, which thin structures complicate here: legs, antennae and setae measure one or two pixels across, are semi-transparent at their edges, and are precisely the structures carrying the discriminative information at order level. If a matte clips them, the resulting specimen is systematically harder to identify than the real animal.

BiRefNet [44] is the tool used for this step. Designed for high-resolution dichotomous segmentation, it follows a bilateral reference design in which the decoder receives high-resolution source patches alongside coarse semantic features, and this preserves thin structures better than a decoder that relies on a feature pyramid alone. There is no need here for a promptable general segmenter such as the Segment Anything Model (SAM) [45]: BIOSCAN imaging is standardised enough that prompting is unnecessary, and SAM masks are less exact at precisely the thin structures that matter in this setting (Section 4.3). The consequences of imperfect mattes are examined in Section 4.3.1 (along with one case in which the mask was correct but the *photograph* itself showed a detached leg instead of a whole animal).

3. Related Work

The literature underpinning the thesis is reviewed in this chapter. The material is a revised and reorganised form of the author's previous independent study [27], restructured around the questions posed in Section 1.2. These ask which detector families are suited to tray imagery, what has been attempted on arthropods specifically, which datasets are available, and what is still missing, because that missing space is the one this thesis occupies.

3.1 General-Purpose Detectors as Transfer Baselines

For a new detection task the usual starting move is to fine-tune a strong model pretrained on COCO; for tray imagery, however, which such model one picks is far from a neutral decision.

DETA and one-to-one assignment. DETA [36] leaves the Deformable-DETR architecture untouched and instead swaps one-to-one Hungarian matching for IoU-based one-to-many label assignment paired with NMS; under matched settings it reports strong COCO Average Precision (AP) and markedly faster convergence. With roughly 0.2 B parameters on a Swin-L backbone, the public `deta-swin-large` checkpoint [46] offers a high-capacity starting point for transfer. Its chief relevance to the present work lies in what it shows about assignment. Early in training, one-to-one matching yields a weak signal, since most predictions are assigned to "no object"; one-to-many assignment brings back a dense signal, at the cost of reintroducing NMS. On a tray holding several hundred tightly packed instances, both sides of the trade-off cause harm: the one-to-one signal is sparse relative to the number of objects present, and NMS is likely to suppress animals that genuinely overlap. Independently of assignment, the parameter count excludes DETA from every deployment setting considered in this work.

RT-DETR and end-to-end real-time detection. RT-DETR [37] makes the case that NMS is more than a post-processing detail: it is a source of latency and of sensitivity to the chosen threshold. The model drops it yet keeps real-time throughput, by way of an efficient hybrid encoder that separates intra-scale interaction from cross-scale fusion, and by way of uncertainty-minimal query selection, which initialises the decoder queries from features that are confident in both classification and localisation. For this thesis the practical lesson is twofold: the quality of query initialisation changes what is found, which matters when the objects are small and many, and the depth of the decoder offers a speed control that needs no retraining.

RF-DETR under occlusion and label ambiguity. Sapkota et al. [47] set an RF-DETR variant against YOLOv12 for detecting green fruit in orchard imagery, a domain sharing the salient difficulties of tray imagery: targets that are small, occluded and camouflaged, and that human annotators themselves label inconsistently. The result they report depends on the metric rather than pointing one way. RF-DETR achieves the best mAP@50 in the single-class (0.9464) and the multi-class (0.8298) settings alike, whereas the YOLOv12 variants stay ahead on the stricter mAP@50:95 (0.7620 for YOLOv12N; 0.6622 for YOLOv12L multi-class). This is precisely the find-versus-place split described in Section 6.2.2, and no other observation in this literature carries over so directly: it predicts that a transformer detector and a YOLO-family detector run on the same tray data ought to be compared at several IoU thresholds, since a comparison at a single threshold can crown either one. Chapter 6 is organised with this in mind.

RF-DETR [39] itself belongs to this lineage: it pairs a self-supervised DINOv2 backbone [30] with deformable attention and end-to-end, NMS-free prediction, and is packaged as a training and deployment pipeline. It is one of the two detectors this thesis evaluates; Section 5.1 gives the reasons for that choice.

3.2 Arthropod-Specific Detection and Segmentation

flat-bug. flat-bug, introduced by Svenning et al. [41], is a general method for detecting and segmenting terrestrial arthropods across different imaging systems. The observation that motivates it is exactly what makes the whole problem difficult. Arthropod appearance, scale and density vary so strongly between monitoring setups that a model trained on one setup transfers poorly to the next; each new deployment therefore triggers another round of manual labelling and retraining. flat-bug addresses this by construction. It brings together 23 sub-datasets covering laboratory and field regimes (6,131 annotated images and 113,550 annotated arthropods) and evaluates under domain shift explicitly by means of stratified cross-validation and leave-one-dataset-out protocols. For its best model it reports a validation F1 score (the harmonic mean of precision and recall) of about 94 %.

Three of its design choices bear directly on this thesis. The first: it builds on YOLOv8 instance segmentation but replaces fixed-size inference with a multi-scale pyramid tiling scheme, so that arthropods covering a wide range of sizes within a high-resolution image are each viewed at a suitable scale. The second: training draws tile-like crops at controlled zoom levels and removes or inpaints instances that are only partly visible, so that the conditions of training match those of tiled inference. The third: deduplication works on polygon IoU rather than box IoU, using a two-stage procedure that keeps polygon intersection tractable, and this matters exactly when specimens touch.

flat-bug acts as the second detector in this thesis, and it is the appropriate foil for RF-DETR for one specific reason: its backbone was pretrained on *real* arthropod imagery, whereas RF-DETR starts from generic COCO/DINOv2 features. If a difference between the two emerges, it is a priori ambiguous between architecture and pretraining domain; for that reason Chapter 6 reports differences between the two models as trained (Section 6.4). A further point is that flat-bug is single-class (“insect”): it localises without identifying, so any order-level label set has to be introduced by fine-tuning.

Robotic and semi-automated sorting. A complementary strand of work automates the physical handling instead of the image analysis. The DiversityScanner [18] employs a robot to pick individual small invertebrates out of a sample, photograph them, classify them to family with a CNN, and move them into a well plate for DNA barcoding. The BIODISCOVER imaging robot [48] follows the same path (specimens imaged and classified one at a time, at 0.98 average accuracy), and its successors estimate biomass from the same image sequences [49]. No existing systems come closer to an end-to-end replacement for the sorting bench, and they are instructive as a contrast: that route dissolves the detection problem by physically isolating a single specimen at a time, paying for it with throughput that scales with the specimen count and with hardware that is far from trivial. The approach taken here, detecting everything in one photograph, has the opposite cost profile.

Tray, bulk and pitfall imagery. A number of systems photograph catches in trays or dishes, yet none reports the quantity measured in this thesis. The nearest neighbour in imaging setup is BeetleFlow [50], with top-down photographs of 1,506 trays of pitfall-derived carabids from the National Ecological Observatory Network (NEON). Its specimens, however, are pinned and spaced apart, and its detector is a single-class open-vocabulary model prompted with “a beetle”. The detection stage is assessed solely by exact-match count agreement (97.81 % of trays); neither AP nor IoU-based precision or recall is reported. Schneider et al. [51] photograph bulk samples of about 20 individuals in trays and dishes (517 images, 13,059 individuals) and sort the crops into 18 functional groups at 86 % top-1 accuracy. Among published work this comes closest to the class granularity adopted here, but it is a crop-classification result rather than a detection one. Blair et al. [52] batch-image pitfall-collected carabids on tiles, 5 to 220 per photograph, and identify 25 species at 84.6 % from 68 hand-engineered morphometric descriptors. That is the earliest demonstration of pitfall imagery as a recognised problem domain, and it reached species level without localising anything at all. Lim et al. [53] release order-level bounding boxes on sticky cards at 1200 dots per inch (24,746 boxes, 7 orders), the closest match to the annotation style of this thesis, but supply no detector. InsectRoleVision [54], from the flat-bug group, localises arthropods in bulk imagery with YOLOv11 at above 93 % and classifies the crops at 80 %; this thirteen-point gap between finding and naming is taken up again in Chapter 6. Boyd et al. [55] detect three beetle species in laboratory arenas at F1 0.99, but with a random 80/20 split across consecutive time-lapse frames of the same individuals and without a held-out test set. Their figure therefore shows why the unit of a split matters (Section 4.7) rather than serving as a performance reference.

Out-of-distribution evaluation. Two studies put a figure on what an arthropod detector loses when the imaging pipeline changes while the metric is held fixed. In the leave-one-sub-dataset-out experiment of flat-bug itself [41], leaving a sub-dataset out of training shifts F1 by −7.1 % (confidence interval −12.1 to −0.4), with recall falling (−10.3 %) and precision rising (+4.5 %): the detector becomes conservative. Ştefan et al. [56] train YOLO detectors on curated citizen-science imagery and apply them afterwards to fixed-camera field time-lapse. On that unseen imagery their best model attains a class-agnostic localisation F1 of 0.702, against the 0.889 the same authors report for citizen-science test images drawn from the training distribution. Because the two figures stem from separate studies, each at its own grid-searched operating point, the 0.187 difference is an indication of the domain change and not a controlled before-and-after. At the level of single arthropods the same model localises 91.21 % of Hymenoptera, 80.69 % of Diptera and just 56.10 % of the remaining flower visitors, which were smaller and blurrier. Classification reaches 80.45 % for Hymenoptera, 66.90 % for Diptera and 92.51 % for the residual group, the last value being high because the model rarely mislabels the two common orders *as* that group. False positives per image are also reported, a quantity that this thesis adopts. Together with the −7.1 % of flat-bug, these numbers set the yardstick against which any synthetic-to-real result in this domain must be placed, with the caveat that both are F1 deltas while this thesis reports a mAP@50:95 delta. They therefore bound the order of magnitude rather than providing a like-for-like comparison.

3.3 Datasets for Insect Monitoring

Three complementary imaging regimes are covered by the datasets available, and the gaps between them define the problem this thesis takes on.

BIOSCAN-1M and BIOSCAN-5M: standardised specimen imagery. BIOSCAN-1M [14] couples about one million microscope images with DNA barcodes and hierarchical taxonomic labels, and is distributed as several derived packages (from the 2.3 TB originals down to cropped 256-pixel images at 7 GB). The cropping is done by a DETR model fine-tuned on 2,000 annotated insect images: the highest-confidence box is taken, expanded and resized. BIOSCAN-5M [15] extends this to 5,150,850 specimens, 47 countries, 10 classes and 55 orders, and adds geographic and size metadata, among it `area_fraction` and `scale_factor`. Both collections are heavily long-tailed and are usually evaluated at coarse ranks or on restricted subsets.

The property that matters most for this thesis is one the BIOSCAN papers do not stress, since it is incidental to their purpose: every image holds exactly one specimen, photographed against a uniform background, with metadata sufficient to recover its physical size. The collection is thereby a library of composable, correctly labelled and correctly scalable object instances, which is exactly what a synthetic detection corpus requires and what no detection dataset supplies directly.

MassID45: bulk trays with tiny instances. MassID45, introduced by Orsholm et al. [43], is constructed around unsorted bulk trap samples. It comprises 49 high-resolution bulk images (45 megapixels, 8192 × 5464) drawn from 45 Malaise-trap samples, together with 17,937 instance masks in COCO format, three metabarcoding replicates for each sample, and 35,510 individual specimen images with barcodes gathered after sorting. The annotation procedure paired watershed initialisation with interactive refinement and expert labelling, where the expert labelling was restricted by sample-specific taxonomies derived from the specimen-level barcodes.

Of the published datasets, MassID45 is the nearest analogue to the target domain, and it departs from that domain in two respects that matter. It is a *Malaise* bulk sample, densely packed and with little debris, while the pitfall imagery treated here is sparser and heavy with debris. Size is the more important difference: 49 bulk images form a benchmark rather than a training corpus for a 22-class detector, and closing exactly that gap is the purpose of synthetic generation. Its methodological findings nonetheless carry over. The authors tile at 512 × 512 with overlap and merge through SAHI and non-maximum merging. They further observe that supervised fine-tuning does substantially better than zero-shot pipelines (mask AP@50:95 of 43.5 for a fine-tuned Mask DINO against 27.1 for a Grounding DINO + Segment Anything Model (SAM) zero-shot pipeline). A margin of that size argues directly against relying on open-vocabulary models in this setting, and in favour of finding a way to obtain in-domain supervision. Two further points come from the benchmark. Its AP@50 of 0.81 to 0.83 alongside AP@50:95 of 0.43 is the state of the art on genuinely dense, overlapping bulk residue; a higher figure obtained on separated specimens in a spread tray, as in Chapter 6, reflects easier imagery rather than a better method. Expert annotation of its 45-megapixel bulk images likewise reached 23 orders and 92 families but only 17 species at high confidence, and for most taxa species-level identification was described as practically impossible. Order level is therefore close to the practical ceiling for tray imagery at this resolution, and that is why it is the level chosen here.

AMI: field imagery and domain shift. The AMI benchmark for fine-grained insect recognition in the wild, introduced by Jain et al. [24], explicitly targets the distribution shift between curated human-captured images and automated camera traps. AMI-GBIF holds roughly 2.5 M curated images spanning 5,364 moth species; AMI-Traps holds 2,893 expert-annotated camera-trap images from five countries, with 52,948 labelled insects. According to the authors, a simple mixed-resolution augmentation raises out-of-distribution accuracy on trap data while leaving in-distribution accuracy intact.

AMI is a classification benchmark, not a detection benchmark, and states as much: object detection is left out of the paper's evaluation and treated as a precursor step. What it does for that step bears directly on the present work. Its localiser is a Faster R-CNN trained on a synthetic copy-paste corpus of 5,000 images built from roughly 4,000 SAM-segmented insect crops pasted onto empty backgrounds. This is the construction of Chapter 4 at a scale more than an order of magnitude smaller, without physical size calibration, and with no detection metric reported for the result. No structural fact in this review carries more weight: a leading benchmark in the field already depends on cut-and-paste synthesis for insect localisation and never measures whether it works. The contribution of this thesis is accordingly the measurement, along with the calibration and the protocol that give the measurement its meaning, rather than the idea itself. The classification transfer gaps of AMI (11 percentage points on the binary task, 6.5 to 36.7 on species, depending on region) are also the best external calibration for how much *naming* degrades off-domain.

3.4 Multimodal Identification

CLIBD [57] maps insect images, DNA barcodes and taxonomic text into one shared embedding space by means of a contrastive image-text objective in the manner of CLIP (Contrastive Language-Image Pretraining), and specimens are identified through nearest-neighbour retrieval against a database of embedded records. Because inference works by retrieval, new specimens can be added to the reference database without any retraining, and both seen and unseen taxa are supported.

CLIBD plays no part in this thesis, but it defines the shape of the larger system of which this work is a component. In a complete pipeline a detector localises and crops the instances in a tray image, and a retrieval model identifies each crop, optionally cross-checked against barcode evidence. That division of labour is the reason order-level classification is a sensible target for the detector: finer determination is a downstream problem, and one with a better-suited tool.

3.5 Field and Edge Deployment

Two systems demonstrate what is feasible on-device. Insect Detect [16], a solar-powered, open-source do-it-yourself camera trap, combines an OAK-1 smart camera (a camera module with an on-board inference chip) with a Raspberry Pi Zero 2 W host. Detection and tracking run on-device at around 12.5 frames per second, only cropped detections and metadata are stored, and classification is deferred to a second, offline stage. The artificial flower platform it uses to standardise background and posture is a hardware solution to the same problem that this thesis addresses in software. AMS-YOLO [58], a lightweight enhancement of YOLOv8n for maize pest detection, reports 94.2 % mAP@50 with about 2.53 M parameters and runs at 12.25 frames per second on a Jetson Nano.

What this thesis takes from these systems is architectural rather than numerical: the consensus in the field is a two-stage split, with a compact detector running where the data is and a heavier identification model running offline. That puts the present work squarely in the offline half (tray images are captured by a stationary imaging station and analysed afterwards), and it is this that allows the use of models far too heavy for a battery-powered trap.

3.6 Synthetic Training Data in Ecological Vision

Apart from the general cut-and-paste literature treated in Section 2.5.1 [19, 20], three strands of later work are relevant to the construction of Chapter 4.

The first strand concerns where instances come from. The supply of segmented instances limits copy-paste augmentation. X-Paste [59] responds by generating instances with a diffusion model and filtering web crawls with CLIP, reporting +2.6 box AP overall and +6.8 on long-tail classes over a strong baseline, on the premise that for rare classes instance diversity is the binding constraint. This thesis relieves the same bottleneck by a third route: an existing specimen library that is taxonomically labelled and physically measured. Chapter 6 (Section 6.5.5) examines whether instance diversity was in fact the constraint across the range of specimen counts in the corpus.

The second is domain randomisation. On this topic the original references [21, 22] have been superseded by a survey [23] and by evidence that, where acquisition is controlled, synthetic-only training can match real-data training: a YOLOv8 trained solely on domain-randomised renders attains 94 to 99.5 % mAP@50 on real industrial parts [60]. The same line of work carries a warning of direct relevance to this thesis. A YOLOv11 detector trained only on synthetic data reached 0.91 mAP@50 on a hidden real test set, yet its synthetic validation metrics were misleading predictors of that real performance [61]. A score on a synthetic held-out split is a quantity internal to the generator and not an estimate of deployment.

The third is the same construction applied outside entomology. Kim et al. [62] extract 79,196 cell crops from diluted brightfield microscopy using rule-based segmentation, composite 50 to 500 of them into each of 5,000 annotation-free synthetic images, train a Mask R-CNN on those images alone, and evaluate on real held-out images. They obtain AP@50 of 0.36 on dense and 0.46 on sparse real samples, with size distributions verified against flow cytometry. Their pipeline is almost identical to the one used in this thesis; the far lower real-data score on dense, overlapping cells is the frank counterweight to any claim that the construction works uniformly.

What all of this work largely lacks, and what this thesis attempts to provide, is a systematic account of which properties of the generator matter. Published applications usually report that synthetic data helped, without distinguishing whether the gain came from the image count, from specimen diversity, from background realism or from the geometric fidelity of the generated labels. As Chapter 4 shows, these carry very different costs and very different ceilings.

3.7 Tiled Inference

Tiling at both training and inference for small objects dates back to Ünel et al. [63], whose 512 to 640-pixel patches became standard practice. Slicing-aided inference and fine-tuning were formalised by SAHI [42], which reports AP gains of +5.1 to +6.8 points across three detectors on standard small-object benchmarks. A resolution-adaptive variant [64] gains a further 1.7 points of mAP@50 over SAHI on VisDrone while cutting inference time. MassID45 uses a standard recipe of 512-pixel tiles at 60 % overlap with SAHI and non-maximum merging at IoU 0.5 [43], and adaptive tiling at both training and inference is the central device of flat-bug [41]. Duplicate detections along tile seams are a known consequence of sliced inference: each of two partial detections of an object straddling a boundary scores lower than a detection of the whole object, and score-ordered suppression discards the complement rather than merging it. In response, the newer work moves towards area- or topology-aware merging [64]. A preprocessing gain of comparable size from a different mechanism (motion-informed enhancement of small insects in time-lapse, which lifts micro-F1 from 0.49 to 0.71 [65]) serves as a reminder that gains of this scale are large yet by no means specific to tiling. The tiling literature treats slicing as independent of architecture; Chapter 6 shows that it is not.

3.8 Positioning of This Work

To the author's knowledge, no published work reports localisation-quality detection metrics on top-down photographs of pitfall-trap trays. The nearest systems evaluate exact-match specimen counts on trays of pinned carabids [50], classify batch photographs of pitfall catches from morphometric descriptors without localisation [52], or classify crops taken from bulk tray images [51]. A dataset with order-level bounding-box annotation of high-resolution trap substrate exists but comes without a detector [53], and a leading moth-monitoring benchmark leaves detection out of its evaluation while it trains its own localiser on a copy-paste corpus and never evaluates that localiser [24]. Strong general detectors exist but need in-domain supervision; flat-bug provides an arthropod-specific model and a tiling methodology but is single-class; MassID45 gives a realistic benchmark but no training corpus; BIOSCAN gives millions of labelled specimens but no scene-level detection data.

This thesis brings the two ends together. From the labelled, physically calibrated specimen library of BIOSCAN it synthesises scene-level detection data in the tray domain. Under a protocol designed so that the answer is not an artefact of the generator, it then asks how far this carries, which architecture and which inference regime exploit it best, and what real annotated imagery still has to contribute. The construction itself is not new; the thesis adds the measurement of what it buys on held-out real trays, the physical calibration and specimen-disjoint protocol that give that measurement its meaning, and the diagnosis of where it stops.

4. Synthetic Corpus Construction

One central claim underlies this thesis: instead of being annotated by hand, training data for insect detection in tray images ought to be generated. A small quantity of real annotated imagery is folded into the generated corpus to provide the domain realism that the generator alone cannot supply.

The construction of the corpus is set out in this chapter stage by stage: selection of source specimens, then segmentation, scale calibration, composition, contamination, annotation and finally export. The real annotated trays are treated next, both their partitioning and the way they are mixed into the synthetic training split. Chapter 5 then covers the two detectors trained on the resulting corpus together with their training recipes.

4.1 Pipeline Overview

The generator takes three inputs and produces a detection corpus in COCO format: a library of specimens, each photographed and identified individually; photographs of empty trap trays; and a small collection of real annotated tray photographs. Its stages, shown in Figure 4.1, are as follows:

1. **Specimen selection** (Section 4.2). Records come from the BIOSCAN collections, limited to the target label set and subsampled within each group under a cap.
2. **Cutout extraction** (Section 4.3). The dish is located in each specimen photograph, BiRefNet segments the specimen (with a retry on diffuse responses and a classical fallback), and post-processing turns the mask into an RGBA cutout. The cutout area is then checked against the pixel area of the specimen recorded in the BIOSCAN metadata. The size metadata and the dish radius travel with the cutout. Cutouts with an area under a third of the metadata value, or over four times it, are discarded, as are the specimens on a short blocklist that inspection showed to be fragments or members of excluded taxa.
3. **Split construction** (Section 4.7). Each group's cutout pool is divided into disjoint training, validation and test pools, at the granularity of the individual physical specimen, *before* a single image is generated.
4. **Tray composition** (Sections 4.4 to 4.5.4). A schedule of per-tray class recipes is constructed for each split, such that every group appears, as does every pair of common groups. Every tray then samples its specimens in the prescribed proportions from the pools of its split, renders each one at physical size, rotates it and places it. A chain of feathering, defringing, shadowing and blending then composites each specimen onto a real tray background, which has been resized to the deployment pixel density.

5. **Contamination** (Section 4.6). On a configurable fraction of trays a procedural soil bed is laid down, its statistics taken from measurements of real soiled trays; specimens are lightly stained and set on top of it in touching clusters.
6. **Annotation and export** (Sections 4.8 and 4.12). Each placement is written to a log, and the logs are turned into COCO annotations.
7. **Mixing with real data** (Sections 4.9 to 4.10). The real annotated trays are partitioned by physical tray, oversampled and folded into the training split; a held-out real test split is written out separately.
8. **Verification** (Section 4.13). The invariants on which the corpus depends are checked by assertions over the generated output rather than over the configuration.

Since each placement is something the generator does deliberately, the position, extent and class of every synthetic object are known precisely. Per image, annotation costs nothing, and the only bound on corpus size is compute.

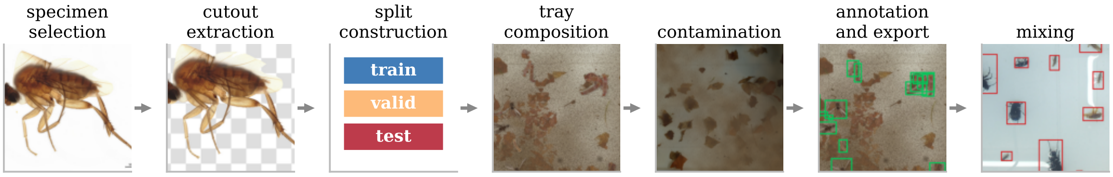


Figure 4.1: The generation pipeline, with one thumbnail per stage. BIOSCAN specimen records are segmented into RGBA cutouts carrying their physical size metadata, partitioned into specimen-disjoint train, validation and test pools *before* any image is generated, composited onto real tray backgrounds at their true physical size under a class-balancing schedule, optionally soiled, exported with boxes taken from the alpha matte, and finally mixed with a small set of real, group-disjoint annotated trays.

4.2 Source Material and Class Selection

4.2.1 The label set

The label set consists of 22 orders, and two design decisions shape it. The first is that every label is an order. Two taxa that the source collection labels at the rank of *class*, Diplopoda (millipedes) and Chilopoda (centipedes), each collect several morphologically unrelated orders under one label (Section 2.1.1). A field ecologist who records “millipede” would appreciate the pragmatism of such a label, yet it would demand that a single detector head learn several distinct appearances at once, so both are resolved to their constituent orders. The second is that any taxon with fewer than ten usable specimens is excluded, because it cannot be evaluated under a specimen-disjoint split (Section 4.7). The specimen counts given below are BIOSCAN records in the source pool before the quality filter of Section 4.2.2; the numbers that actually reached trays are smaller and are listed in Table 4.7.

- **Diplopoda** is represented by **Polyxenida** (50 specimens), its dominant constituent order; the minority orders Julida (1 specimen), Chordeumatida (2) and Polydesmida (7) are excluded.

- **Chilopoda** is represented by **Lithobiomorpha** (71 specimens) together with **Scutigeromorpha** (31), while Geophilomorpha (3) is excluded; together these account for the 105 specimens that BIOSCAN holds under Chilopoda. A house centipede bears no resemblance to a stone centipede.
- **Raphidioptera** (2 specimens, one adult and one larva) and **Siphonaptera** (6 specimens) are excluded, since a group of two specimens would be trained on one animal and tested on the other.

A search of the source collection for further material on the excluded taxa found none; the counts above are everything BIOSCAN holds for those groups. This is a limitation of the source, not of the pipeline, and it is reported as such.

The label set is:

Lepidoptera	Psocodea	Odonata	Zoraptera
Phasmatodea	Zygentoma	Thysanoptera	Strepsiptera
Archaeognatha	Dermaptera	Diptera	Coleoptera
Hymenoptera	Hemiptera	Orthoptera	Trichoptera
Neuroptera	Araneae	Opiliones	Polyxenida
Lithobiomorpha	Scutigeromorpha		

Taxa absent from the label set. The set reflects what the source collection can provide at usable specimen counts rather than what a pitfall tray contains. The groups that dominate epigeic catches in Section 2.2.1 are springtails (Collembola), woodlice (Isopoda), mites (Acari) and ants (Formicidae, which as Hymenoptera are absorbed into a label dominated by Malaise-caught wasps and bees). These are either absent from the BIOSCAN pool at the counts the schedule needs or cannot be separated at order rank. A detector trained on this set has never seen them; on a real pitfall tray they will be missed, or else boxed and assigned the nearest order it knows. Since the real annotated set (Section 4.9) holds five orders and does not record whether other animals were present, how such animals are treated on deployment imagery is a question for the label set rather than for the generator.

4.2.2 Sampling, cutout cap and quality filter

Abundance in the source collections spans six orders of magnitude: Diptera contributes millions of records, while several target groups contribute fewer than twenty. A per-group cap is therefore applied: at most N_{cap} records are taken from any group, and a group holding fewer than that contributes all of its records. The corpus sets $N_{\text{cap}} = 20{,}000$. Even at that cap the abundant groups use well under 1 % of what is available, while the rare groups are left entirely untouched by it.

The pool of 177,718 cutout records then passes through a quality filter that rests on a piece of BIOSCAN metadata most pipelines ignore: each record carries `image_measurement_value`, the measured pixel area of the specimen in its own photograph. The area of the segmented mask divided by that value is a direct test of the segmentation. A cutout is discarded whenever its ratio falls outside $[0.30, 4.0]$ (that is, the mask captured less than a third of the animal, or more than four times it). This leaves 148,285 cutouts (29,433 dropped, or 16.6 %), of which 144,914 distinct specimens were placed on at least one tray. The bounds are deliberately loose: they catch gross segmentation failures and nothing finer, not inaccurate mattes, and they were set by inspection rather than derived. Where the loss fell matters for the per-class analysis of

Chapter 6. It did not fall on the rare orders as a whole (the fourteen orders of the source pool with fewer than 200 records, counted before the exclusions of Section 4.2.1, lost 10.4 %, the rest 16.6 %), but it was far from uniform. Polyxenida lost 40 % of its records (50 → 30), Odonata 24 % (17 → 13) and Opiliones 28 %. Across the capped orders the loss ranged from 1 % (Coleoptera) to 38 % (Psocodea, 20,000 → 12,441), with Hemiptera, Orthoptera and Neuroptera each losing about a quarter. The filter therefore reshaped the pool one taxon at a time, according to how well BiRefNet copes with each group, and this is one reason Polyxenida ranks among the most starved orders in the per-class results of Chapter 6. Of the kept pool, 17.6 % is STRICT, 63.5 % LOOSE-OK, 15.6 % LOOSE-MISS (a mask outside $[0.5, 2.0]$ of the metadata area but within the wider bounds of the filter), and 3.3 % carries a NO-TARGET or SUSPECT-TARGET tag. The 2,532 NO-TARGET records have no metadata area against which to be checked and so pass the filter by default. Taken group by group, the specimens available to the training split range over three orders of magnitude, from 11,881 down to 6, and the numbers placed across all three splits from 19,082 down to 10 (Table 4.7).

This asymmetry is the structural limitation of the whole approach. Raising the cap costs only more segmentation compute, but no cap setting can conjure specimens that were never collected: for a group represented by six individuals, ten thousand generated images still show the model the same six animals.

4.3 Specimen Segmentation

Each selected specimen photograph is converted into an RGBA cutout. Given the BIOSCAN imaging protocol (one specimen, viewed dorsally, on the uniform floor of a small dish, under controlled lighting) the task is a two-class foreground/background problem, but not a trivial one. Among the structures that discriminate at order level are legs, antennae and wing venation just one or two pixels wide; the rim of the dish and its optical reflections are salient in themselves; and a scale bar is often present in the frame. Across the pool the procedure comes down to BiRefNet on the full photograph for 96.3 % of records, BiRefNet on the dish crop for 2.1 %, and the classical dish-subtraction fallback for 1.6 %. There are five stages, and the fifth is what makes the quality filter of Section 4.2.2 possible.

1. Dish detection. The circular field in which every specimen is photographed, referred to here as the dish, is the second of the generator's two size references: its diameter is taken as 8 mm, so that its radius in pixels converts the photograph into millimetres (Section 4.5.2). BIOSCAN's metadata does not name the vessel, the constant is the project's own, and the empirical check of Section 4.11 is the only test of its value. An ensemble of Hough circle transforms, run at three blur radii across a grid of detector parameters, locates the radius. Every candidate is scored by its proximity to the image centre and by how close its radius comes to a nominal value of 30 % of the shorter image side, and the best-scoring candidate is accepted if its score exceeds 0.5. If no circle is found, a colour-based fallback thresholds a map built from the chroma and lightness channels of the CIELAB colour space, weighted by a central Gaussian, and takes the centroid of the blob that results. If that also fails, the image centre is used. Both fallbacks return the nominal radius, 230 pixels for the 1024 × 768 distribution images. Over the pool a circle was genuinely detected on 34,957 records (19.7 %); the other 142,761 (80.3 %) carry the nominal 230 pixels, which is why the pooled median is exactly 230 with an interquartile range of zero. Because the nominal records are one constant by construction, the pooled figure says nothing about the vessel. The detected subset does carry information. Its radius runs from 187 at the fifth percentile, through quartiles of 211, 221 and 231, to 253 at the ninety-fifth, an interquartile range of 9 % of the median, consistent with a single vessel size photographed at a single magnification. The nominal value lies inside that interquartile range, 4 % above the detected median, so using it wherever detection failed shifts the dish route by roughly 4 % and, with the route weighted at 0.3, the rendered size by roughly 1 %. On four records in five, then, the dish route is a nominal scale; the substitution is bounded; and the physical calibration of both routes rests on Section 4.11.

2. Saliency. The high-resolution configuration of BiRefNet [44] is applied to the full photograph upsampled to 2048 × 2048 (the distribution images measure 1024 × 768), and its output is resized back to source resolution as a saliency map in $[0, 1]$. The choice of BiRefNet follows the reasoning of Section 2.5.3: a decoder that receives high-resolution source patches keeps thin structures better than one upsampling from a feature pyramid alone. On this imagery its single characteristic failure is a *diffuse* response, either the whole dish lighting up or no confident peak anywhere, and two tests detect it (a maximum saliency below 0.5, or more than 30 % of pixels above 0.5). In that case the model is run again on a crop of 1.15 dish radii about the dish centre, so the specimen fills a larger share of the input, and the cropped result is adopted provided it passes the same two tests.

3. Mask extraction and post-processing. A threshold of 0.30 is applied to the saliency map. The connected component nearest the dish centre, with area as a weight, is retained, along with any component of at least 5 % of its area whose bounding box touches that of the main component; this rule is what preserves wings and legs that look detached. A 3×3 morphological closing is then applied. The mask is intentionally *not* clipped to the detected dish circle, since the detector may be slightly off and clipping would cut away wings and legs that genuinely reach beyond the rim.

4. Rejection and fallback. A mask is discarded when it holds fewer than 30 pixels, or when it is a *rim crescent*, the optical artefact of the dish edge, which its radial statistics betray. Such a mask has a mean distance from the dish centre above 0.85 radii combined with a standard deviation below 0.08 radii, i.e. pixels crowded into a thin shell along the rim. A genuine specimen, even one resting against the rim, is spread more widely in radius. Once a mask is rejected, a classical *dish-subtraction* segmenter takes its place: the colour of the dish floor is sampled from an annulus between 0.55 and 0.85 radii, and each pixel within 0.92 radii is scored by its CIELAB distance from that colour. The 82nd percentile of these distances is used as the threshold, and after an opening and a closing the largest component closest to the centre is retained. If both routes fail their checks, whichever mask exists is kept and tagged as unverified.

5. The alpha matte and quality tagging. After a 5×5 Gaussian blur (standard deviation 1), which feathers the edge by one pixel, the mask becomes the alpha channel of the cutout, and the cutout is cropped to the mask's bounding box plus a five-pixel pad. Its area is then set against the record's `image_measurement_value` and a tag is assigned: STRICT when the ratio lies within $[0.90, 1.05]$, LOOSE-OK within $[0.5, 2.0]$, LOOSE-MISS beyond that, SUSPECT-TARGET when the metadata value is itself implausible (below 300 pixels, above 1.5 dish areas, or below 0.1 % of a dish area), and NO-TARGET when there is no value at all. A log-Gaussian agreement score is stored next to the tag.

Together with its matte, each cutout carries the following forward: the `processid`, the stable identifier of the individual physical specimen, which is what makes specimen-disjoint splitting possible and is the key on which the blocklist below is enforced. It carries as well the detected dish radius in pixels and the `scale_factor`, the two size references of Section 4.5.2 (`area_fraction` travels with them but is never used). Lastly it carries the quality tag, tier, area ratio and the route by which the mask was produced, retained so that segmentation quality can be treated as an explanatory variable rather than assumed to be adequate.

4.3.1 Cutout quality, fragment specimens and the blocklist

A cutout that ought not to be used can arise from two distinct failure modes, and each calls for a different remedy.

The first is a *segmentation failure*: the photograph shows an intact animal, but the matte is wrong. Re-segmenting is the remedy.

The second is a *fragment specimen*: the matte is clean, tight and correct, yet the photograph does not show a whole animal. Detached legs and loose body segments do occur in the source imagery, and a segmentation pipeline will cut out such a fragment flawlessly. Here the remedy is to exclude the record at source, because re-segmentation can do nothing. Leaving it in costs more than a noisy box: the model learns that a detached leg is a valid instance of the order.

An automatic detector sees the two cases as the same thing: low mask fill, or an area anomalously small relative to the median of the group. A flag therefore means *inspect this*, and it is the image itself that settles which case applies. Three Odonata records were flagged automatically; inspection of the group's montage confirmed all three and revealed a fourth, so that four fragments were identified in total (legs with the femur-tibia joint visible, and loose body segments). Two of these four also lie outside the area-ratio bounds of the quality filter (Section 4.2.2) and are already removed by it. The remaining two pass the filter and are removed by the blocklist alone, which is why the thirteen Odonata records passing the filter give the eleven placed specimens of Table 4.7.

A **blocklist of 25 specimens** is applied before generation, made up of the 21 specimens from the excluded taxa of Section 4.2.1 and the 4 Odonata fragments (Figure 4.2). Dropping an excluded taxon from the label list does not suffice; it must be blocklisted as well, because a specimen simply absent from the label set would still be pasted onto trays without a label and would teach the model that arthropods are background. Enforcement is verified on the generated output rather than on the configuration: every placement log is read back and it is asserted that no blocklisted `processid` occurs in it (Section 4.13).

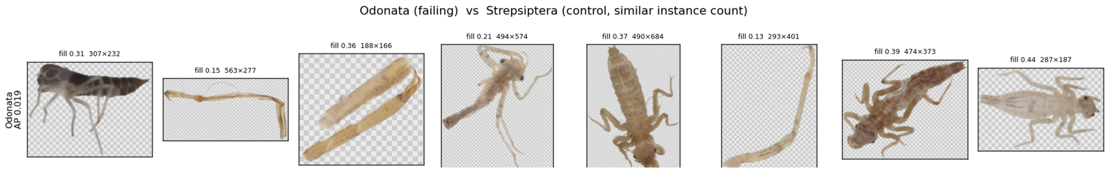


Figure 4.2: Fragment specimens in the Odonata source records. The segmentation mattes are tight and clean, yet what the photographs show is not whole animals but detached legs (with the femur-tibia joint visible) and loose body segments. In the source records each carries the label Odonata, and without the blocklist each would be pasted onto trays as a valid instance of that order.

4.4 Tray Backgrounds

The backgrounds are 146 photographs, 2644 × 2108 pixels each, of real empty trays from the target imaging station. They carry the debris that comes with a pitfall catch (twigs, needles, leaf fragments) but no arthropods, so the background delivers genuine hard negatives at no cost. Real backgrounds also bring the vignetting, rim shadow, surface texture and residual staining of a used tray, none of which a procedural background would reproduce and none of which the model should be permitted to exploit as a shortcut.

These photographs have a native resolution lower than that of the deployment frames; the point is taken up again in Section 4.5.2.

4.5 Placement and Physical Scale Calibration

4.5.1 Motivation for scale calibration

In this problem object size is evidence, not a nuisance parameter. A 3 mm springtail and a 25 mm ground beetle can be told apart by size alone in a large share of cases, so a detector trained on trays whose specimens were pasted at arbitrary sizes learns that size carries no information, which is true of that training data and false of the world. It acquires, in addition, a size prior matching no real imaging geometry, leaving its behaviour on a real tray unpredictable.

For that reason the generator derives a physical size for every specimen and renders it at the number of pixels a real specimen of that size would cover in the tray image. The size comes from metadata; it is not copied from the pixel dimensions of the cutout itself.

4.5.2 The size computation

A `scale_factor` s is present in every BIOSCAN record; BIOSCAN-5M documents it as the ratio between the specimen crop and its resized version [15] (Section 2.1.2). Because the crop is fitted to the specimen at a fixed magnification, s grows with the extent of the specimen in the photograph, and the generator uses $s \cdot 8\,\text{mm}$ as the major axis of the specimen in millimetres. The calibration constant is a choice of the project rather than a documented property of the metadata; Section 4.11 puts it to an empirical test. A pixels-per-millimetre factor follows from the imaging geometry of the tray (a physical width of 320 mm and a pixel width $W_{\text{tray,px}}$). Combining the two gives:

$$\rho = \frac{W_{\text{tray,px}}}{320\,\text{mm}} \qquad \text{(tray pixels per millimetre),} \tag{4.1}$$

$$\ell_{\text{px}} = s \cdot 8\,\text{mm} \cdot \rho \qquad \text{(rendered specimen length in pixels).} \tag{4.2}$$

The cutout is scaled so that its longer side equals ℓ_{px}, with the shorter side scaled by the same factor, which keeps the aspect ratio exact.

In practice Equation 4.2 is not applied by itself. A second, independent physical estimate comes from the detected dish (Section 4.3): dividing the pixel diameter of the dish by its 8 mm physical diameter yields the pixels-per-millimetre of the *source* photograph. Dividing the tray's pixels-per-millimetre by the source's then yields a scale factor that maps the pixel dimensions of the cutout directly onto the tray. If both the metadata and the dish detection are sound, the two estimates should agree; if either is not, they should not. The generator blends them, giving the metadata route a weight of 0.7 and the dish route a weight of 0.3. Record by record the two routes disagree considerably: over a sample of 5,510 cutouts covering every order, the metadata route asks for a size 1.26× that of the dish route at the median (interquartile range 0.48, fifth to ninety-fifth percentile 0.65 to 1.70). The two agree within 20 % on only 37 % of records. Yet what the blend does with that disagreement is close to uniform. Relative to the metadata route on its own, its output for a ratio r is $0.7 + 0.3/r$, i.e. a shrink of about 6 % at the pooled median. Across the eighteen orders having at least thirty sampled cutouts the shrink falls between 0 and 9 % (Thysanoptera 0.997, Araneae 0.908), with no trend against body size ($r = -0.23$ over those orders). Records on which the dish route fell back to its nominal radius (Section 4.3) differ from the detected ones by 1.4 % in the effect of the blend. A factor so nearly uniform is absorbed by the global calibration of Section 4.11, which validates the blended size rather than either route alone; the fixed weight is therefore close to harmless and treats all classes alike. The result is clamped to a minimum of 3 pixels and a maximum of 15 % of the corresponding tray dimension. The clamp is a guard against a bad `scale_factor` and is never triggered on this corpus, since the largest box in a 15,884-box sample reaches 54 % of the cap. The blend is a robustness measure, and one further consequence must be stated. The size actually rendered is not precisely what Equation 4.2 predicts, so every analysis of specimen size relies on the sizes recorded in the placement logs rather than on sizes recomputed from the metadata.

The correctness of Equation 4.2 depends entirely on ρ, and ρ is fixed by the pixel width of the tray image. At a tray width of 320 mm, the background photographs of Section 4.4, which are 2644 pixels wide, would give $\rho = 8.26\,\text{px/mm}$. The real annotated frames from the imaging station, on which the models must ultimately perform, measure $\sim 3473 \times 2570$ pixels instead, that is $\rho = 10.85\,\text{px/mm}$ (Section 4.9 describes a second, larger frame format). Backgrounds are therefore upscaled to a width of 3,473 pixels *before* compositing begins, which yields 3473×2769 trays at $\rho = 10.85\,\text{px/mm}$, matching the station frames of the real set to two decimal places. The interpolated background gains no genuine detail of its own, while each insect is rendered at its true deployment size from the full-resolution cutout. Cutout resolution is not sacrificed: of a sample of 150 cutouts, none would require upscaling at this tray size (the median render is 0.23× the size of the cutout), so every insect pixel carries real source detail.

4.5.3 Composition

Composing a clean tray proceeds as follows. One background is drawn uniformly from the 146 photographs of Section 4.4 and resized to 3,473 pixels in width using Lanczos resampling. A target instance count is drawn uniformly between 30 and 50, and a per-tray pool of 300 cutouts is gathered from the split's specimen pools in the class proportions that the schedule of Section 4.5.4 lays down for that tray, sampling without replacement and topping up with replacement where a group is too small to supply its share. Every placement then takes one cutout at random from the pool and carries out, in this order:

1. **Resize**, isotropically and with Lanczos resampling, to the size given by Section 4.5.2.
2. **Rotate** by an angle drawn uniformly from $[0°, 360°)$, with bicubic resampling and the canvas expanded to hold the rotated image. The rotation is uniform because no orientation is preferred on a drained tray; it is the one deliberate departure from the source imagery, where every specimen is photographed in a standard pose, and it prevents the model from learning that orientation carries information.
3. **Measure** the bounding box of the rotated alpha channel (threshold 10); this is recorded

as the annotation (Section 4.8), with the expanded canvas logged alongside it.

4. **Place.** A position is drawn uniformly within a margin of 5 % of the tray dimensions and redrawn, as many as fifty times, for as long as the canvas rectangle would overlap one already placed. If fifty attempts yield no clear position, the last draw is taken anyway, so on clean trays overlap is rare but not ruled out.
5. **Composite**, via a chain built to eliminate the seam artefacts a detector would otherwise pick up as a shortcut [19]. The chain comprises five steps, applied in the following order:
    (a) a Gaussian of 0.8 pixels feathers the alpha channel;
    (b) a two-pixel weighted blur spreads the colours of the fully opaque pixels into the semi-transparent border, so the feathered edge takes on insect colour instead of the black of the cutout's transparent corners;
    (c) a drop shadow is laid beneath the specimen, formed from its alpha channel shifted two pixels along each axis, blurred with a six-pixel radius, at 22 % opacity;
    (d) the semi-transparent edge pixels (those with alpha between 0.05 and 0.9) are blended 70:30 with the tray colour beneath them;
    (e) the outcome is alpha-pasted.

Clean trays receive no per-instance jitter of colour, brightness or contrast; each specimen appears exactly as it does in its source photograph. The generator implements three further appearance-harmonisation steps, none of which was enabled for the corpus reported here. The first two are a per-tray CIELAB match of luminance and white balance between the pasted specimens and the substrate, and a camera-realism pass that adds sensor noise (sampled from a clean patch of the background) and a slight optical blur to the foreground. The third relights the composite with IC-Light, a diffusion relighting model. Leaving all three out was a practical decision, not a judgement of quality: relighting needs a GPU and diffusion weights, and the corpus was composited on a machine with neither, from pre-existing cutouts. Since the other two passes are written to run after relighting, they were skipped along with it. They are described because the behaviour of the generator is otherwise easy to over-describe, and because a measurement of their effect is a direct extension of the present design (Chapter 7).

Figure 4.3 shows the output of the composition, two trays from the test split at full frame with windows at 1:1, and Figure 4.4 shows the annotation precisely as the detector receives it, on a generated tray displayed next to its placement log.

4.5.4 Class-balanced generation schedule

If the specimens for each tray were drawn uniformly from the split pool, the corpus would inherit the pool's imbalance: with specimen counts differing by 2,000× between groups, the rarest groups would turn up on hardly any trays. Class balance is instead governed by the generation schedule, not by pool size.

Within each split, groups are assigned to tiers by the number of specimens the pool of that split holds for them: *rich* ($\geq$ 200), *medium* (50 to 199), *sparse* (10 to 49) and *rare* ($<$ 10). Every tray is generated from a *recipe*, a set of groups paired with the fraction of the tray's instances each is to supply, and the schedule of recipes for a split is built in two phases. Coverage is secured by the *systematic* phase: two single-group trays for each rich group, one 50:50 tray for each pair of rich groups, and two "spotlight" trays for every group that is not rich. On a

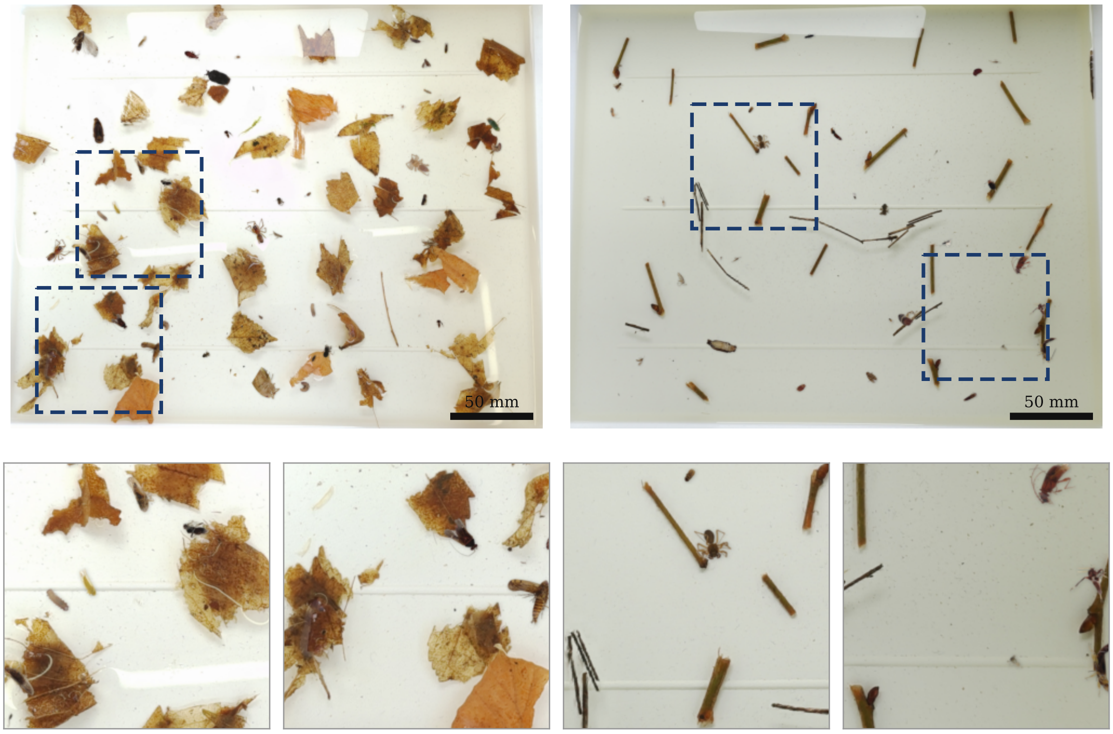


Figure 4.3: Two generated trays from the *test* split, 3473 × 2769 pixels at 10.85 px/mm, with 50 mm scale bars: a dense tray with leaf debris on the left and a sparser tray with twigs on the right. In the lower row are four 75 mm windows at 1:1, which is the scale at which the network receives a tile. No boxes are drawn here; the annotation appears in Figure 4.4.

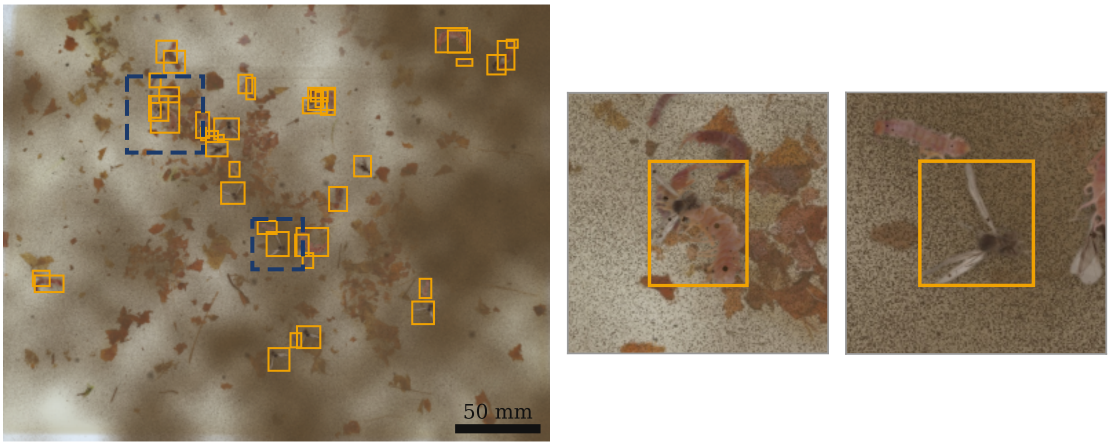


Figure 4.4: The annotation as the detector receives it: the 42 boxes of a single generated tray, taken from its placement log as the extent of the alpha matte (the convention of Section 4.8). Two specimens are shown enlarged: a Lithobiomorpha box measuring 181 × 230 pixels and a Zoraptera box measuring 140 × 153 pixels. The tray follows a rare-order recipe (Lithobiomorpha 30, Zoraptera 12) at a heavy soil level.

spotlight tray that group provides 35 % of the instances while three rich or medium companions share the rest. The *random* phase then uses up the remaining tray budget by drawing from ten recipe types with fixed weights. The types and their weights are as follows: a single rich group alone (8 %); a dominant pair at 70:30 (12 %); a dominant triple at 60:25:15 (10 %); and two, three or five groups in equal shares (12 % each). The remaining types are eight rich-or-medium groups in equal shares (10 %); all rich groups in equal shares (8 %); a rare-group spotlight at 30 % with three companions (10 %); and five to eight groups weighted by pool size (6 %). The schedule is shuffled afterwards so that systematic and random trays interleave. Clean trays are split 12,000 / 4,000 / 4,000 across train, validation and test, and the background of each tray is drawn independently. The result (Figure 4.5) is that annotation counts differ by only 2.2× between groups (from 69,318 for Coleoptera down to 31,702 for Zygentoma). Specimen counts, by contrast, differ over three orders of magnitude (training pool 11,881 to 6; placed across all splits 19,082 to 10).

Rather than a flat, frequency-weighted draw from the pool, which would leave the rare groups almost absent from a quarter of the corpus, the soiled-tray subset (Section 4.6) follows the same balanced schedule; on soiled trays each group accounts for 4.1 to 4.9 % of the instances.

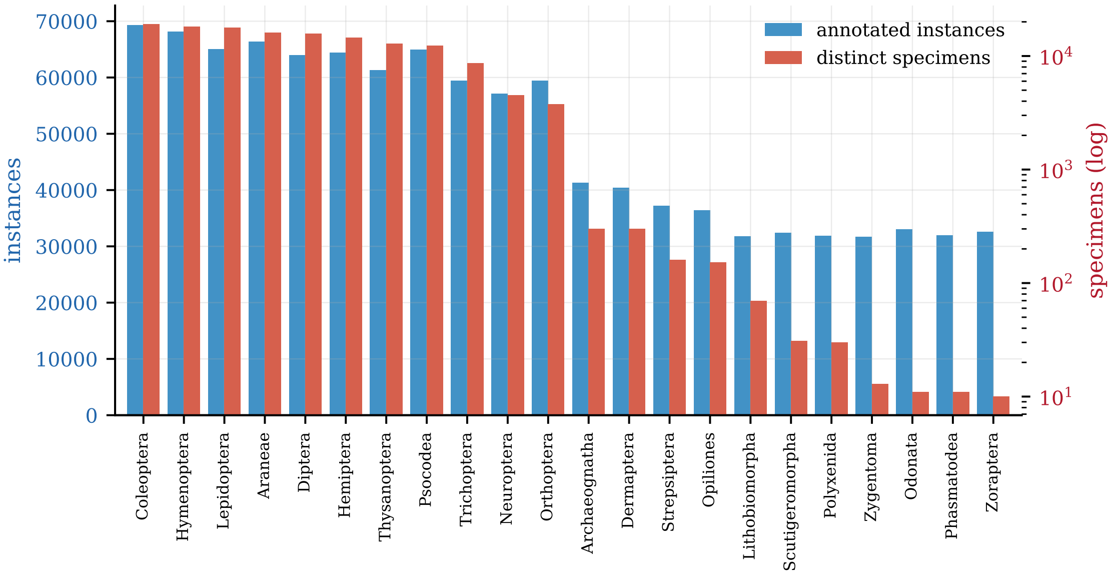


Figure 4.5: Class balance by construction. For every one of the 22 orders, the number of annotated instances in the corpus (left axis, linear) is plotted against the number of distinct source specimens it draws on (right axis, log). Specimen counts range over three orders of magnitude, from 10 for Zoraptera to more than 10,000 for the common orders, whereas instance counts range over a factor of 2.2. The sampler evens out what lies within its control; specimens it lacks it cannot create.

The corollary of Figure 4.5, namely that the sampler cannot invent specimens it does not have, holds for all that follows, subject to one qualification. Instance count is much more even than specimen count, yet it is not even: the eleven orders with over a thousand training specimens sit at roughly 35,000 to 40,000 training instances, and the other eleven at 19,000 to 25,000 (Table 4.7). The rare orders thus hold about half as many instances as the rich ones, and this divide coincides with the divide in specimen count, so instance count is only partly controlled.

That caveat aside, whatever separates a failing group from a healthy one lies upstream of the generator: in the number of distinct animals it has, the size at which they render, and their appearance.

4.6 Modelling Tray Contamination

4.6.1 Characteristics of real soiling

Soil arrives in a pitfall trap along with the animals, and visually a soiled tray poses a different problem from a clean one. Instead of guessing what soiling looks like, the model was derived from measurements taken on real soiled trays from the target imaging station and set against clean trays from the same station. The contrast measured is large: interior brightness falls from roughly 171 on a clean tray to roughly 109 on a soiled one (median 77), the share of dark pixels (intensity below 100) climbs from about 5 % to about 43 %, and the colour cast warms markedly, the red-minus-blue channel difference going from around 9 to around 29. The soiled trays themselves span a wide range (from a lightly soiled tray at brightness 165 with 11 % dark pixels to the heaviest at brightness 64 with 92 % dark pixels), so soiling must be a continuous parameter rather than a binary flag.

The *structure* is every bit as important. Real soil is a continuous, fine-grained bed of sediment ranging from tan to brown, gathered organically into patches with genuinely clean zones between them; it is not a scattering of discrete round dark blobs across an otherwise bright tray. A model trained on trays of discrete blobs encounters clutter that shape alone distinguishes from insects, learns nothing about rejecting look-alike debris, and is then confronted with real sediment at deployment.

4.6.2 The soil model

Soiled trays are rendered by a procedural model governed by one scalar, the soil level $\ell \in [0, 1]$, drawn afresh for each tray (Figure 4.6):

1. A density field is built as the weighted sum of three smooth value-noise fields whose feature sizes are 33 %, 11 % and 4 % of the tray's longer side (weights 0.55, 0.30, 0.15); it is normalised to $[0, 1]$, raised to a gamma of $2.6 - 1.9\ell$, and then multiplied by $0.42 + 0.85\ell$. It is the gamma that keeps large areas clean when the soil level is low and lets the bed spread when it is high.
2. The sediment bed takes its colour from local density: it is interpolated between a tan $(158, 140, 112)$ and a brown $(112, 90, 62)$, both sampled from real soiled trays, and blended onto the tray wherever density exceeds 0.05.
3. In the denser regions only, a weak, large-scale stain towards a dark tone $(72, 57, 40)$ is added at 18 % strength.
4. Fine grain is laid down in three speck layers of decreasing density and increasing darkness (coverage up to 62 %, 30 % and 18 % of a gate that is itself the density field modulated by a clumping field), so clean zones remain free of grain while soiled zones become a granular bed. The darkness originates here: if darkness were instead derived from smooth stain clouds, the trays would read as out-of-focus smudges rather than as sediment.
5. In the densest regions, between six and fifteen soft Gaussian crumb aggregates, scaled by $0.4 + \ell$, are added.

Specimens are then placed *on top* of the finished bed, fully visible, and this is where the soiled trays differ from the clean ones in a second respect: the specimens are laid out in *clusters*. A cluster centre is drawn inside the tray margin, a cluster size between two and five is drawn, and each member is placed at the centre plus Gaussian jitter whose standard deviation is 1.5 to 5 % of the shorter side of the tray. With no collision avoidance, members touch and overlap as specimens in a drained catch do. Before rotation each member receives light soil staining at an intensity i drawn from $[0.12, 0.35]$. Its colour is blended by i towards one of five soil tones, its saturation reduced by $0.4i$ and its brightness by $0.15i$, and a few small soil specks and a faint sheen are added inside the body. In every other respect the compositing chain is that of Section 4.5.3, and the recorded box is again the alpha extent. Specimens are not buried under the clutter, for the reason given next.

Soiled trays number 0.35 of the clean ones (7,000 soiled to 20,000 clean), and on them the soil level is drawn from the upper half of its range, because the clean trays already occupy the clean end of the spectrum.

4.6.3 Occlusion by debris

Covering specimens with clutter looks attractive on the surface, since real specimens are partly buried, but it sits badly with an annotation that consists of boxes alone. When clutter is drawn over a specimen and its recorded box left as it was, the model is being trained to predict a box around an object of which only fragments are visible, a different task from the one on which it will be evaluated. The generator accordingly retains the dense sediment *background*, which is what teaches debris rejection, without corrupting the outlines of the specimens.

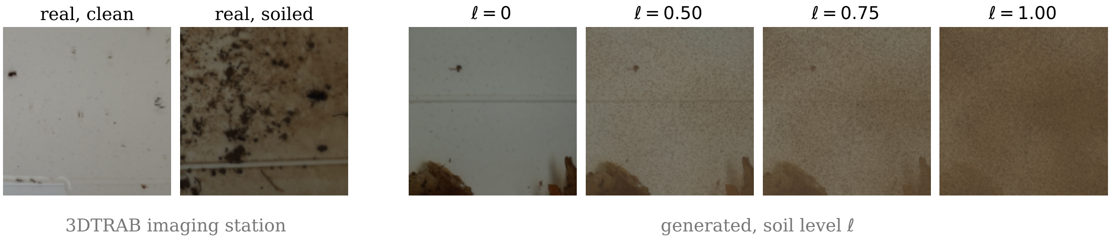


Figure 4.6: Tray soiling, real and generated. Left pair: a clean and a heavily soiled tray from the imaging station. Right: the generated soil ladder at $\ell = 0$, 0.5, 0.75 and 1.0. Every panel is the same 60 × 60 mm of physical tray, resampled to a common size, so the speckle is comparable across the two different pixel densities. The ladder spans from a clean tray to a cast heavier than the real soiled example; the generated debris is finer and more uniform than real detritus, which clumps.

4.7 Specimen-Disjoint Splitting

4.7.1 Leakage through shared specimens

The obvious way to partition a generated corpus is to generate all the images first and divide them afterwards. With this generator that leaks, and it is wrong without any warning sign. In general terms the point is well established for structured data, where blocked or grouped cross-validation is the standard remedy [25] and leakage of this kind is a documented cause of irreproducible results [26]. What is specific here is that the leaked entity is a source specimen, which the generator reuses thousands of times over.

Take a group with two source specimens, from which the generator makes ten thousand trays. Draw the specimens from a single global pool and then split the resulting *trays* 60/20/20, and both specimens land in the training set and both in the test set, at different positions, rotations and soil levels, but still the same two animals. The model is not being asked to generalise to a specimen it has never seen; it is being asked to recognise an animal it has already met thousands of times under augmentation. It will manage that, and the AP reported for the group will be high and meaningless.

What makes this dangerous is that it does not touch all classes alike. For an abundant group with thousands of distinct source specimens, each on only a handful of trays, splitting at image level is roughly honest; for a rare group it is memorisation pure and simple. The per-class table that results thus shows its *highest* scores for its *least* trustworthy classes, and the aggregate mean Average Precision (mAP), which gives every class equal weight, is inflated precisely by the classes with the least evidence behind them. Anyone reading such a table would conclude the opposite of the truth about where the model is weak.

4.7.2 Splitting protocol

The corpus is accordingly partitioned by individual physical specimen, before a single image is generated:

1. Independently for each group, the cutout pool is shuffled under a fixed seed and divided 60/20/20 into training, validation and test *specimen* pools. Division is by `processid`, so every cutout of one physical individual ends up on the same side. Small cases are treated explicitly: a group holding one specimen goes entirely to training and cannot be evaluated; a group holding two gets one training and one test specimen; otherwise the training and validation shares are rounded and at least one test specimen is guaranteed.
2. Trays, clean and soiled alike, are generated split by split, drawing solely on that split's pools. Each generated file carries its split in its filename, and the COCO export routes images by that name instead of re-slicing a shuffled list.
3. The realised specimen count of each group in each split is recorded and reported next to every per-class result, so that a class evaluated on very few specimens can be seen for what it is.

Each of the 144,914 specimens in the corpus is confined to a single split, and the assertion that no specimen appears in more than one split forms part of the verification suite (Section 4.13). Every split contains all 22 orders.

4.7.3 Measured effect of leakage

What this subsection reports is a measurement, not a method; it appears here because it justifies the protocol. A preliminary experiment ran a reduced configuration of the generator (input 800, a cap of 5,000 cutouts, 3,780 trays, and a label set that still contained groups absent from that of Section 4.2.1). Within it, a single RF-DETR configuration was trained both on an image-level split and on a specimen-disjoint regeneration of the same generator configuration. With the image-level split the test mAP@50:95 came to 0.467 (mAP@50 0.775, mAP@75 0.491); with the specimen-disjoint split it came to **0.308** (0.594 / 0.285). Leakage thus accounted for up to a third of the headline figure. The specimen-disjoint corpus, however, also trains on just 60 % of the specimens of each order (one specimen for an order of two), so the drop mixes the removal of memorisation with a loss of training diversity; each figure is from a single run.

It is the per-class pattern that is diagnostic. The abundant groups barely moved (Lepidoptera 0.389 → 0.388, Coleoptera 0.383 → 0.378), since with thousands of specimens each on a handful of trays, image-level splitting had already been roughly honest for them. The rare groups collapsed: Raphidioptera 0.79 → 0.00, Siphonaptera 0.59 → 0.002, Diplopoda 0.75 → 0.35, Phasmatodea 0.66 → 0.32. The leaky table had given its best scores to exactly those groups with the least evidence behind them.

Two further points follow. First, under the leaky split the validation and test scores were close (0.464 and 0.467), and that closeness had been read as evidence that the model was not overfitting. It was nothing of the kind: validation was leaked in exactly the same way as test, so the two agreed with each other while both overstated generalisation. Agreement between two contaminated splits carries no information. Second, these figures are the framework's native metric; the comparison is valid because it stays within one convention and one framework, but the absolute values cannot be compared with the results of Chapter 6.

What the protocol costs is worse numbers, and much worse ones for the rare classes. That is precisely the intention, because the image-level numbers measured a different quantity.

4.8 Annotation Geometry

4.8.1 The box as the extent of the specimen

The log records, for every placement, the specimen's `processid`, its position, its rendered width and height, its rotation angle and its scale metadata, and the COCO export rebuilds the annotations from these logs. The recorded width and height are those of the bounding box of the rotated alpha matte: once rotated, the matte is thresholded at an alpha value of 10 and its bounding box is taken.

The alternative, not used here, is the size of the rotated image buffer. When a raster is rotated with expansion enabled in the imaging library in use, the buffer grows until the whole rotated rectangle fits, and the resulting corners are filled with transparent pixels. The extent of that *expanded canvas* is not the extent of the *specimen*, and a box derived from it is systematically inflated by an amount that varies with orientation. For a source cutout of width w and height h rotated by θ, the expanded canvas has dimensions

$$w' = w\,|\cos\theta| + h\,|\sin\theta|, \tag{4.3}$$

$$h' = w\,|\sin\theta| + h\,|\cos\theta|, \tag{4.4}$$

while the specimen's true extent is the bounding box of the rotated alpha matte, which is smaller at every θ that is not a multiple of 90°. As θ is drawn uniformly from $[0°, 360°)$ (Section 4.5.3), the expected per-side inflation factor for a roughly isotropic shape is

$$\mathbb{E}\big[|\cos\theta| + |\sin\theta|\big] = \frac{4}{\pi} \approx 1.273. \tag{4.5}$$

The expectation is different for a body that is not round; for real specimens, which are elongated and carry appendages that do not fill their box, it must be measured rather than derived.

For each placement both boxes are logged, with the canvas box held under separate column names for provenance, so that the inflation can be measured on the generated output. Over 15,884 boxes from 400 trays, the per-side ratio of canvas box to alpha box (the square root of the area ratio, which is the quantity that maps onto IoU) is **1.294** in aggregate and 1.316 at the median. The generator's own self-check on the same corpus gives 1.295, so per side the recorded boxes are 1.295× tighter than the canvas boxes. In area terms, about **40 %** of each canvas box would be tray rather than insect. The measured value lies above the round-body $4/\pi$ because real specimens are elongated; the two figures describe different bodies and are not in conflict. The spread is wide, because a rotation close to a multiple of 90° inflates almost nothing, and this is the reason a canvas-box convention would survive visual inspection: a fair share of boxes would be barely inflated, and a handful of crops can easily look correct.

Taken together, Equation 4.5 and the IoU arithmetic of Section 6.2.1 show what is at stake. A box that is correctly centred but 27 % too large on each side has IoU $\approx 1/1.273^2 \approx 0.62$ against the true box, which clears a 0.5 threshold and falls well short of a 0.75 one. A model trained on labels of this kind would reproduce them faithfully; against tightly annotated real data it would record a largely intact mAP@50 alongside a near-zero mAP@75, its predictions correctly centred yet systematically too large. On tight real annotation, then, mAP@75 is the metric that measures an error of convention; on the synthetic split it cannot, for the reason set out in the next section.

4.8.2 Verification against real annotation

An error in the annotation convention is invisible in synthetic evaluation, and the precise reason for its invisibility determines what such an evaluation can establish.

Training and test annotations follow the same convention. A model trained on inflated boxes predicts inflated boxes and is scored against inflated boxes, so it is correct by its own standard. Every *relative* comparison made entirely within the synthetic corpus (one corpus size against another, one class against another) is therefore unaffected by the convention, which is constant across all of them. What does depend on the convention is any *absolute* claim about localisation quality and any *cross-architecture* claim about localisation quality (architectures may differ in how faithfully they reproduce an inflated target). So does any transfer to data annotated under the ordinary convention, that is, to any real data.

The hazard applies to generated corpora in general, which is why the label gap was placed alongside the appearance and content gaps in Section 2.5.2. No matter how large a synthetic benchmark is or how carefully it is split, it cannot detect an error in its own annotation convention. The check must be made against annotation produced independently of the generator. That is supplied by the real set of Section 4.9, whose boxes are drawn by hand and sit tight to the specimen (Figure 4.8): the box-scale sweep of Chapter 6, which rescales the predicted boxes on the real photographs, is that check, and its peak lies at unity.

4.9 Real Annotated Trap Trays

The corpus mixes a small set of hand-annotated real tray photographs into the synthetic training data. This section describes that set and how it is partitioned; the mixing itself is treated in Section 4.10.

4.9.1 The annotated set

The real set comprises 92 photographs of pitfall-trap trays from the target imaging station; one is excluded below, which leaves 91. Two native formats are present: 65 station overview frames of 3473 to 3484 × 2570 to 2574 pixels, and 27 frames from a second capture series (three camera heights per tray) of 3947 to 4051 × 2970 to 3032 pixels. The 10.85 px/mm that the generator matches (Section 4.5.2) is the pixel density of the station frames; no separate calibration exists for the second format, and if its frames cover the same 320 mm tray its density is roughly 12.5 px/mm. The frames show clean white and grey trays holding large dark beetles and no soil, so they sit at the clean end of the range spanned by the synthetic soil model. The ecologists of the project annotated them with polygons and bounding boxes at order level. Annotations exist for five orders (Table 4.1). The annotation files constitute the entire record of the set as delivered; they do not state whether every arthropod on the trays belongs to one of those five orders, or whether animals from other groups were present but left unlabelled. The 41 unlabelled regions of Section 4.9.2 are the only evidence in either direction. Every precision figure reported on these frames in Chapter 6 accordingly treats each unlabelled object as a true negative, and those figures are lower bounds if labelling was confined to the five orders. The composition of the set is given in Table 4.1.

Table 4.1: Composition of the real annotated set. Median box is the median of $\sqrt{wh}$ over the boxes, in pixels at native resolution, over all 91 kept frames; every "median box" figure in this thesis uses that definition.

Order	Boxes	Median box (px)
Coleoptera	2,638	130
Orthoptera	156	143
Diptera	111	136
Hemiptera	103	72
Araneae	94	75
All	3,102	129

Coleoptera account for 85 % of the boxes; that is biologically right, since ground beetles dominate pitfall catches, and it means the real set says something about only five of the 22 orders.

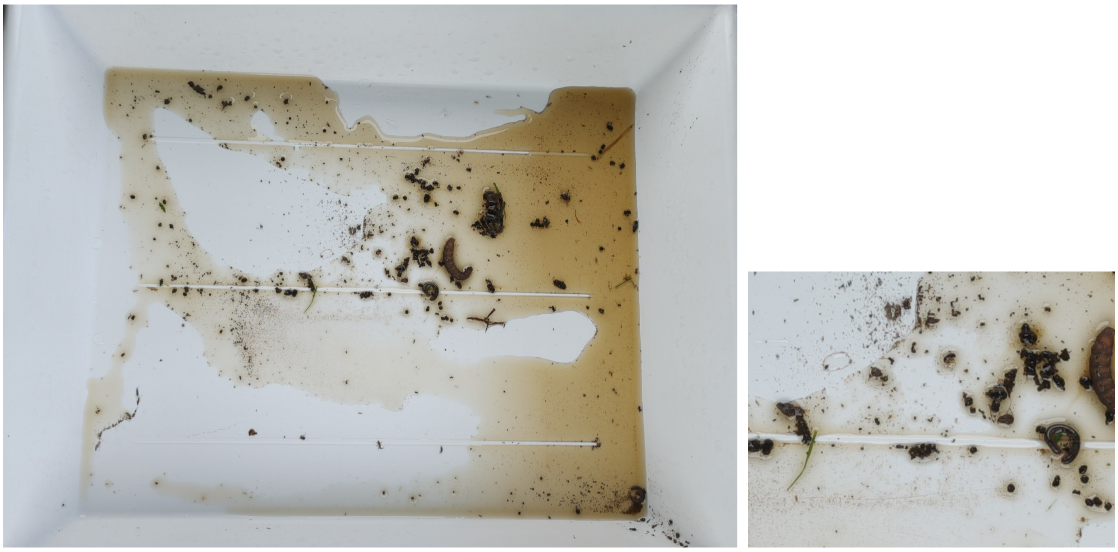


Figure 4.7: A real pitfall-trap tray from the imaging station (left, 3476 × 2571 pixels, shown downscaled) and an 800 × 600 pixel crop of its centre at native resolution (right). Specimens are a few tens to a hundred-odd pixels across in an image several thousand pixels wide. This frame is from the station's pitfall series and carries light debris; the annotated set of Section 4.9.1 has no soil bed.

4.9.2 Annotation recovery

The annotations were exported in the native format of a labelling tool, in which one insect may carry a polygon, a rectangle or both, and the order label may hang on either. A first converter read only polygons and recovered 3,072 labelled insects. Looking through the raw export revealed regions the annotators had added by hand as rectangles with no polygon, and polygons whose label sat on a sibling rectangle. Accounting for every region gives:

3,072	polygon, labelled
28	rectangle only, labelled: *recovered*
2	polygon, label on sibling rectangle: *recovered*
35	polygon, no label anywhere: trains as background
32	rectangle only, no label: trains as background

The recovery is free of ambiguity: no region carries two different labels, and across the 2,340 insects labelled with both polygon and rectangle the two labels never disagree. One photograph, on which all 25 insects were unlabelled, is dropped entirely, since keeping it would teach the detector that real insects are background. The 41 unlabelled insects (1.3 %) remaining in the kept frames are accepted as a source of label noise (the table counts regions: the 67 unlabelled regions minus the 25 on the excluded photograph leave 42 regions on 41 insects, one insect having both an unlabelled polygon and an unlabelled rectangle).

The boxes of the real set were drawn and checked by eye (Figure 4.8); they are tight and correct. This is important because the real set is the sole check on the annotation convention of the synthetic side (Section 4.8), and it establishes the set as a sound benchmark.

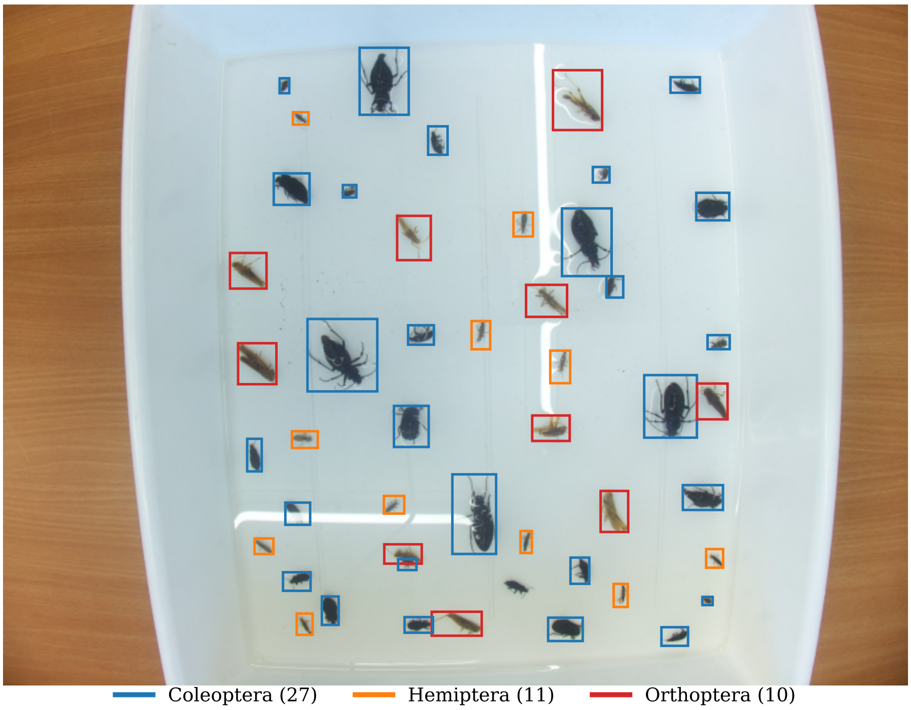

Coleoptera (27)
Hemiptera (11)
Orthoptera (10)

Figure 4.8: Ground-truth annotation on a real tray: a single frame with every annotated box drawn and coloured according to order. Because the boxes fit the specimen tightly, the set serves as a check on the annotation convention used on the synthetic side (Section 4.8).

4.9.3 Group-disjoint splitting

The 91 photographs are not 91 independent samples. They come from **19** physical trays, eleven of them photographed repeatedly at different camera heights and exposures (Figure 4.9); one tray alone contributes 31 frames and 1,474 of the 3,102 insects (47.5 %). A random split by photograph would place the same beetles on both sides, which is the error guarded against in Section 4.7 in another guise.

Whole tray groups are therefore moved as units. Hill climbing from sixty random starts finds the assignment of the 19 groups to three splits; each start is refined by single-group moves until no move reduces the cost further. The cost sums, over the three splits, the squared relative deviation of each order's insect count from its 60/20/20 target (normalised per order, so that a rare order weighs as much as Coleoptera). It adds to this three times the squared relative deviation of the total insect count of the split, plus a penalty of 25 for every order missing from a split although enough tray groups hold it to reach all three. This last term encodes a deliberate priority: a split without an order cannot report AP for it at all, which outweighs a few percent of balance. The result is shown in Table 4.2:

Table 4.2: Group-disjoint split of the real set.

Split	Tray groups	Images	Insects
train	7	46	1,890
valid	6	31	733
test	6	14	479

Two shortcomings of this split are matters of arithmetic rather than failures of the optimiser, and they are reported instead of concealed. Diptera appears in just two of the nineteen tray groups, so across three splits one must go without Diptera, and that is the validation split. Coleoptera, for their part, cannot be balanced, since the single largest group holds 56 % of them and cannot be divided.

The 479 boxes of the test split overstate its size, for precisely the reason this section exists. Its fourteen frames consist of two single station frames and four trays each photographed three times at different camera heights, so most insects in the split carry three annotations. Taking each order's largest per-frame total on each tray as its number of distinct animals gives about 196 (Table 4.3), and four of the five orders originate from a single tray apiece: all Diptera and Araneae are on one tray, all Orthoptera and Hemiptera on another. Each per-order real-data figure for those four orders in Chapter 6 is thus a measurement on six to eleven animals from one tray, and those four groups carry four fifths of the weight of the five-order averages. The chapter says so wherever it matters, and the tray-level bootstrap of Section 6.5.6 reports the width of the resulting intervals.

Table 4.3: What the real test split contains. Boxes count the annotations across the fourteen frames; animals give the estimated number of distinct individuals, obtained by taking, on each tray, the largest per-frame count for each order. One tray holds both Orthoptera and Hemiptera; a second holds both Diptera and Araneae.

Order	Boxes	Animals	Trays
Coleoptera	374	161	six
Hemiptera	33	11	one
Orthoptera	30	10	one
Diptera	24	8	one
Araneae	18	6	one
All	479	196	six

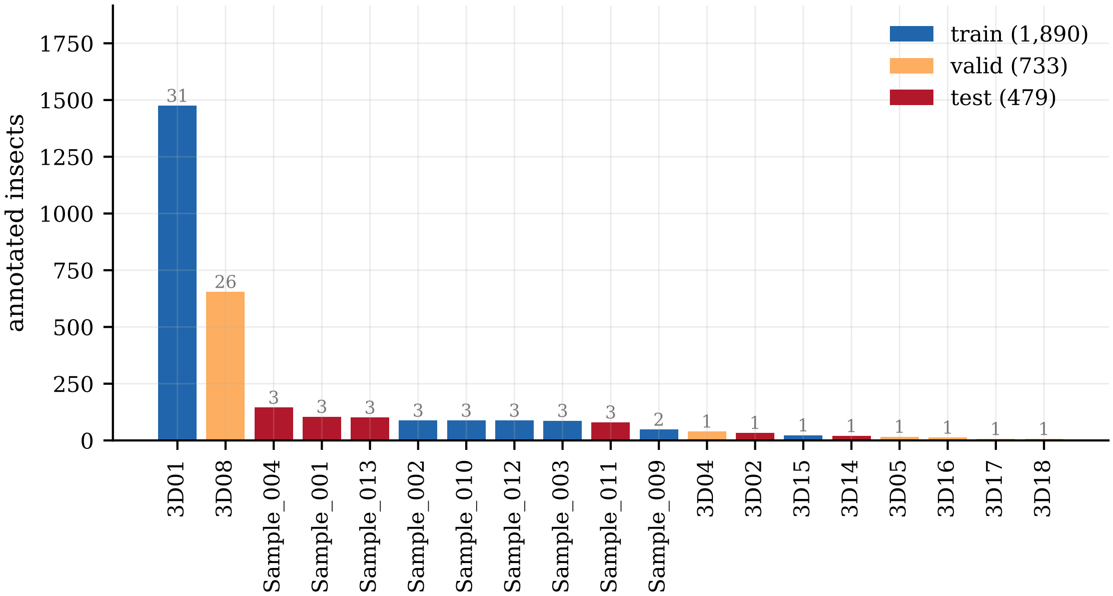


Figure 4.9: Group structure of the real set. Every bar represents one physical tray group; the figure above it gives the number of photographs the group contributed, and its colour marks the split to which the entire group was assigned. On its own the largest group supplies 31 frames and 47.5 % of all annotated insects in the set, and this is the reason the split must be group-disjoint rather than image-disjoint. A split made on photographs would place 31 views of a single tray on both sides of the partition. The axis labels are the tray identifiers used by the station.

4.10 Mixing Real and Synthetic Data

A preliminary experiment supplied the evidence for choosing mixing over sequential fine-tuning. Models pretrained synthetically and then fine-tuned on the real frames alone, in three group-disjoint folds at 800 pixels under the same reduced configuration, collapsed on the synthetic split for which they had originally been trained. This is the classic catastrophic forgetting of a fine-tune on a small target. Real data are accordingly mixed into the synthetic training set instead of being trained on afterwards.

Set against 648,000 synthetic boxes, the 1,890 real training boxes would make up 0.29 % of the training signal. The real training images are therefore repeated 18 times, as symlinks that occupy no disk space, raising the real share to **4.99** % of training boxes (Table 4.4). Oversampling is applied to the training split only; validation and test keep single copies. The share must be read for what it is: 46 distinct photographs, each seen 18 times (828 image instances), five percent of the training signal but nowhere near five percent of its variety, an exchange of influence for the memorisation of those photographs. The real frames enter at native size, and deliberately so, since the generator already renders at the pixel density of the station frames (Section 4.5.2) and resizing them would undo that match. The split fractions apply to real insects rather than frames, which is why the frame counts are 46/31/14 rather than a 60/20/20 division of 91.

Table 4.4: Composition of the mixed corpus, expressed in annotated instances.

Split	Synthetic, clean	Synthetic, soiled	Real	Real share
train	479,997	168,248	34,020	4.99 %
valid	160,155	56,005	733	0.34 %
test	159,933	55,822	479	0.22 %

The fourteen real test photographs are held back completely and written to their own annotation file, and the 77 real photographs that occur in training or validation are listed explicitly. To score against the whole real set would be to measure memorisation; every real-data number in Chapter 6 is computed on those fourteen photographs.

4.11 Residual Scale Mismatch

Because synthetic and real trays are at matched resolution (10.85 px/mm on each side), median box size per order can be compared head to head (Table 4.5, Figure 4.10). Two orders agree to within 1 %, and three do not:

Table 4.5: Median box size (px) of the five annotated orders, real versus synthetic, at matched tray resolution.

Order	Real	Synthetic	Ratio
Araneae	67	67	0.99
Hemiptera	67	66	1.01
Orthoptera	146	84	1.75
Coleoptera	131	70	1.88
Diptera	136	53	2.58

The interpretation adopted here is that the source data, not a fault in the generator, account for this. A wrong calibration constant would move every order by the same amount, and the selectivity points to pitfall traps catching larger-bodied individuals than a taxonomic reference collection holds. The two-order check does not exclude one alternative. The metadata route depends on a crop ratio rather than a physical measurement (Section 4.5.2). Were the specimen crop fitted differently around elongated or appendage-rich bodies than around compact ones, the constant would be right for some orders and wrong for others, which would be selective as well. Separating the two would require the scale bar in the source photographs themselves, which lies outside the pipeline. BIOSCAN's Coleoptera have a median length of 7.9 mm, and just 23 % of them exceed 12 mm, the average of the real trap catch. The real column differs from Table 4.1 because it is measured over the boxes as they stand in the mixed corpus, in which the training frames are repeated 18 times. The difference amounts to at most 8 pixels (Araneae 75 and Hemiptera 72 on the unweighted 91-frame set, against 67 and 67 here) and leaves the ordering of the offsets unchanged.

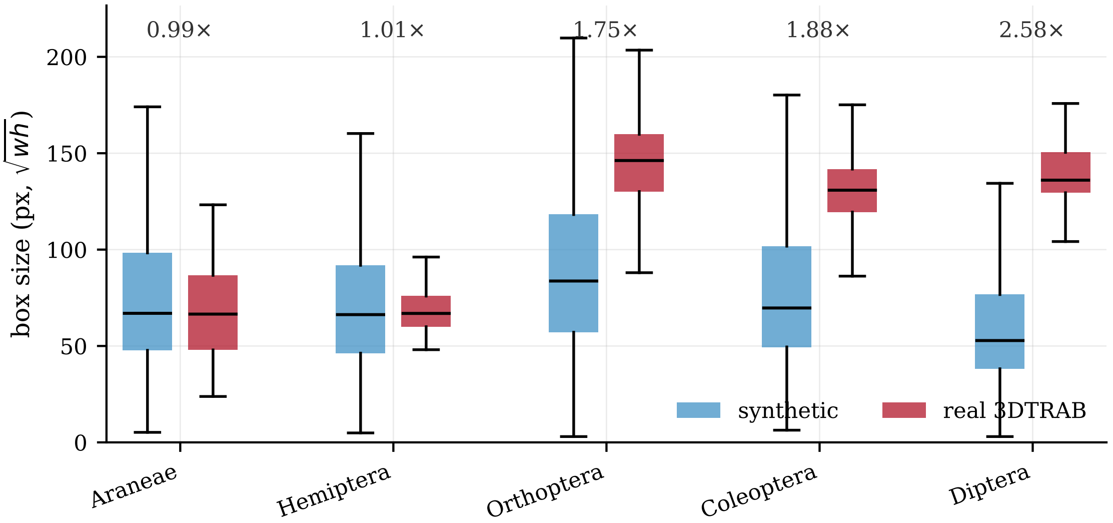


Figure 4.10: Rendered box size, real versus synthetic, for the five orders shared by both sets, measured at matched tray resolution. The boxes mark the interquartile range and the whiskers the 5 to 95 percentiles; the figure above each pair is the ratio of medians. Araneae and Hemiptera agree to within 1 %, whereas Orthoptera, Coleoptera and Diptera come out 1.75 to 2.58 times larger on real trays. The real distributions are narrow since the real boxes of each order come from a small number of animals photographed again and again. Table 4.1 gives the box counts; in the mixed corpus the training frames are repeated 18 times.

Figure 4.11 places 90 mm of generated tray beside 90 mm of real photograph at the same pixels per millimetre. Two of the three gaps from Section 2.5.2 are visible at once, and neither concerns scale. The generated panel carries a warm amber cast while the real photograph is a cool blue-grey (the appearance gap). Leaf and twig debris dominate the generated tray while the real frame shows standing water and specular highlights (the content gap). The generator, as configured for the corpus reported here, addresses neither gap (Section 4.5.3).

A question raised in the preceding sections is also settled by the two matched orders. Since the 8 mm reference dimension multiplies every rendered size, an error in it by a factor k would put every order off by k. Araneae and Hemiptera nevertheless render within 1 % of their real annotated sizes on the training-weighted set and within about 10 % on the unweighted 91-frame set (Table 4.1), compared with 1.75 to 2.58× for the other three orders. Because a wrong global

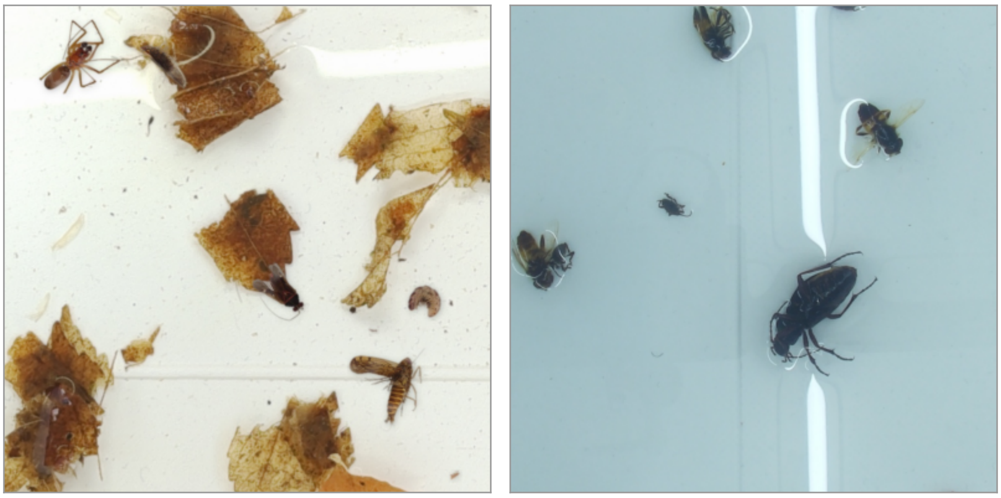


Figure 4.11: Ninety millimetres of generated tray (left) beside ninety millimetres of real photograph from the imaging station (right) at matched pixels per millimetre. Specimen scale agrees; the colour cast and the kind of debris do not.

multiplier cannot act selectively, a global error in the constant is excluded as the explanation for the offsets; the per-order crop-fitting alternative described above still stands.

The body lengths implied across the orders (from 2.8 mm for Thysanoptera up to 18 mm for Phasmatodea, most of them between 6 and 10) are likewise biologically plausible for a reference collection of small arthropods; had the constant been wrong by a factor of two, they would sit at 6 to 36 mm.

What the constant cannot guarantee is per-order accuracy for groups whose `scale_factor` rests on just a few specimens. When the rendered major-axis sizes of the corpus are converted back to millimetres and set beside general taxonomic body-length ranges (a coarse check, not a measurement), all the abundant orders appear plausible: Diptera 5.5 mm, Araneae 6.8, Hemiptera 7.3, Coleoptera 7.6. These are rendered major axes, whereas the 7.9 mm above is the metadata length. Two of the rarest orders fail the check: Scutigeromorpha renders at roughly 7 mm for an animal of 25 to 50, and Zoraptera at 13 mm for an animal of 2 to 4. In each case the rendered size is the one given by the metadata of the few source specimens. Their synthetic scores are not affected, since train and test share the rendering, and neither order occurs in the real set. The issue is one of deployment validity, because a detector trained on this corpus has learned the wrong physical size for those two orders, and it is recorded as a limitation.

No correction of the offset by downscaling the real images is attempted, because that would shrink the two orders that already match and degrade the very data on which evaluation takes place. The prediction recorded before the real-data evaluation was that, should the offset matter, Coleoptera, Orthoptera and Diptera in particular would lag in per-order accuracy on real data while Araneae and Hemiptera held up. The outcome was different; the ordering came out close to the reverse (Section 6.6.7). Since Coleoptera account for 85 % of real boxes, about 94 % of real annotations belong to offset orders. The corrective is the five percent real share in training, as the model does see 34,020 real boxes at their native scale (the real frames enter the mixed corpus unresized, and hence at the two pixel densities of Section 4.9.1).

4.12 The Corpus

Table 4.6: The corpus in summary. Every row is described in the section to which it refers.

Label set	22 orders (Section 4.2.1)
Clean / soiled trays	20,000 / 7,000 (Sections 4.5.4 and 4.6)
Tray resolution	3473 × 2769 at 10.85 px/mm (Section 4.5.2)
Box convention	alpha bounding box (Section 4.8)
Blocklist	25 specimens (Section 4.3.1)
Soiled-tray sampling	class-balanced (Section 4.5.4)
Splits (trays)	16,200 / 5,400 / 5,400 (60 / 20 / 20 %), specimen-disjoint (Section 4.7); clean trays 12,000 / 4,000 / 4,000
Annotations (train / valid / test)	648,245 / 216,160 / 215,755
Real data	46 train frames (18×) + 31 valid (once); 14 test frames held out (Section 4.10)
Real share of training boxes	4.99 % (Table 4.4)
Export format	JPEG q95

Annotations are written from the placement logs in COCO detection format, one box per placement, with category identifiers drawn from a single ordered class list shared by the generator and the training configurations, so that label indices cannot be permuted between them. Boxes are clamped to the image. Before any image is written, the taxon of every placement is checked against the class list, and the export refuses to continue if any fails to resolve: a converter that silently mapped unresolved taxa to the first category would relabel every specimen of an excluded order as Lepidoptera. The generator writes trays as JPEG at quality 95, and they are copied into the export unchanged, so each image is compressed only once. For the models that require it, a YOLO-format export is derived from the same source. The conversion writes the five-token box lines explicitly instead of going through a general converter, because a general COCO-to-YOLO converter run with segmentation enabled emits normalised polygons in the same position in the file. Any reader that expects boxes silently misreads those as centre-and-size. A test asserts the token count.

Table 4.6 summarises the corpus and Table 4.7 sets it out class by class. Its totals match the verification suite (144,914 specimens and 1,080,160 instances), and it shows the decoupling that the generation schedule is designed to produce: instance counts vary by a factor of 2.2, specimen counts across three orders of magnitude, from 19,082 Coleoptera down to 10 Zoraptera. *Specimens* here denotes those actually placed on a tray; the pool a group draws from can be somewhat larger (Coleoptera placed 11,397 of 11,881 available training specimens), and that pool size is what Table 6.5 reports.

Table 4.7: Per-class composition of the synthetic corpus. *Spec.* is the number of distinct BIOSCAN specimens actually placed in that split; *Inst.* is the number of annotated instances. The split is specimen-disjoint, so no individual insect appears in more than one column. Median box is $\sqrt{\text{box area}}$ in pixels at the rendered tray resolution of 3473×2769. Instance counts vary by a factor of 2.2 across classes by construction, while specimen counts span three orders of magnitude.

	Train		Valid		Test		Median
Class	Spec.	Inst.	Spec.	Inst.	Spec.	Inst.	box (px)
Coleoptera	11,397	40,354	3,845	14,410	3,840	14,554	70
Hymenoptera	10,899	40,510	3,638	13,833	3,646	13,781	55
Lepidoptera	10,717	38,089	3,554	12,507	3,612	14,400	94
Araneae	9,671	39,014	3,245	13,806	3,236	13,573	67
Diptera	9,398	37,754	3,171	13,570	3,154	12,617	53
Hemiptera	8,750	38,083	2,923	13,468	2,913	12,844	66
Thysanoptera	7,737	35,811	2,584	12,795	2,580	12,725	25
Psocodea	7,411	38,806	2,466	12,790	2,476	13,380	57
Trichoptera	5,210	34,807	1,737	12,084	1,736	12,521	81
Neuroptera	2,709	34,018	903	11,573	903	11,535	67
Orthoptera	2,257	35,043	752	12,372	752	12,042	84
Archaeognatha	181	24,498	60	8,507	61	8,258	95
Dermaptera	180	23,627	60	8,526	60	8,282	106
Strepsiptera	97	24,446	32	6,750	32	5,975	44
Opiliones	92	24,444	31	6,021	30	5,920	96
Lithobiomorpha	42	19,551	14	6,617	14	5,654	122
Scutigeromorpha	19	19,938	6	6,027	6	6,476	73
Polyxenida	18	19,951	6	5,822	6	6,074	34
Zygentoma	8	19,269	3	6,195	2	6,238	87
Odonata	7	20,468	2	5,998	2	6,593	107
Phasmatodea	7	19,991	2	6,046	2	5,919	157
Zoraptera	6	19,773	2	6,443	2	6,394	111
All	86,813	648,245	29,036	216,160	29,065	215,755	

4.13 Verification Built into the Pipeline

Each invariant on which the corpus depends is asserted over the *generated output*, not over the configuration that was meant to produce it. Every assertion has been negative-tested, by introducing on purpose the fault it guards against and confirming that it fires:

- no specimen turns up in more than one split, checked across every placement log;
- no blocklisted specimen made it onto a tray;
- the recorded boxes are the insect and not the rotated canvas: each recorded box covers strictly less than the canvas from which it was cut;
- each declared class was placed on at least one tray, and no tray file is routed to two splits;
- each annotation points to an existing image whose stored dimensions match the file, all image and annotation identifiers are unique, all category identifiers belong to the declared list, and no box is degenerate or out of bounds;

- on the real side, no tray group straddles two splits, each image is placed exactly once, and no evaluation image occurs in the training list.

The verification suite also reports, without asserting anything, how many distinct individuals each class is built from in each split. It additionally flags any class with fewer than fourteen training individuals or fewer than three test individuals, thresholds set when the corpus was designed.

The motivation is that a fault in generation code which runs without error and produces plausible-looking images leaves no trace in the configuration. Such a fault is caught only by an assertion over the generated output.

5. Detection Models and Training

The corpus of Chapter 4 is the training material; the two detectors trained on it are introduced in this chapter. Section 5.1 presents the two models, a set-prediction transformer (RF-DETR) and an arthropod-specific convolutional detector (flat-bug), together with the axes on which they differ. Section 5.2 gives the training recipe of each in the whole-tray regime and in the tiled regime, along with the construction of the tiled training data. Section 5.3 closes with the software environment and the released repository. Section 6.4 sets out how the comparison between the two detectors is to be read.

5.1 Detection Models

Two detectors take part in the comparison. They were chosen not as the two strongest models available but because they differ along the axes with which this thesis is concerned, prediction paradigm and pretraining domain, while still being trainable on the same data.

5.1.1 RF-DETR

RF-DETR [39] is a member of the DETR family of set-prediction detectors [34, 35, 38]; it pairs a self-supervised DINOv2 backbone [30] with deformable attention and an end-to-end head that needs no NMS. Every experiment uses the Large variant. Weights are initialised from the COCO-pretrained checkpoint; the classification head is re-initialised to fit the label set, while the backbone and the box regression head retain their pretrained values.

Three properties make it relevant here. Global attention lets each query see the whole tray, which is the right structure for distinguishing an insect from a soil clump of similar size on evidence that is not purely local. Dispensing with NMS removes a failure mode that matters exactly when specimens touch. And since the DINOv2 backbone was pretrained without labels, it suits a setting in which in-domain labelled data is the scarce resource.

Two practical constraints apply. The input resolution must be a multiple of the effective patch stride. Augmentation is supplied by an external library whose configuration format changed between major versions, so the library is pinned to the version for which the configured transforms were written.

5.1.2 flat-bug

flat-bug [41] is an arthropod-specific instance segmentation model built on YOLOv8 [33] and pretrained on real arthropod imagery gathered from 23 sub-datasets covering laboratory and field regimes (Section 3.2). It has one class: it localises arthropods without identifying them.

In this work it provides a warm start for a YOLOv8 *detection* model: the state dictionary is intersected by name and shape, so the backbone and neck are carried over while the segmentation head is dropped, and a new multi-class detection head is then fitted on the synthetic boxes. The Large variant is used. At the level of the task this places the comparison with RF-DETR on an equal footing; the cost is that flat-bug's segmentation capability is left unused, because the box-only synthetic corpus cannot supervise it.

The pairing is deliberate. flat-bug's backbone has seen real arthropods and RF-DETR's has not, so any difference between the two is a priori ambiguous between architecture and pretraining domain. How the comparison is read in consequence is set out in Section 6.4.

The two are placed side by side in Table 5.1.

Table 5.1: The two detectors compared. Both were trained at 1536 px input, either on whole trays (downscaled 0.44× from 3473 px) or on 1536 px tiles fed 1:1. The two differ in prediction paradigm, and therefore in what post-processing they need: RF-DETR emits a fixed set of queries matched one-to-one to ground truth and requires no NMS, whereas YOLOv8 emits dense predictions that must be suppressed.

	RF-DETR-Large	flat-bug (YOLOv8-L)
Parameters	36,484,074 (36.5 M)[a]	43,623,570 (43.6 M)
GFLOPs @ 640	n/a	164.9
Architecture	LW-DETR, transformer	CNN, anchor-free
Backbone	DINOv2 (ViT)	CSPDarknet
Initialisation	pretrained RF-DETR (COCO)	published flat-bug Large checkpoint (published arthropod model)
Detection head	reset to 22 classes	reset to 22 classes
Prediction	set prediction, 300 queries	dense, anchor-free
Matching	Hungarian, one-to-one	task-aligned assigner
Post-processing	top-k over queries; no NMS	NMS, IoU 0.5
Input resolution	1536 (multiple of 32)	1536 (multiple of 32)
Resize	square, aspect distorted	letterbox, aspect kept
Optimiser	AdamW (decoupled weight decay)	SGD
Learning rate	10^{-4} decoder, 1.5×10^{-4} encoder	$0.01 \rightarrow 10^{-4}$ linear
Weight decay	10^{-4}	5×10^{-4}
Batch × accum.	4 × 4 (effective 16)	8 whole / 4 tiled, accumulated to 64
Epochs	100 whole, 30 tiled	100 whole, 30 tiled
Multi-scale aug.	yes	mosaic, closed last 10 epochs

[a] Counted directly from the trained checkpoint (rf-detr 1.7.1, RFDETRLarge). The count depends on the input resolution, the growth being entirely in the backbone: the same configuration holds 33.8 M at 768, 33.9 M at 800, 34.5 M at 1024 and 36.5 M at the 1536 used here. The 800 figure matches the 33.9 M Roboflow documents for RF-DETR-L (quoted at its native 704 × 704), so the model is the published Large variant and the larger figure above is the resolution difference, not a different architecture. Roboflow's ≈126 M models are the separate XL and 2XL variants.

GFLOPs are quoted at 640 × 640, the figure Ultralytics reports by default; at the 1536 px input actually used they scale by $(1536/640)^2 = 5.76$, i.e. roughly 950 GFLOPs for flat-bug. The Roboflow release publishes no GFLOPs figure for RF-DETR-L at this input, so the cost comparison of Section 6.5.8 rests on measured inference time.

5.2 Training

5.2.1 RF-DETR

RF-DETR-Large is trained on the full-resolution corpus at an input resolution of **1536**: up to 100 epochs, a batch size of 4 with gradients accumulated over 4 steps, giving an effective batch of 16 on a single GPU. The base learning rate is 10^{-4}, the encoder has its own learning rate of 1.5×10^{-4}, warm-up lasts five epochs, the learning rate drops at epoch 50, and multi-scale training is enabled. The exponential moving average of the weights offered by RF-DETR is disabled, so the scored weights are the raw ones. Early stopping monitors the validation detection metric with a patience of 20 epochs and a minimum improvement of 0.001. No seed was set for these runs: the framework seeds its random number generators only when a seed is present in its configuration, and none was, so the RF-DETR runs are reproducible in configuration but not bit for bit; each configuration was trained once (Section 6.1.1).

The tiled RF-DETR model is trained on 1536-pixel tiles at 1:1 under the same recipe, except that the learning-rate drop falls at epoch 25 and the patience is 10, with a cap of 30 epochs. It was later continued for ~ 80 further epochs ("+cont") to check the training budget (Section 6.5.4).

5.2.2 flat-bug

The YOLOv8-L detection model, warm-started as described, is trained with Stochastic Gradient Descent (SGD): the initial learning rate of 0.01 decays to 0.01 of that value, momentum is 0.937, weight decay is 5×10^{-4}, and there are three warm-up epochs. The batch size is 8 (4 for the tiled model), and in both cases the trainer accumulates gradients up to a nominal batch of 64. Mosaic augmentation is disabled for the final ten epochs, and a fixed seed (42, with the deterministic mode of the trainer enabled) is used for every run. The remaining augmentation keys keep the Ultralytics defaults: horizontal flip with probability 0.5, translation of up to 10 %, and rescaling with a gain of ±0.5 (each object therefore appears at between 0.5 and 1.5 times its rendered size). Jitter in hue, saturation and value is 0.015, 0.7 and 0.4; mosaic is applied to every image until the last ten epochs, and mixup is not used. The whole-tray model has an input size of 1536, a cap of 100 epochs and a patience of 20, so that the two detectors train at the same input resolution (Section 6.3.1). The tiled flat-bug model trains on 1536-pixel tiles at 1:1 with the optimiser settings unchanged (learning rate 0.01, three warm-up epochs), capped at 30 epochs with a patience of 10. The run reported here was stopped after 15 epochs while its validation metric was still rising (Section 7.2).

5.2.3 Tiled training data

In both families the tiled models train on tiles cut from the same trays with the grid that inference also uses (Section 6.3.1): 1536-pixel tiles at a stride of 1280, so that adjacent tiles overlap nominally by 256 pixels. The last column and row are placed flush with the tray edge. Tiles without any annotated box are skipped. A box is retained in a tile only when at least 85 % of its area falls inside it, and it is then clipped to the tile boundary; below that threshold the box is dropped. An insect that is, for instance, half inside a tile is thus present in the pixels of that tile but has no label there, and the network learns to treat it as background. The complete copy of the same insect in the neighbouring tile does carry the label. Both tiled models were therefore taught under this rule to treat an object cut at a tile border as background. This stands, alongside the detectors themselves, as one candidate explanation for their behaviour at seams (Sections 6.5.7 and 6.6.2); the published flat-bug recipe removes or inpaints partially visible instances instead [41], a step not adopted here. The tiled validation run of the RF-DETR training framework applies its edge penalty within 16 pixels of a tile border, whereas the comparison pipeline uses 31 pixels, one identifiable source of the gap between the two evaluation paths of Section 6.5.1. The tiled set contains 166,850 tiles (102,203 for training, 32,331 for validation, 32,316 for test) with 1,645,182 boxes. That is 1.52× the 1,080,160 boxes on the whole trays, where the overlap geometry would predict 1.47×; validation and test amount to six tiles per tray, exactly the three-by-two grid. The two detectors train on identical tiles, flat-bug on a YOLO-format conversion of the same set.

5.3 Implementation and Reproducibility

Training environment. Installing the Ultralytics package pulls in a non-headless build of OpenCV, which cannot be imported on a machine without a display server; the headless build must be substituted, and reinstalling any package that depends on it silently reintroduces the problem. The Ultralytics trainer also enables an experiment-tracking callback automatically whenever the corresponding library is installed, and in this environment that aborts training during setup, because the tracker rejects its default file-system store; the trackers must be disabled explicitly and the cached callback module removed before training starts.

The generator is written as a set of Python Jupyter notebooks, with the Python Imaging Library (PIL) and OpenCV handling compositing and PyTorch the segmentation model; every draw is governed by a single global random seed. Given the seed, the list of source records and the configuration, the corpus can be reproduced: the sorted specimen identifiers determine the split deterministically, and tray composition draws on a seeded generator.

Certain figures quoted in Chapter 4 or Chapter 6 do not appear in the comparison outputs: the soil measurements, the split experiment of Section 4.7.3 and the per-class analysis. They come from the run logs and output files (per-class and per-view CSVs) of the corresponding Jupyter notebooks in the released repository.

Generation runs on a machine other than the one used for training and needs no GPU: the segmentation stage runs once and the generator reuses its cutouts, so composition is CPU-bound. At the full 3473×2769 resolution, the 20,000 clean trays took 5 h 04 min (0.92 s per tray, about 15 minutes per thousand) and the 7,000 soiled trays 5 h 41 min (2.92 s per tray, about 49 minutes per thousand); the procedural soil bed makes a soiled tray about three times as costly as a clean one. On disk the 27,000 trays occupy about 182 GB as JPEG at quality 95, the COCO export adds another 130 GB, and the peak during the build is roughly 312 GB.

The generator, the real-data mixing step, the four training configurations, the comparison notebook and the scripts behind every figure in this document are released together as a single repository [66], with notebook outputs stripped.

6. Experiments

The detectors of Chapter 5, trained on the corpus of Chapter 4, are evaluated in this chapter. The experimental protocol and the evaluation metrics come first, the two inference regimes follow, and the results are then reported and discussed. All results claimed here rest on the mixed corpus of Section 4.10: the synthetic trays into which the five percent real share has been mixed.

6.1 Experimental Setup

6.1.1 Protocol

Unless stated otherwise, the experiments use the mixed corpus of Section 4.10. Model selection uses the validation split alone, and each configuration is scored exactly once on the test split. The metrics are mAP@50:95, mAP@50 and mAP@75 (Section 6.2.2), reported per class as well as averaged across classes. In the comparison-pipeline tables, the precision and recall columns are class averages of each class's precision and recall at its own best-F1 confidence, with that confidence chosen on the very set being evaluated. They thus describe the operating point a class could reach on that set, not a threshold fixed in advance, and on the real test frames this makes them test-selected. Predictions are subject to a confidence floor of 0.05; lowering it to 0.005 moved mAP@50:95 on the real frames by at most -0.002 for flat-bug and by -0.02 for RF-DETR. For flat-bug the floor therefore does not bind, while for RF-DETR the score depends on it by 0.02, and the floor is accordingly reported as part of the protocol.

Three aspects of the protocol limit how much weight the results can bear.

The first is that metrics are not exactly comparable across frameworks. RF-DETR is scored natively by `pycocotools` and flat-bug by the Ultralytics implementation, and the two report precision and recall at different operating points. For that reason every cross-model figure is recomputed by a single independent implementation, applied identically to the cached predictions of both models. For each class and each IoU threshold, that implementation forms the precision envelope over the confidence-ranked predictions and integrates it by the trapezoidal rule across 101 evenly spaced recall points, with a final precision of zero. This is the Ultralytics form of the metric rather than the `pycocotools` one, so a perfect detector scores 0.995 under it instead of 1.000. The implementation was verified numerically rather than merely executed: given perfect predictions it returns that 0.995 ceiling, given a two-of-three-classes case it returns exactly two thirds of that, and a sweep of pure translations shows the strict-threshold entry responding at the expected IoU value. In the five-order view of the real frames, a prediction carrying one of the seventeen orders not annotated there is dropped rather than counted as a false positive, because a class without ground truth is skipped. The precision of that view is optimistic by the same amount.

The second is that each configuration was trained a single time; differences on the synthetic split below ±0.01 mAP are consequently not interpreted, and neither are cross-model differences smaller than the evaluator-level spread of Section 6.5.1 (0.02 to 0.03). On the real frames the resolution is given by the tray-level bootstrap of Section 6.5.6.

The third is that comparisons are set up so that a single factor changes at a time. Where a comparison is nonetheless confounded, the confound is named at the point where the comparison is used rather than deferred to a limitations section at the end.

6.1.2 Hardware and environment

Every training and inference run was carried out on a single server equipped with NVIDIA A100 80 GB PCIe GPUs. One run occupies one GPU, and the four training runs were placed on separate devices. Corpus generation takes place on a different machine and needs no GPU (Section 5.3). The flat-bug environment is Python 3.11 with PyTorch 2.13 (CUDA 13.0) and Ultralytics 8.3.124; the RF-DETR environment is Python 3.10 with PyTorch 2.9.1 (CUDA 12.8) and `rf-detr` 1.7.1.

Because the two detectors need environments that cannot coexist (different Python and PyTorch versions), a checkpoint is only ever loaded by the interpreter that produced it. Each evaluation accordingly writes its predictions to disk in a shared format, and scoring is a separate step over those caches that does not depend on either backend. Though an implementation detail, this one carries weight: it is what allows a single metric implementation to score both models.

The two environment settings that the runs required are recorded in Section 5.3.

6.2 Evaluation Metrics

6.2.1 Intersection over Union

The similarity of a predicted box B_p to a ground-truth box B_{gt} is quantified by IoU:

$$\mathrm{IoU}(B_p, B_{gt}) = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}. \tag{6.1}$$

A prediction counts as a true positive if its class matches and its IoU with a ground-truth box of that class that is still unmatched exceeds a threshold τ; otherwise it is a false positive. Every ground-truth box left unmatched is a false negative. Because a ground-truth box can be matched at most once, repeated detections of the same object are counted as false positives.

Much of the analysis that follows hinges on how unforgiving IoU is, and the point deserves to be made explicit. Consider two boxes of equal size: displacing one by a quarter of the box width already reduces the IoU to 0.6. Consider two concentric boxes with linear size ratio $r > 1$: their IoU is exactly $1/r^2$ (Figure 6.1). A perfectly centred box that is 27 % too large in each dimension therefore scores $\mathrm{IoU} = 1/1.27^2 \approx 0.62$, comfortably above a 0.5 threshold and well below a 0.75 one. A systematic size bias is thus nearly invisible at $\tau = 0.5$ but catastrophic at $\tau = 0.75$, and the gap between the two values is diagnostic. Section 4.8 derives precisely this signature for a box convention that inflates every side by a constant factor.

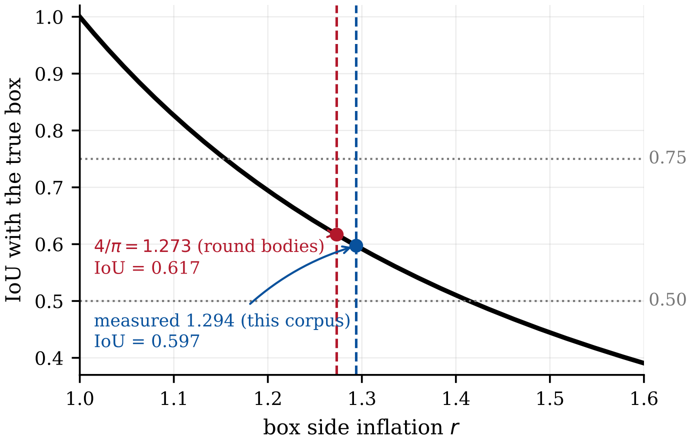


Figure 6.1: IoU of two concentric boxes plotted against their linear size ratio r, $\text{IoU} = 1/r^2$. Two values of r are marked: $4/\pi$, which is the ratio a rotated bounding canvas yields for a round body, and 1.294, the inflation measured in practice over this corpus in Section 4.8. Each lies above 0.5 and below the 0.75 threshold: scored against tight annotation, a box inflated by that factor would forfeit nearly all of AP@75 and leave AP@50 unchanged.

6.2.2 Precision, recall, and Average Precision

Sorting all predictions of a class by confidence and sweeping the confidence threshold downward from high to low yields a precision-recall curve, where $\text{Precision} = TP/(TP + FP)$ and $\text{Recall} = TP/(TP + FN)$. The area under that curve is the AP of the class. The framework-native RF-DETR figures use the COCO convention [67], in which the area is computed by 101-point interpolation: precision is sampled at the recall levels $0, 0.01, \ldots, 1.00$, each sample being the highest precision reached at any recall $\geq$ that level, and the mean of the samples is taken. The framework-native flat-bug figures, and with them every cross-model and real-data figure, rely on the trapezoidal variant of Section 6.1.1, which has a ceiling of 0.995. On the prediction files of this thesis the trapezoidal variant is higher by a few hundredths (0.02 to 0.10 on the real frames), since it interpolates the precision tail down to zero rather than cutting it off at the highest recall attained; under neither convention does any ranking in any view change. The mAP is the mean of AP over classes.

Three summary figures are reported:

- mAP@50: mAP at $\tau = 0.5$. The criterion is loose; the question it asks is whether the object was *found*.
- mAP@75: mAP at $\tau = 0.75$. The criterion is strict; the question it asks is whether the box was placed *tightly*.
- mAP@50:95: the mean over $\tau \in \{0.50, 0.55, \ldots, 0.95\}$; as the primary COCO metric, it is the headline number used here.

Reporting all three involves no redundancy. At a low threshold mAP measures detection, while at a high threshold it measures localisation, so the *difference* between the two separates two failure modes that the summary metric merges. Several findings of this thesis rest on that separation.

Two further points are relevant here. First, mAP integrates over the entire confidence range and is thus indifferent to the choice of a reporting threshold; it is *not*, however, indifferent to the inference-time confidence floor below which a model emits nothing, and that floor caps recall hard. Second, mAP averages across classes without weighting by frequency, so a single collapsed rare class costs exactly as much as a single collapsed abundant class. Under the severe class imbalance of Section 2.1.2 the mean is therefore fragile, and per-class AP tables accompany it throughout.

6.3 Whole-Tray versus Tiled Inference

6.3.1 The two regimes

Whole-tray. The tray is resized in one step to the input resolution of the network, and the predictions are mapped back afterwards. That amounts to a single forward pass per tray; it is simple; and, as Section 2.4 argues, it destroys small-object evidence in proportion to the downscale factor. At input 1536 a tray 3473 pixels wide is downscaled by 0.44×.

Tiled. The tray is split into overlapping square crops; every crop is processed at the native resolution of the network, its predictions are mapped back into tray coordinates, and the union of all of them is merged. The procedure is governed by three parameters. The tile size matches the network resolution, so a crop is neither enlarged nor reduced (1536, fed 1:1). The stride is 1280 pixels, giving a nominal overlap of 256. That overlap is larger than all but a handful of objects (among 15,884 sampled synthetic boxes the largest measures 280 pixels, and six exceed 256), so scarcely any object is cut by every tile containing it. The last column and row lie flush with the image edge, and their overlap with the neighbouring tile is correspondingly wider (Figure 6.2). The third parameter is the merge rule. Any box within 31 pixels of an interior tile border receives a confidence penalty (its confidence is multiplied by 0.85), the reasoning being that a truncated object should not compete on equal terms with a complete one. Class-agnostic NMS over the union at an IoU threshold of 0.6 then follows. A tray 3473 pixels wide yields six tiles (three columns, two rows); the frames of roughly 4,000 pixels in the second format of the real set (Section 4.9.1) yield nine.

Since tiled models are trained on tiles and evaluated on tiles alike, a tiled configuration is a distinct model, not merely a different inference procedure applied to the whole-tray one. Section 6.5.7 separates the two contributions by running each checkpoint in the regime for which it was not trained.

The four combinations. The training regime and the inference regime are chosen independently; Table 6.1 lists what the network is shown under each of the four combinations and where each one is reported. With whole-tray *training*, every training image is a complete tray resized to the network input (1536 pixels along the long side, letterboxed in the case of flat-bug and square-resized in the case of RF-DETR, Section 5.2), and the usual augmentation is applied to that image; with tiled training, the training images are the 1536-pixel tiles of Section 5.2.3, fed at 1:1. With whole-tray *inference*, the test tray is resized in the same manner and scored in a single pass; with tiled inference it is cut by the grid described above, every tile is scored at 1:1, and the union is merged. The in-regime pairs (whole/whole and tiled/tiled) constitute the five models of the main results; the two cross-regime pairs are evaluated on the same checkpoints, with no retraining, in Section 6.5.7.

Table 6.1: Training regime against inference regime: what the network is shown, and where each combination is reported. The whole-tray input is the 3473-pixel tray downscaled to 1536 (0.44×); a tile is a 1536-pixel crop at native scale.

	Whole-tray inference	Tiled inference
Trained whole-tray	one pass on the downscaled tray; the "whole" models of Section 6.5	six to nine passes on native tiles, merged; cross-regime, Section 6.5.7
Trained on tiles	one pass on the downscaled tray; cross-regime, Section 6.5.7	six to nine passes on native tiles, merged; the "tiled" models of Section 6.5

6.3.2 Expected benefit and cost of tiling

A specific prediction follows from the argument of Section 2.4: *the benefit of tiling is proportional to the downscaling it avoids, and it is a benefit only insofar as it moves the object scale at inference towards the object scale at training.* Where the imagery is only slightly larger than the network input, tiling ought to help a little; where it is many times larger, a great deal. And where tiling would present objects *larger* than those seen in training, it ought to do harm.

A second, less obvious prediction concerns the rare classes. A tile shows a specimen with less context than a whole tray does, and a class with very few distinct specimens depends on seeing them in many arrangements. The tiled model, capped at 30 epochs, also meets fewer distinct tray-level arrangements than the whole-tray model, whose best checkpoint is from epoch 34 under a 100-epoch cap. That remains true although a tiled epoch costs about six times as many gradient steps (from the checkpoints: 6,644 against 1,096 steps per epoch, a ratio of 6.1). The rare classes could therefore lose under tiling while the abundant classes gain from the resolution. This prediction was recorded before the tiled runs; Section 6.5.4 tests its sign and the mechanism proposed for it separately.

Three costs stand against the benefit. Inference time scales with the number of tiles. So does the number of emitted predictions, which AP can absorb (it integrates over the full confidence range) but which matters at any fixed operating point, and therefore for deployment. And an object straddling a tile boundary depends on the merge rule rather than on the detector.

6.3.3 Confidence floors

Two things that are easily run together are distinguished here, since the distinction bears on how the results are read. The *scoring* threshold picks an operating point on a curve that already exists: it exchanges precision for recall and can do nothing to mAP, which integrates over every prediction in any case. The *inference* confidence floor determines which boxes are emitted in the first place and sets a hard ceiling on recall. Whenever a model's predicted confidences bunch up against that floor, the floor binds, and the recall the model appears to have is an artefact of it.

For deployment, where a fixed threshold converts a ranked list into a count, the operating point matters, so the threshold sweep over the held-out real frames is reported here (Table 6.2). Each model's threshold is selected by the sweep on the same fourteen frames on which it is scored, which makes the best-F1 values test-selected and optimistic. In practice a deployment threshold would be chosen on the 31 validation frames instead. Four of the five models attain their best F1 within a narrow band of 0.96 to 0.97, the optimum falling anywhere from 0.15 to 0.20; the threshold is not a delicate choice. The fifth model, flat-bug tiled, is the outlier on both axes: it has the lowest best-F1 and is the only one requiring a threshold as high as 0.40, even though it holds the best class-aware AP of the five. This is not noise; Section 6.6.2 explains it.

Table 6.2: Best F1 and the confidence threshold that achieves it, class-agnostic, on the fourteen held-out real frames (resampled copy; threshold selected on the same frames).

Model	Best F1	at confidence
RF-DETR tiled +cont	0.972	0.15
flat-bug whole	0.971	0.20
RF-DETR tiled	0.970	0.20
RF-DETR whole	0.964	0.20
flat-bug tiled	0.939	0.40

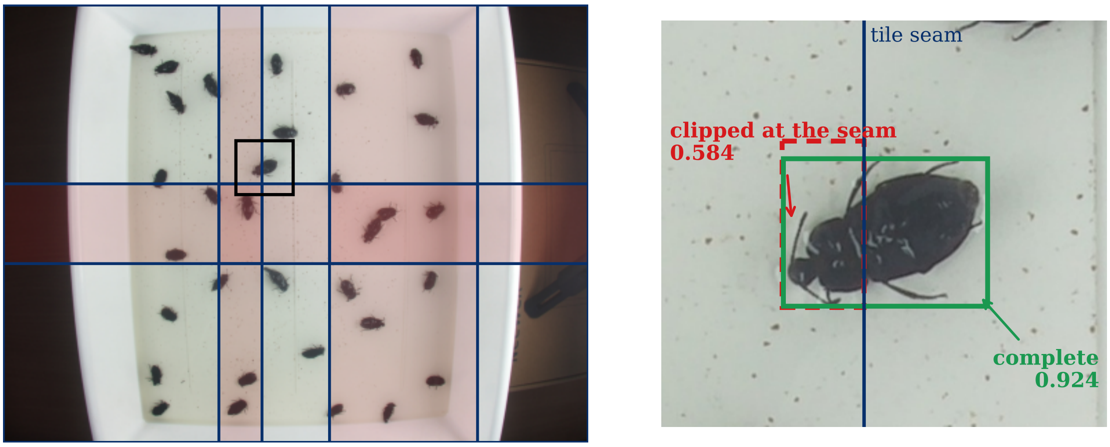


Figure 6.2: Tiled inference. Left: the 1536-pixel tile grid laid over a real frame resampled to 3473 × 2598 pixels (the copy used in Section 6.5.6), three columns by two rows at a stride of 1280, with shading on every region that more than one tile covers. The final column and row are flush with the edge, so they overlap their neighbour by more than the nominal 256 pixels. Right: what follows from this. An insect lying on the seam is detected twice, complete in one tile and clipped at the tile border in the other. The edge penalty lowers the clipped copy to 0.584, yet the two boxes overlap at only IoU 0.37, under the 0.6 merge threshold, and cross-tile NMS therefore retains both. The boxes and confidences shown are the actual flat-bug tiled predictions on this frame.

6.4 Pretraining Domain versus Architecture

flat-bug's backbone was pretrained on real arthropod imagery and RF-DETR's was not (Section 5.1), so any difference between the two detectors combines architecture with pretraining domain. What this chapter compares is therefore the two models as trained, initialisation included, and the term *architecture* is kept for the design of the detectors themselves. None of the differences reported below (the size of the tiling gain, the rare-class penalty under tiling, the behaviour at tile seams, the disagreement on Zoraptera) is ascribed to one factor on its own.

The control that would separate them, RF-DETR pretrained as a single-class detector on the flat-bug data and then fine-tuned on the corpus with a re-initialised classification head, extends the present design directly (Section 7.3).

6.5 Results

6.5.1 Two evaluation paths

Two sets of numbers are reported in this section, and on the same models they differ slightly. The *framework-native* figures are the test evaluation that each framework performs itself at the end of training: for flat-bug the Ultralytics implementation, for RF-DETR `pycocotools`, each with its own inference settings. Both the per-class breakdown of Table 6.5 and the whole-versus-tiled comparison within a framework are drawn from them. The *comparison-pipeline* figures send the checkpoint of every model through one inference and one scoring implementation (Section 6.1.1); these underpin every cross-model claim and every claim about real data.

On the same two whole-tray models the two paths give 0.694 and 0.684 (framework-native) against 0.671 and 0.707 (comparison pipeline): differences of 0.02 to 0.03 that reverse the order of the pair. That is exactly the cross-framework noise the protocol anticipated, and it is why no conclusion below rests on a cross-model difference of that size. A third scoring of the same cached predictions, written independently to produce Figure 6.8, agrees with the comparison pipeline's real five-order mAP@50 to within roughly 0.03 for every model and preserves the ordering. When three implementations agree on the order and differ by a few hundredths in absolute value, the difference sits at the evaluator level rather than at the data level. Under every path the tiled RF-DETR model is ahead of both whole-tray models by a margin far beyond that noise; flat-bug tiled is reported only through the comparison pipeline.

The comparison-pipeline results that follow are on the synthetic test split (5,400 trays) at a model input of 1536 pixels; the mixed test split, which adds the fourteen real photographs (5,414 images, 216,234 instances), reproduces the synthetic-split figures to three decimals, which confirms that the two measure the same quantity.

6.5.2 Synthetic test split, all five models

In all the tables, "flat-bug" refers to the YOLOv8-L detection model that was warm-started from the flat-bug checkpoint and fitted with a 22-class head (Section 5.1.2); the pyramid tiling and polygon deduplication of flat-bug itself are not used.

Table 6.3: All five models on the synthetic test split, comparison pipeline. "+cont" denotes the tiled RF-DETR model trained on for a further ~ 80 epochs past its 30-epoch cap. Bold marks the maximum of each column, here and in every table that follows.

Model	mAP@50:95	mAP@50	mAP@75	Precision	Recall
flat-bug tiled	**0.823**	**0.875**	**0.851**	**0.905**	0.800
RF-DETR tiled +cont	0.756	0.833	0.790	0.901	0.780
RF-DETR tiled	0.756	0.833	0.790	0.896	0.781
RF-DETR whole	0.707	0.866	0.765	0.872	**0.801**
flat-bug whole	0.671	0.808	0.737	0.841	0.747

Tiling at native scale is the dominant factor: it moves flat-bug from last place to first (+0.152) and raises RF-DETR by +0.049. For the two whole-tray models, the comparison pipeline gives RF-DETR a lead of 0.036 where the framework-native evaluation had put flat-bug ahead by 0.010; the safe reading is that the pair are close. Extending the RF-DETR run changed nothing (0.7559 → 0.7561), a result returned to in Section 6.5.4.

6.5.3 Whole-tray results under the framework-native evaluation

Table 6.4: Whole-tray results on the mixed test split, each framework's own evaluation; bold marks the row maximum. The corresponding rows of Table 6.3 are the ones to compare against.

	flat-bug	RF-DETR
mAP@50:95	**0.694**	0.684
mAP@50	0.802	**0.859**
mAP@75	0.738	**0.740**
Precision	0.850	**0.879**
Recall	0.749	**0.801**

On this path (Table 6.4) the two sit within 0.01 of each other on the headline metric and within 0.002 at the strict threshold; RF-DETR locates more objects (mAP@50 +0.057, recall +0.052) while flat-bug is slightly the more precise. The per-class breakdown is given in Table 6.5 and Figure 6.3.

Table 6.5: Per-class whole-tray results on the mixed test split, both detectors at 1536-pixel input, framework-native evaluation, ordered by the larger of the two detectors' AP@50:95. Training specimens are the distinct source individuals *available* in the training pool (those actually placed are given in Table 4.7); the median box is $\sqrt{wh}$ in pixels at full tray resolution. Bold marks either fewer than twenty training specimens or a median box below 40 pixels.

	Train	Median	flat-bug		RF-DETR	
Class	spec.	box (px)	AP	Recall	AP	Recall
Lithobiomorpha	42	122	0.929	0.935	0.859	0.910
Phasmatodea	**7**	157	0.907	0.973	0.797	0.949
Dermaptera	180	106	0.854	0.837	0.784	0.884
Odonata	**7**	107	0.787	0.742	0.841	0.884
Araneae	9,910	67	0.836	0.908	0.761	0.893
Coleoptera	11,881	70	0.830	0.896	0.752	0.894
Zoraptera	**6**	111	0.096	0.013	0.820	0.841
Lepidoptera	11,150	94	0.813	0.889	0.706	0.857
Orthoptera	2,257	84	0.809	0.893	0.701	0.850
Scutigeromorpha	**19**	73	0.799	0.897	0.690	0.841
Opiliones	92	96	0.784	0.773	0.789	0.869
Trichoptera	5,221	81	0.775	0.887	0.700	0.866
Strepsiptera	97	44	0.763	0.779	0.690	0.859
Archaeognatha	181	95	0.543	0.882	0.743	0.863
Hymenoptera	11,309	55	0.738	0.812	0.643	0.822
Hemiptera	8,902	66	0.735	0.814	0.662	0.804
Psocodea	7,465	57	0.691	0.809	0.662	0.803
Neuroptera	2,709	67	0.641	0.830	0.674	0.805
Diptera	9,675	53	0.626	0.786	0.601	0.795
Thysanoptera	7,827	**25**	0.523	0.634	0.447	0.743
Polyxenida	**18**	**34**	0.464	0.397	0.335	0.358
Zygentoma	**8**	87	0.326	0.092	0.397	0.228
All (22 classes)			0.694	0.749	0.684	0.801

6.5.4 Tiled models

Training of the tiled models uses 1536-pixel tiles fed at 1:1. The tiled RF-DETR model ran to its 30-epoch cap, but its best checkpoint comes from epoch 25, so improvement had stopped before the cap. The whole-tray model had a cap of 100 epochs and never reached it: its best checkpoint comes from epoch 34, and with a patience of 20 early stopping closed the run well short of the limit. The tiled model was subsequently continued for ~ 80 additional epochs ("+cont").

Table 6.6: RF-DETR, whole-tray against tiled, mixed test split, framework-native evaluation.

	Whole	Tiled	Change
mAP@50:95	0.684	**0.741**	+0.057
mAP@75	0.740	**0.785**	+0.045
mAP@50	0.859	0.848	−0.012

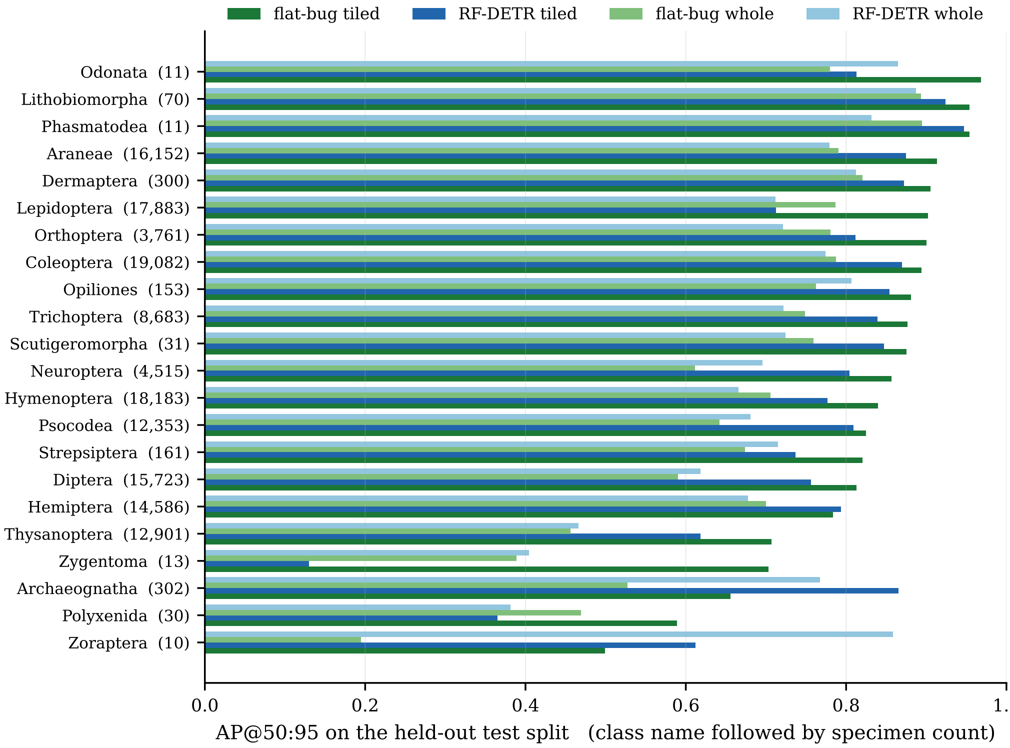


Figure 6.3: Per-class AP@50:95 of the four principal models on the held-out synthetic test split, ordered by flat-bug tiled. The number in each label gives how many distinct specimens were placed for that order over all three splits (Table 4.7); roughly 60 % of that is the training pool. Three of the four weakest rows (Zoraptera, Polyxenida, Zygentoma; Archaeognatha is the fourth) are the specimen-poorest, yet Odonata and Phasmatodea at the top are specimen-poor as well, and specimen count alone therefore fails to explain the ordering (Section 6.5.5).

The gain (Table 6.6) shows at the strict threshold and in the headline figure, while the loose threshold falls slightly. For RF-DETR, tiling improved where the box is placed rather than whether the object is found. Under the comparison pipeline the same pattern holds for RF-DETR (Table 6.3); flat-bug gains at every threshold, and overall by three times as much.

Underneath the aggregate is a split by specimen count (Figure 6.4). Its sign is the one the second prediction of Section 6.3.2 anticipated; the training-budget part of the mechanism proposed there does not withstand the continuation test that follows:

Table 6.7: Mean per-class change in AP@50:95, tiled versus whole-tray RF-DETR, grouped by the number of distinct training specimens (Table 4.7). Comparison pipeline, synthetic test split; the same data as Figure 6.4.

Classes	n	Mean change
> 1,000 training specimens	11	+**0.105**
< 20 training specimens	6	−0.058

The two mechanisms at work can be seen in three of the weakest classes (Table 6.8):

Table 6.8: The three weakest classes under whole-tray and tiled RF-DETR, framework-native evaluation, which is why the values depart from the comparison-pipeline figures of Tables 6.7 and 6.10 by the evaluator spread described in Section 6.5.1. Median px denotes the median rendered box size at full tray resolution.

Class	Specimens	Median px	Whole	Tiled	Change
Thysanoptera	7,827	**25**	0.447	**0.597**	+**0.150**
Polyxenida	18	34	0.335	0.351	+0.016
Zygentoma	**8**	87	0.397	**0.093**	−**0.304**

Thysanoptera has specimens in plenty and is simply too small: under whole-tray downscaling of 0.44× a 25-pixel insect reaches the network at about 11 pixels, beneath the level of reliable detection, and presenting it at native resolution gains more than for any other class. Zygentoma is of middling size and perfectly visible, with eight specimens; resolution can do nothing for it, and tiling left it worse off. Polyxenida suffers from both problems, and neither was remedied.

Independence from the training budget. The obvious explanation for the rare-class loss (that a tiled budget of 30 epochs was too short for classes that need repeated exposure) was tested by continuing the run (Table 6.9). Roughly 80 further epochs changed almost nothing:

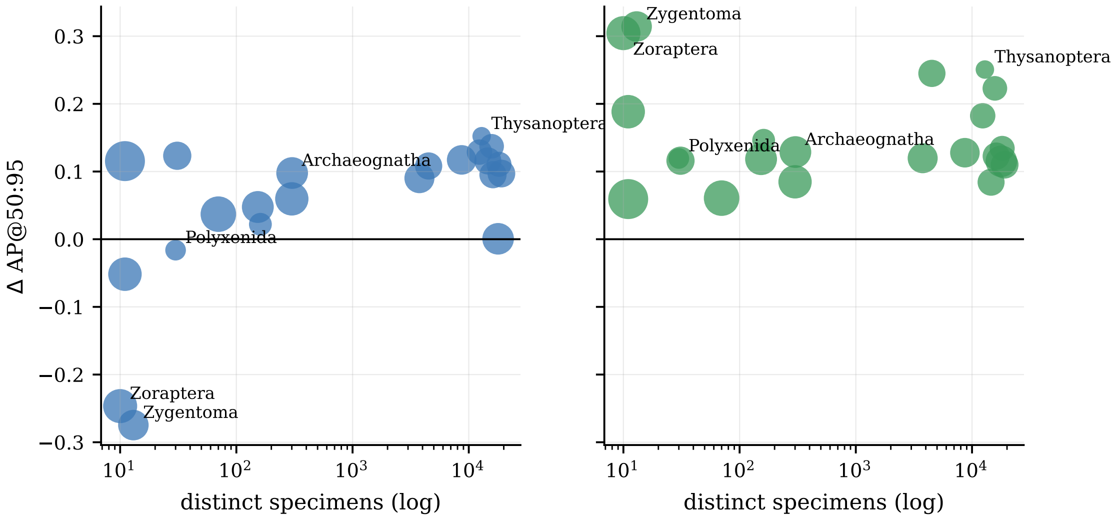


Figure 6.4: Per-class change from whole-tray to tiled training and inference (comparison pipeline, synthetic test split) against distinct specimens placed across all splits (log scale, Table 4.7); marker area is proportional to median rendered box size. For RF-DETR (left) the change is positive for the specimen-rich orders and strongly negative for the poorest: Zoraptera and Zygentoma lose more than 0.24 AP. For flat-bug (right) the same two orders *gain* the most.

Table 6.9: RF-DETR tiled at its 30-epoch cap and after continuation, comparison pipeline, synthetic test split.

	Tiled, 30 ep	+cont	Change
Zygentoma	0.130	0.150	+0.021
Zoraptera	0.612	0.616	+0.004
Polyxenida	0.365	0.368	+0.003
Overall	0.7559	0.7561	+0.000

The 30-epoch model had already converged, so the penalty is real.

Detector dependence of the penalty. No such penalty appears for flat-bug tiled. The difference between the two tiled models is attributed neither to the architecture nor to flat-bug's pretraining on real arthropods (Section 6.4). On exactly the classes where RF-DETR tiled collapses, flat-bug tiled performs well:

Table 6.10: The rare and small classes under the two tiled models, comparison pipeline, synthetic test split.

Class	RF-DETR tiled	flat-bug tiled
Zygentoma (8 specimens)	0.130	**0.703**
Polyxenida (18)	0.365	**0.589**
Thysanoptera (25 px)	0.618	**0.707**

Here lies the strongest case for flat-bug tiled as the deployment model: it comes out ahead on average and suffers no per-class collapse. It does carry one failure mode that AP cannot see, which Section 6.6.2 examines.

6.5.5 Per-class accuracy and specimen count

As a mean over classes, the aggregate mAP is dragged down by whichever classes fail, and the obvious candidate explanation for a failing class is that it has too few distinct specimens. The per-class results do not support this. Six orders have fewer than twenty training specimens (Table 6.5; four of them are tested on two held-out specimens and two on six, Table 4.7). Under the framework-native evaluation, Phasmatodea reaches 0.91/0.80 for the two detectors from seven training specimens, Odonata 0.79/0.84 from seven, and Scutigeromorpha 0.80/0.69 from nineteen. Zoraptera, with six, reaches 0.82 under RF-DETR whole-tray (0.86 under the comparison pipeline) but only 0.10 under flat-bug, which is a disagreement between the models and not a property of the data (Section 6.6.6). The remaining two rare orders are the weakest in the corpus: Polyxenida at 0.46/0.34 from 18 training specimens rendered at 34 px, both specimen-poor and small, and Zygentoma at 0.33/0.40 from eight, which Section 6.6.4 shows is found and misnamed rather than missed. Of the abundant orders, Thysanoptera at 0.52/0.45 is the weakest, and its difficulty is size (25 px) rather than supply.

The same picture emerges when the relation is measured across all classes. For each of the five models, the comparison-pipeline per-class AP@50:95 is correlated with the distinct specimens placed across all splits (Table 4.7). Under a 60/20/20 specimen-disjoint split that count and the training count share identical ranks and differ by a near-constant factor, so using the training count instead leaves every Spearman ρ below unchanged and moves every Pearson r by less than 0.01. Pearson's r ranges from -0.13 (RF-DETR whole) to $+0.29$ (RF-DETR tiled), none of them significant ($p \geq 0.19$). Spearman's ρ remains within ± 0.13 of zero for four of the five models and is weakly negative (-0.38, $p = 0.08$) for the fifth (Table 6.11, Figure 6.5). By neither measure, and for no model, does specimen count predict per-class AP. Since the correlation covers 22 single-run points, four scored on two held-out specimens and two more on six, the result describes this range of specimen counts and is not a general law. Of the six orders with fewer than twenty training specimens, three (Phasmatodea, Odonata, Scutigeromorpha) score above 0.74 under flat-bug and 0.69 to 0.84 under RF-DETR; the other three are Zoraptera, a disagreement between models, and Polyxenida and Zygentoma, the two groups that are genuinely starved. The number of specimens below which an order fails is therefore order-specific, and for Zoraptera it differs between the two detectors.

Table 6.11: Correlation between per-class AP@50:95 and log(distinct specimens placed across all splits) over the 22 classes, comparison pipeline.

Model	Pearson r	p	Spearman ρ	p
RF-DETR tiled +cont	+0.29	0.19	+0.07	0.77
RF-DETR tiled	+0.28	0.21	+0.02	0.95
flat-bug tiled	+0.21	0.34	+0.05	0.84
flat-bug whole	+0.20	0.39	+0.12	0.58
RF-DETR whole	−0.13	0.57	−0.38	0.08

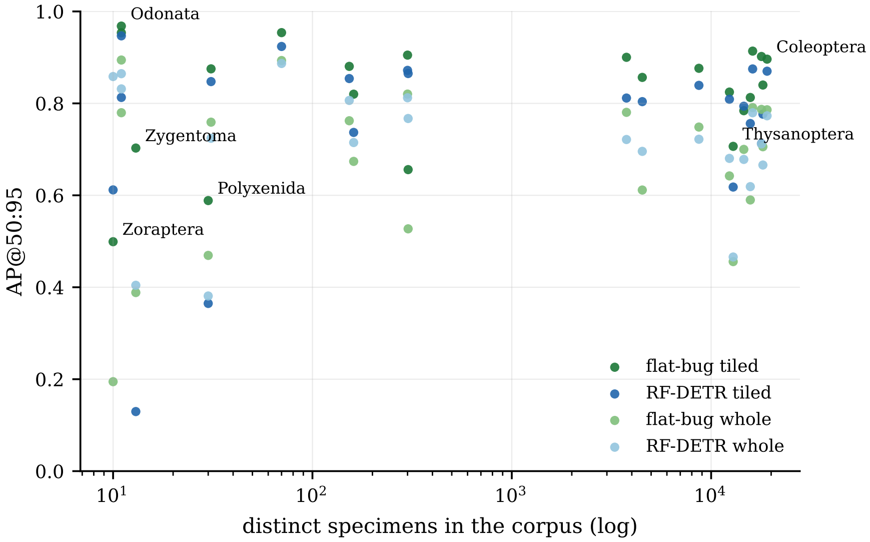


Figure 6.5: Per-class AP plotted against the number of distinct specimens placed across all splits (log scale) for the four principal models, with labels on the specimen-poorest orders. The correlation is weak ($r = +0.21$ for flat-bug tiled): the specimen-poorest orders attain AP of 0.5 to 0.9, and the spread at a given specimen count exceeds the trend across counts.

6.5.6 Real photographs

The fourteen real photographs withheld from training (479 annotated boxes on roughly 196 distinct animals across six trays; Table 4.3) are scored under two views. The detection-only view disregards class and applies class-agnostic NMS, so that a wrong label is not penalised twice in a metric that concerns detection alone. The five-order view is limited to the five annotated orders and scores the class. Alongside these two real-data views stands the synthetic 22-class evaluation of the preceding subsections. The comparison pipeline holds two copies of the fourteen frames: the native files, twelve of which are in the roughly 4,000-pixel format, and a copy resampled to 3,473 pixels wide. The native copy underlies the tables of this subsection, the box-scale sweep, the false-positive table and the cross-regime experiment; the resampled copy underlies the threshold sweep, the inference timing, and the qualitative and precision-recall figures. In the detection-only view the two copies agree to within 0.01 for every model. That is enough to exchange the two leading models (RF-DETR tiled 0.888 against flat-bug tiled 0.879 on the native copy, 0.879 against 0.889 on the resampled one), and for that reason no ordering of the top models on real detection is claimed.

Table 6.12: Held-out real photographs, detection only, comparison pipeline.

Model	mAP@50:95	mAP@50	mAP@75	Precision	Recall
RF-DETR tiled	**0.888**	0.977	0.945	0.979	**0.969**
RF-DETR tiled +cont	0.886	**0.978**	0.944	**0.983**	**0.969**
flat-bug tiled	0.879	0.968	0.932	0.933	0.906
flat-bug whole	0.870	0.977	**0.949**	0.975	**0.969**
RF-DETR whole	0.863	0.975	0.938	0.962	0.962

RF-DETR tiled, the top row of Table 6.12, produces 499 boxes above the 0.05 floor for 479 ground-truth boxes, and at its test-selected best-F1 point both precision and recall exceed 0.96. On the clean trays of this station, class-agnostic detection is near saturation for every model (mAP@50:95 between 0.86 and 0.89 on roughly 196 animals across six trays), with the five models within 0.025 of one another. This agreement shows that these frames make the task easy, not that the task is solved in general. A control confirms the reading. Applied zero-shot to the resampled copy of the same fourteen frames through the comparison pipeline, with no training on this corpus, the released flat-bug checkpoint [41] scores 0.891 in the detection-only view (precision 0.965, recall 0.990, 493 boxes). Since that value falls inside the tray-level interval of every model in Table 6.14 (0.834 to 0.922 for flat-bug tiled, 0.841 to 0.940 for RF-DETR tiled), the models trained here and an off-the-shelf arthropod detector are indistinguishable on detection at this sample size; the advantage of the released model lies in recall. As a single-class model it cannot enter the five-order view. The contribution of the corpus is the order-level label, which that control cannot provide at all.

Table 6.13: Held-out real photographs, the five annotated orders, comparison pipeline.

Model	mAP@50:95	mAP@50	mAP@75	Precision	Recall
flat-bug tiled	**0.777**	**0.892**	**0.829**	0.913	**0.829**
RF-DETR tiled +cont	0.725	0.854	0.789	0.902	0.805
RF-DETR tiled	0.713	0.833	0.772	**0.926**	0.741
flat-bug whole	0.705	0.816	0.755	0.869	0.730
RF-DETR whole	0.600	0.734	0.648	0.730	0.740

When class is scored (Table 6.13) the models separate. The top three places go to the three tiled models, flat-bug tiled is 0.05 ahead of the next model, and RF-DETR whole falls to 0.60. Most of that lead comes from the four rare orders with six to eleven animals each, so it is a difference within this test set and not an established one. A bootstrap over the six physical trays (Table 6.14) shows how much these figures can bear: trays, never frames, are resampled with replacement, because the three frames of a tray are not independent, and 4,000 draws are taken. In the detection-only view, where the five models span 0.025, the intervals are roughly 0.10 wide. In the five-order view, where they span 0.18, the intervals are 0.17 to 0.33 wide. Paired over the same trays, no difference discussed in this chapter excludes zero. In the five-order view flat-bug tiled is ahead of RF-DETR tiled by +0.065 (interval −0.036 to +0.068, positive in 91 % of draws). The tiling gain of flat-bug is +0.072 (−0.022 to +0.076, 93 %) and that of RF-DETR is +0.112 (−0.017 to +0.121, 95 %). The continuation run contributes +0.012 (82 %), and in the detection-only view RF-DETR is ahead of flat-bug by +0.009 (84 %). The signs are nonetheless stable: leaving out any one tray keeps the five-order lead of flat-bug tiled between +0.046 and +0.064. Each difference therefore has a consistent direction, but six trays do not resolve its magnitude. With six clusters there are only 462 distinct resamples, so the intervals are coarse and asymmetric (upper bounds near 0.90 for models whose point estimate is 0.60). They state the order of magnitude of the precision rather than exact limits, and the third decimal printed in the real-data tables is not supported by them. For any one model, the gap between the two views (0.888 against 0.713 for RF-DETR tiled) is the cost of naming rather than of finding, as Section 6.6.4 shows directly. Table 6.15 gives the per-order breakdown.

Table 6.14: Tray-level bootstrap intervals (2.5 and 97.5 percentiles, 4,000 draws over the six test trays) for mAP@50:95 on the fourteen held-out photographs, comparison pipeline, native copy. The point estimates are those of Tables 6.12 and 6.13, reproduced to four decimals before the intervals were drawn.

	Detection only			Five orders		
Model	Point	2.5 %	97.5 %	Point	2.5 %	97.5 %
flat-bug tiled	0.879	0.834	0.922	**0.777**	0.712	0.885
RF-DETR tiled +cont	0.886	0.836	0.939	0.725	0.671	0.908
RF-DETR tiled	**0.888**	0.841	0.940	0.713	0.668	0.909
flat-bug whole	0.870	0.818	0.920	0.705	0.661	0.890
RF-DETR whole	0.863	0.814	0.913	0.600	0.570	0.901

Table 6.15: Per-order AP@50:95 on the real held-out photographs. Boxes is the number of annotated boxes of that order in the fourteen frames; animals is the estimated number of distinct individuals behind them, and each of the four rare orders comes from a single tray (Table 4.3). The four rare rows are therefore measurements on six to eleven animals of one tray, and their ordering should not be read. “fb” is flat-bug.

Order	Boxes	Animals	fb tiled	RF-DETR tiled	fb whole	RF-DETR whole
Diptera	24	8	**0.860**	0.729	0.742	0.429
Orthoptera	30	10	**0.840**	0.738	0.752	0.697
Coleoptera	374	161	0.820	**0.864**	0.836	0.854
Hemiptera	33	11	**0.740**	0.694	0.658	0.405
Araneae	18	6	**0.625**	0.537	0.535	0.616

Box-scale sweep. The box-scale sweep of the comparison pipeline tests the predicted boxes on their own. Before scoring on the real held-out photographs, every predicted box is scaled about its centre by a uniform factor; the factor at which mAP@50:95 peaks is the factor by which the model’s boxes are systematically wrong in size. All five models peak at 1.00 (three are listed in Table 6.16; the other two, flat-bug whole at 0.870 and RF-DETR tiled +cont at 0.886, also peak at 1.00; Figure 6.6). No systematic scale bias against the real annotation is present in the predicted boxes, confirming that the recorded boxes represent the extent of the specimen (Section 4.8.2).

Table 6.16: Box-scale sweep on the held-out real trays for three of the five models, detection-only mAP@50:95. The peak of every model lies at 1.00.

Scale factor	1.10	**1.00**	0.90	0.80
flat-bug tiled	0.639	**0.879**	0.621	0.300
RF-DETR tiled	0.665	**0.888**	0.616	0.295
RF-DETR whole	0.668	**0.863**	0.595	0.285

The synthetic-to-real gap at fixed task. Placing the synthetic 22-class score beside either real-data view does not measure the domain gap. The detection-only view drops class scoring,

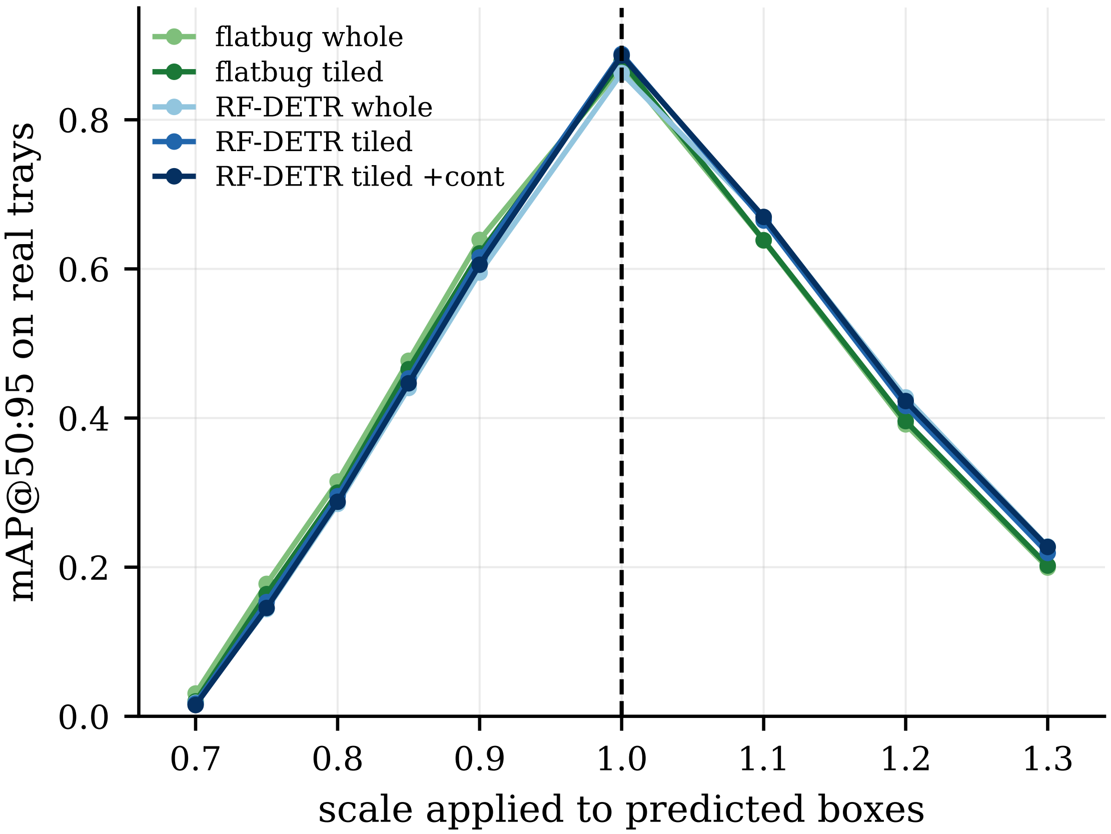


Figure 6.6: The box-scale sweep: real-data mAP@50:95 plotted against a multiplicative scale factor applied to every predicted box before scoring. Every one of the five models peaks at 1.0, direct evidence that the predicted boxes carry no systematic scale bias relative to the real annotation.

which raises mAP mechanically (every well-placed box that was mislabelled now counts), and Section 6.6.4 shows the effect to be large in this corpus; the five-order view changes the class set. The comparison that holds the task fixed scores the synthetic test split on the same five orders, macro-averaged identically, and sets it against the real five-order result (Table 6.17). The drop lies between 0.02 and 0.11 depending on the model, and is 0.084 for the best one; the four principal models keep the same order on both sides. On the real side of this comparison four fifths of the weight rests on four orders of six to eleven animals from one tray each (Table 4.3). The uncertainty on the size of the drop is therefore far larger than the ±0.01 assumed on the synthetic side. What is robust is its sign and its order of magnitude.

Table 6.17: The synthetic-to-real gap with the class set held fixed: mAP@50:95 over the five annotated orders, macro-averaged, on the synthetic test split and on the fourteen held-out real photographs. Comparison pipeline; changes are from unrounded values.

Model	Synthetic, five orders	Real, five orders	Change
flat-bug tiled	**0.861**	**0.777**	−0.084
RF-DETR tiled	0.821	0.713	−0.109
flat-bug whole	0.729	0.705	−0.025
RF-DETR whole	0.714	0.600	−0.114

False positives. On the fourteen frames at a confidence of 0.30, the five models produce between 7 and 46 false positives (Table 6.18). flat-bug tiled produces by far the most, and 35 of its 46 are *duplicates of a true positive*: a lower-scoring box that overlaps a higher-scoring detection which did match ground truth, at an IoU between 0.15 and 0.6. That overlap is too large for a different animal and too small for the merge to catch. Of these, 24 are clipped exactly at an interior tile border. The other 11 duplicates arise from the flush final column and row, whose overlap with their neighbours far exceeds the nominal 256 pixels (879 and 474 pixels on the frame of Figure 6.2). flat-bug whole-tray, the same architecture run without tiles, has no duplicates whatsoever; the two RF-DETR tiled models share one. Section 6.6.2 examines the mechanism.

Table 6.18: False positives (FP) on the fourteen held-out real frames at confidence 0.30, together with the number of them that are duplicate boxes on an insect already detected (a true positive, TP), overlapping it at IoU 0.15 to 0.6.

Model	FP	Duplicate of a TP	. . . of which at a seam	Other
flat-bug tiled	46	**35**	**24**	11
flat-bug whole	13	0	0	13
RF-DETR tiled	9	1	0	8
RF-DETR tiled +cont	7	0	0	7
RF-DETR whole	16	0	0	16

6.5.7 Training-time versus inference-time tiling

Because the tiled models were trained on tiles and run on tiles, the tiled rows above merge two things: what the model learned from tiles, and what native-resolution inference offers any model whatsoever. The experiment that separates them takes the four trained checkpoints as

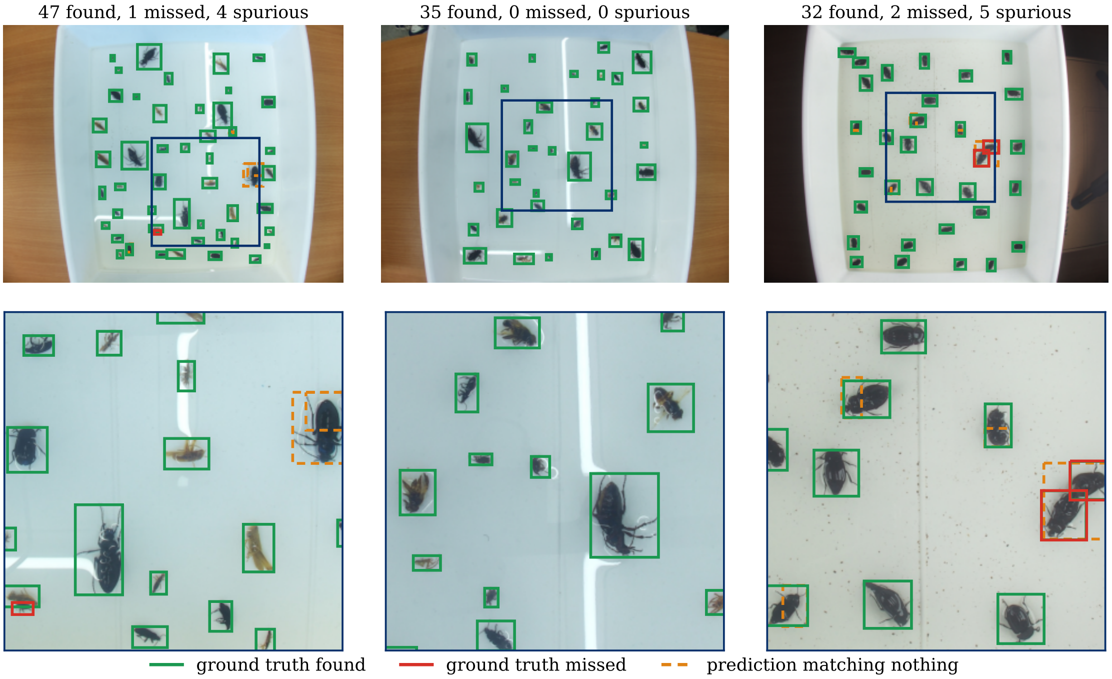

47 found, 1 missed, 4 spurious
35 found, 0 missed, 0 spurious
32 found, 2 missed, 5 spurious
ground truth found
ground truth missed
prediction matching nothing

Figure 6.7: Qualitative results for flat-bug tiled on three held-out real frames at a display threshold of 0.30, rather than the best-F1 operating point used in the tables. Top row: the whole tray, with the region enlarged beneath it outlined in blue. Green marks ground truth found, red ground truth missed, and orange dashed a prediction that matches no ground truth. Two of the orange boxes merit attention: in one frame an orange box lies on a genuine beetle absent from the annotation, and in another an orange box overlaps two red ones, a localisation failure on touching specimens rather than a spurious detection.

they are and runs each in the *other* regime, whole-tray-trained models under tiled inference and tile-trained models under whole-tray inference, on the fourteen held-out photographs over the five annotated orders, through the comparison pipeline. Nothing is retrained. If a whole-tray model run on tiles came close to its tiled counterpart, the lever would be an inference-time one, and the cost of tiled training (along with, for RF-DETR, the rare-class penalty of Section 6.5.4) could be avoided in deployment.

Table 6.19: Training regime against inference regime for the five annotated orders on the fourteen held-out real photographs, comparison pipeline, native copy. The rows are the four trained checkpoints, each run both in its own regime (in-regime, the results of Table 6.13) and in the other one (cross-regime); no model was retrained. Precision and recall are per-class best-F1 operating points at IoU 0.5, macro-averaged (Section 6.1.1), rather than a fixed threshold; n_{pred} counts the boxes emitted above the 0.05 floor against 479 annotated.

Model	Trained	Run	mAP@50:95	AP@50	AP@75	Precision	Recall	n_{pred}
flat-bug	whole	whole	**0.705**	0.816	0.755	0.869	0.730	520
flat-bug	whole	tiled	0.262	0.357	0.287	0.538	0.356	683
flat-bug	tiled	tiled	**0.777**	0.892	0.829	0.913	0.829	613
flat-bug	tiled	whole	0.617	0.712	0.689	0.754	0.750	510
RF-DETR	whole	whole	**0.600**	0.734	0.648	0.730	0.740	999
RF-DETR	whole	tiled	0.397	0.478	0.430	0.853	0.372	1,170
RF-DETR	tiled	tiled	**0.713**	0.833	0.772	0.926	0.741	519
RF-DETR	tiled	whole	0.521	0.652	0.563	0.637	0.732	762

Run on tiles, the whole-tray models come nowhere near their tiled counterparts. Every model performs best in the regime it was trained in; the diagonal beats the off-diagonal in all four cases (Table 6.19). A whole-tray model run on tiles loses 0.443 (flat-bug) and 0.203 (RF-DETR); a tile-trained model run on whole trays loses 0.160 and 0.192. Tiling is therefore not an inference-time device that can be attached to a model trained in another way: the model must be trained in the regime it will run in. The gain of Section 6.5.4 belongs to training and inference as a pair, and in deployment they cannot be separated. That settles the question of training-time against inference-time tiling, and the answer is more useful than the one the experiment was set up to produce, since it rules out a shortcut that would otherwise have been recommended.

The forward direction. Two contributions are considered, object scale and the tile boundary, and object scale is taken first. Whole-tray inference reduces a 3473-pixel frame by 0.44×, while a tile is fed at 1:1, so a whole-tray model applied to tiles sees objects roughly 2.3× larger than those it learned from. Measured against ground truth, its cross-mode boxes come to 0.81× (flat-bug) and 0.91× (RF-DETR) of the true size, whereas every in-regime row gives 0.99 to 1.03×. Boxes that are too small are consistent with a scale mismatch fragmenting objects larger than the learned size prior. A box at 0.81× per side around the correct centre has an IoU of about 0.66 with its target; on its own that removes every match at the strict thresholds and thus accounts for a large share of mAP@50:95. The test carried out here (the prediction that the boxes would be 2.3× too large) does not rule out scale as a mechanism. What the full set of metrics in Table 6.19 does settle is *which* quantity collapses, and the two directions fail in different ways. In the forward direction a whole-tray model applied to tiles fails on recall, which halves for both models (0.730 → 0.356 for flat-bug, 0.740 → 0.372 for RF-DETR). Precision shifts in opposite directions (0.869 → 0.538 for flat-bug, 0.730 → 0.853 for RF-DETR, which outputs 1,170 boxes for 479 annotated but places its confident ones well). In the reverse direction a tile-trained model applied to a whole tray fails on precision: recall hardly changes (0.829 → 0.750, 0.741 → 0.732) whereas precision drops (0.913 → 0.754, 0.926 → 0.637). A model misses animals on tiles it was not trained for and invents them on a whole tray it was not trained for. Localisation, expressed as AP@75 over AP@50, is poorest for flat-bug forward (0.805 against 0.88 to 0.93 in-regime) and best of all eight cells for flat-bug reverse (0.967), so poor box fit belongs to the forward direction specifically and is not a general effect. These precision figures are best-F1 operating points; the seam counts that follow are taken at a fixed confidence of 0.30 and measure a different quantity. The second contribution is the tile boundary, the single artefact that tiled inference introduces and whole-tray inference lacks. At a confidence of 0.30, Table 6.20 counts the detections whose box has an edge within two pixels of an interior boundary of the 1536-pixel tile grid (as a control, the same grid is imposed notionally on the whole-tray rows). It also counts how many of the detections matching no ground-truth box at IoU 0.5 lie on such a boundary.

Table 6.20: Detections on tile seams, fourteen held-out real photographs, confidence 0.30. A detection is *on a seam* if a box edge lies within 2 pixels of an interior boundary of the 1536-pixel tile grid; a *false positive* here is any detection of any class with no ground-truth box at IoU ≥ 0.5. This is a looser count than the class-restricted, one-to-one matching of Table 6.18, so the absolute numbers differ between the two tables; the comparison within this table is what matters.

Model	Trained / run	Detections	On seam	FP	Precision	FP on seam
flat-bug	whole / whole	492	3.5 %	12	97.6 %	0 %
flat-bug	whole / tiled	457	19.0 %	**128**	**72.0** %	48 %
flat-bug	tiled / tiled	546	19.4 %	65	88.1 %	**92** %
RF-DETR	whole / whole	518	3.9 %	16	96.9 %	0 %
RF-DETR	whole / tiled	619	24.1 %	**141**	**77.2** %	76 %
RF-DETR	tiled / tiled	477	5.0 %	9	**98.1** %	56 %

Running a whole-tray model on tiles multiplies its false positives by 10.7× (flat-bug, 12 → 128) and 8.8× (RF-DETR, 16 → 141), lowers its precision to 72 % and 77 %, and puts between half and three quarters of the new false positives on tile seams. Part of the forward collapse is explained by this: a model that has never seen a cut object treats the fragments at every boundary as insects. Not all of it is, since the off-seam false positives rise too (from 12 to 66 for flat-bug), and the undersized boxes discussed above are a loss of their own. On the twelve native 4,000-pixel frames the tile grid is three by three rather than three by two, so the seam counts of Table 6.20 refer to that grid. In the reverse direction seams can play no part, as whole-tray inference has none. That direction is the precision failure of Table 6.19, invented detections at nearly unchanged recall; the candidate mechanism is the scale change on its own, a model trained on 1:1 crops run at 0.44× scale, which is the mismatch of Section 6.3.2 reversed (Section 7.3).

Detector dependence. From Section 6.5.6 one would have expected a high seam share for flat-bug and a low one for RF-DETR. Under cross-mode inference both are high (19.0 % and 24.1 %): the two models differ not in how they fail on tiles they never trained on, but in what tiled *training* does to repair the failure. Trained and run on tiles, flat-bug still places 19.4 % of its detections on seams and produces 65 false positives, 92 % of them on seams; RF-DETR trained and run on tiles has 5.0 % and 9. Tiled training teaches boundary robustness to RF-DETR and not to flat-bug, a difference that traces the seam-duplicate finding of Section 6.6.2 to training rather than inference. Which of the three is responsible, the architectures, flat-bug's pretraining, or the visibility rule under which both were trained on cut instances (Section 5.2.3), is left to the control of Section 7.3. One more number is relevant here: RF-DETR tiled has the best precision of all six rows (98.1 %, nine false positives), yet its five-order mAP (0.713) is below that of flat-bug tiled (0.777), which has seven times as many false positives. This agrees with the finding of Section 6.6.4 that the remaining errors are errors of naming rather than of detection.

6.5.8 Inference cost

Inference time was measured one model at a time on an idle A100 80 GB at batch size 1, with device synchronisation on either side of the timed region and three warm-up passes discarded. The figure per frame is the median of five repeats, taken across the fourteen held-out photographs in their resampled 3,473-pixel copy, in which each frame yields six tiles; model loading is excluded (Table 6.21, Figure 6.10). The accuracy column of Table 6.21 is the five-order result of Table 6.13, scored on the native copy, in which twelve frames yield nine tiles. The two columns therefore come from the two copies of the same frames (Section 6.5.6), and the tiled costs are those of the six-tile geometry.

Table 6.21: Inference cost against real five-order accuracy, fourteen held-out photographs. Timing on the resampled copy, on which tiled models make six forward passes per photograph; accuracy from the native copy (see text). "Per tray" denotes one photograph of a tray.

Model	Passes / tray	s / tray	s / pass	Real mAP@50:95
flat-bug whole	1	**0.166**	0.166	0.705
RF-DETR whole	1	0.284	0.284	0.600
flat-bug tiled	6	**0.572**	0.095	**0.777**
RF-DETR tiled	6	1.016	0.169	0.713
RF-DETR tiled +cont	6	1.017	0.170	0.725

flat-bug whole and flat-bug tiled form the Pareto frontier. Each of the three RF-DETR configurations is dominated, being at once slower and less accurate than some flat-bug configuration. Per forward pass the two architectures are 1.8× apart (0.095 s against 0.169 s on a tile), so what handicaps RF-DETR is the cost per pass, not the tiling; the two tiled models make the same six passes. The per-pass figure for flat-bug whole-tray exceeds its own per-tile figure because the whole-tray call resizes the entire 9-megapixel image within the timed region, something a 1536-pixel tile does not require.

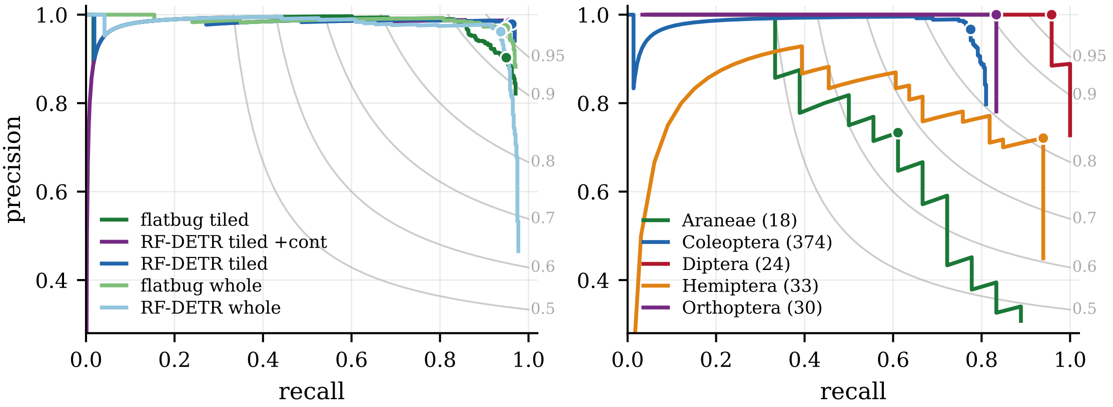


Figure 6.8: Precision-recall curves on the fourteen held-out real frames (479 boxes, IoU 0.5), with iso-F1 contours in grey and the best-F1 point of each curve marked. Left: detection only, class-agnostic, all five models; for every model precision remains above 0.95 up to a recall of about 0.9, and the models part only in the last few percent of recall. Right: the same set split by order for flat-bug tiled. Coleoptera, holding 374 of the 479 boxes, is close to saturation; Araneae, with 18 boxes from six animals on a single tray, accounts for the ragged curve. Computed independently from the cached predictions; the class-averaged AP@50 matches Table 6.13 to within roughly 0.03.

6.6 Discussion

6.6.1 Effect of tiling by detector

Table 6.22: The four principal models under the three evaluations, comparison pipeline, mAP@50:95. Because the columns are distinct tasks (22 classes, five orders, and class-agnostic), a difference between two columns does not measure domain transfer; that measurement is given in Table 6.17.

	Synthetic	Real, five orders	Real, detection
flat-bug tiled	**0.823**	**0.777**	0.879
RF-DETR tiled	0.756	0.713	**0.888**
RF-DETR whole	0.707	0.600	0.863
flat-bug whole	0.671	0.705	0.870

Of all the levers in these experiments, tiled training and inference at native scale is the largest (Table 6.22). For RF-DETR the gain sits at the strict threshold and in the headline figure, while the loose threshold drops slightly (Table 6.6); flat-bug gains at every threshold, +0.067 at mAP@50 included. Because only one downscale factor (0.44×) was tried, the dependence of the gain on that factor lies outside what these runs show; at that single factor the two architectures gain +0.05 and +0.15, so the factor by itself does not determine it. By way of comparison, slicing-aided inference lifts AP by 5 to 7 points on standard small-object benchmarks [42], and adaptive slicing adds a further 1.7 points of mAP@50 [64]; the +0.15 of flat-bug here lies at the upper end of that range. That is plausible rather than anomalous, since this is the extreme case

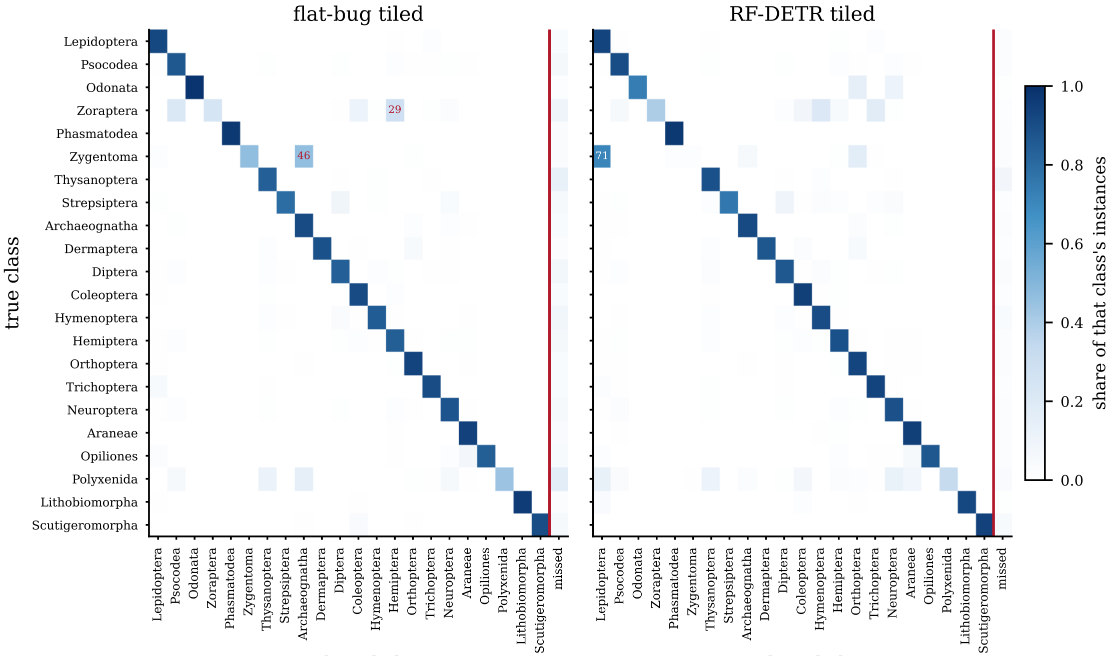


Figure 6.9: Row-normalised confusion matrices for flat-bug tiled and RF-DETR tiled on the synthetic test split, matched class-agnostically at IoU 0.5, with a final “missed” column for ground truth that drew no box at all. The missed column is nearly empty everywhere: the weak orders are found and then named wrongly. Under the tiled models shown, Zygentoma is called Archaeognatha 46 % of the time by flat-bug and Lepidoptera 71 % of the time by RF-DETR, and flat-bug calls Zoraptera Hemiptera 29 % of the time (the whole-tray model’s top confusion, Table 6.23, is Diptera). This is the evidence for Section 6.6.4: the failure is naming, not detection.

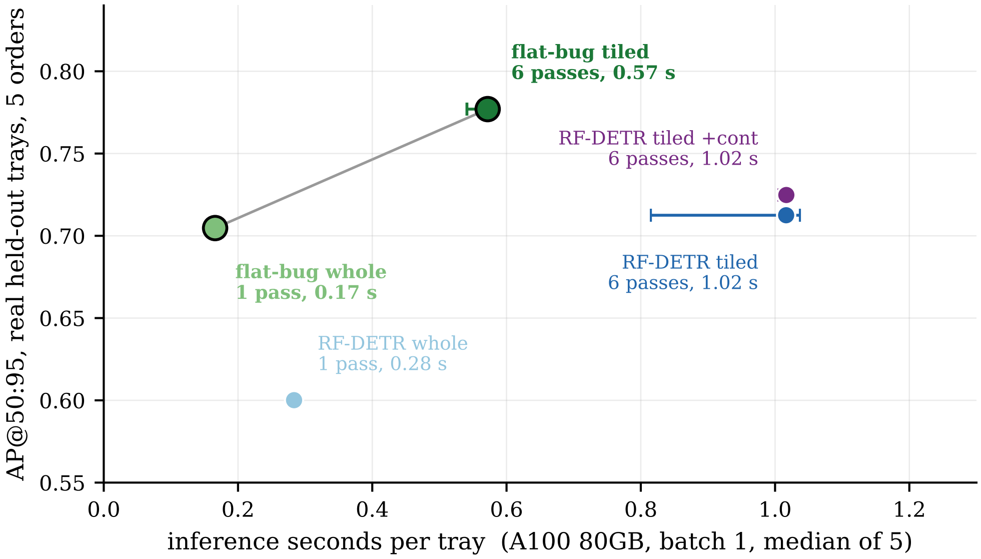


Figure 6.10: Accuracy against inference cost on the fourteen held-out real photographs. Timings are one model at a time on an idle A100 80 GB at batch size 1, with device synchronisation around the timed region, three warm-up passes discarded and the median of five repeats per frame; bars are the interquartile range across frames, and model load time is excluded. The grey line is the Pareto frontier. Tiling costs flat-bug 3.4× the time for +0.072 AP, which is the only genuine trade-off in the figure: all three RF-DETR points are dominated, each being both slower and less accurate than a flat-bug configuration. Per forward pass the two architectures differ by 1.8× (0.095 s against 0.169 s), so RF-DETR's disadvantage is per-pass cost, not the tiling.

tiling was made for: specimens a few millimetres across inside a 320 mm frame, downscaled by 0.44× on top of an already small relative size. The gain is also one measured within a single generator, not a general one. When class is scored on real data, the three top places belong to the tiled models. On the synthetic split flat-bug tiled leads by a margin well beyond the assumed noise floor. On the real five-order split its lead over the next model is 0.05, a margin whose sign survives the removal of any single tray but whose tray-level bootstrap interval includes zero (Table 6.14). On real detection the two tiled models lie within 0.01 of each other and swap order according to which copy of the frames is scored (Section 6.5.6), so neither is ahead. The synthetic lead of flat-bug tiled is real on AP, but its price is duplicate detections that AP does not charge (Section 6.6.2).

The gain is uneven across classes, and here the two detectors diverge (Tables 6.7 to 6.10, Figure 6.4). Slicing is treated as architecture-agnostic in the tiling literature [63, 42, 64], and no published counterpart to the interaction described next was found. Because the two detectors differ in pretraining domain as well as in architecture (Section 6.4), the interaction is reported as one between the two models as trained and attributed to neither factor. Trained on tiles, RF-DETR loses on the classes with the fewest specimens (Zygentoma drops from 0.40 to 0.09 under the framework-native evaluation), and the continuation run shows this to be a converged property of the model, not an artefact of the budget. On exactly those classes flat-bug tiled gains. Combined with its lower cost per pass (Section 6.6.3), this makes flat-bug tiled, a model that wins on average without any per-class collapse, the one to deploy, subject to the qualification of Section 6.6.2. The gain does not carry over between regimes: a whole-tray model applied to tiles loses 0.20 to 0.44, and a tile-trained model applied to whole trays loses 0.16 to 0.19 (Section 6.5.7), so tiled training belongs to the recipe and cannot be replaced by inference.

6.6.2 Duplicate detections at tile seams

On the real frames, three quarters of the false positives of flat-bug tiled are duplicate boxes that tiling brought into being (Table 6.18). flat-bug whole-tray is the control (same architecture, same frames, no tiles, zero duplicates), so this is a property of tiling as it was applied, not of an architecture that over-fires. It is equally a property of what tiled training taught each architecture: run on tiles, both whole-tray models place a fifth to a quarter of their detections on seams, and tiled training reduced RF-DETR to 5 % while leaving flat-bug at 19 % (Table 6.20).

The mechanism is the one illustrated in Figure 6.2. An insect sitting on a tile border is detected twice, once whole and once clipped; the edge penalty lowers the clipped copy to 0.58 (against 0.92 for the complete box) without removing it; and since the clipped box is far smaller, the IoU of the pair is only 0.37, under the 0.6 merge threshold, so cross-tile NMS never puts them side by side.

This is what F1 can see and AP cannot. Computed over the full ranking, AP hardly registers two dozen additional low-scored boxes; F1 at a fixed threshold registers each one. That is why flat-bug tiled combines the best class-aware AP of the five models with the worst best-F1, and why it is the only model needing a threshold of 0.40 (Table 6.2). For a deployment that reports a *count* instead of a ranked list, this is the failure mode that matters, since it inflates abundance; it is also the cheapest one to remove. Three candidate remedies exist (Section 7.3): replacing the merge criterion by intersection over the *smaller* box, which comes to 0.87 in the worked example and would correctly suppress the clipped copy; discarding outright any box touching an interior tile border; or masking the cut instances during training. Only the first of these is principled.

6.6.3 Cost against accuracy

Section 6.5.8 placed every configuration on a cost-accuracy plane: the frontier is formed by flat-bug whole-tray and flat-bug tiled, the difference sits in the forward pass, and the only real trade-off left is within one architecture. For deployment this decides matters in a way the accuracy tables alone do not. Where a throughput budget forces a single model, flat-bug whole-tray at 0.166 s per tray gives 0.705, six times cheaper than RF-DETR tiled at an accuracy within 0.01 of it (0.705 against 0.713), a difference beyond what this test set can resolve. Where accuracy is the constraint, flat-bug tiled is the best available, subject to the duplicate boxes of Section 6.6.2.

6.6.4 Naming errors versus detection errors

For deciding what comes next, this is the finding that matters most. The weak classes show low recall in the per-class recall column of Table 6.5, which alone would suggest models that are right when they fire and rarely fire: a failure of detection. Low per-class recall, though, means *not detected as that class*; it does not distinguish "no box at all" from "a box with the wrong label". The confusion breakdown does distinguish them, and it is overwhelmingly the latter:

Table 6.23: Confusion breakdown for the weakest classes on the synthetic test split: the share of ground-truth instances that were boxed with the correct label, boxed with a wrong label, or not boxed at all, and the most frequent wrong label. Upper block: the specimen-poor classes, where the failure is naming. Lower block: the two smallest classes, where a substantial share is missed outright.

Model	Class	Correct	Wrong class	Missed	Top confusion
flat-bug whole	Zoraptera	0.014	**0.940**	0.046	Diptera 28 %
RF-DETR tiled	Zygentoma	0.014	**0.966**	0.021	Lepidoptera 71 %
flat-bug whole	Zygentoma	0.081	**0.880**	0.039	Archaeognatha 58 %
RF-DETR tiled	Polyxenida	0.328	**0.630**	0.043	Lepidoptera 13 %
flat-bug whole	Thysanoptera	0.672	0.085	**0.243**	Hymenoptera 3 %
RF-DETR whole	Thysanoptera	0.770	0.072	**0.158**	Hymenoptera 2 %
flat-bug tiled	Thysanoptera	0.835	0.036	**0.129**	Hymenoptera 1 %
RF-DETR tiled	Thysanoptera	0.884	0.038	0.078	Hymenoptera 1 %
flat-bug whole	Polyxenida	0.398	0.425	**0.177**	Archaeognatha 16 %
RF-DETR whole	Polyxenida	0.356	0.547	**0.097**	Thysanoptera 17 %
flat-bug tiled	Polyxenida	0.440	0.398	**0.163**	Archaeognatha 15 %

In the specimen-poor, mid-sized classes (the upper block of the table) the models *find* the insects (missed rates of 2 to 5 %) and then give them the wrong name. In the two smallest classes both failures occur at once: under the whole-tray models and under flat-bug tiled (lower block), Thysanoptera and Polyxenida are missed 10 to 24 % of the time. RF-DETR tiled alone reduces their missed rates to 4 to 8 %, which is the size effect of Section 6.5.4 viewed from the other side. Several confusions make taxonomic sense: Zygentoma (silverfish) and Archaeognatha (bristletails) are two orders of primitive wingless insects of very similar habitus. The same fact accounts for the gap between the two real-data views: for RF-DETR tiled the same photographs score 0.888 when class is ignored and 0.713 when class is scored. The boxes are correct; the names are what go wrong.

The asymmetry itself is not a discovery of this thesis; it is the consensus of the field. A contemporaneous two-stage system from the flat-bug group reports localisation above 93 % against classification at 80 % [54], and under distribution shift the two common taxa are localised at 91 % and 81 % but classified at 80 % and 67 % [56]. AMI treats localisation as settled and benchmarks naming only [24], and MassID45 finds species-level naming limited by optics [43]. The contribution of this thesis is the asymmetry measured with one metric family on identical detections (class-ignored against class-scored mAP@50:95 on the same boxes), together with the confusion-matrix decomposition: 2 to 5 % missed against 63 to 97 % mislabelled for the specimen-poor classes. That locates the error in a way the arthropod literature reviewed here does not.

The implication for remedies is specific, and it divides by class. For the naming group, additional specimens of the *confused pairs* would help; additional tray images would not, because composition is not the limiting factor; nor would additional resolution, because those insects are already being found. For the two smallest classes the remedy is resolution, and the tiled results already bear this out. That reduces the future-work request of Chapter 7 to a short list of named orders.

6.6.5 Agreement between detectors on weak classes

Across the runs the weak classes are the same for both architectures (Table 6.5), and the failure takes the same shape in each: found, then misnamed. When two independent architectures fail in the same way, the cause lies in the data rather than in either model: apart from the exceptions noted below, the failing groups are those with too few specimens or too few pixels. The behaviour of the recommended model on real trays is shown in Figure 6.7.

Across the range present in this corpus, from six to 11,397 training specimens, specimen count alone does not predict per-class accuracy (Section 6.5.5). Since that correlation covers 22 single-run points, the result describes this range of specimen counts and says nothing about the long-tail regime where X-Paste [59] reports its largest gains. The naming-failure finding above shows what a starved class is missing: not exposure but *contrast* with the classes it is confused with.

6.6.6 Disagreement between the detectors on Zoraptera

Zoraptera is the largest exception. From the same six training specimens the whole-tray models reach 0.858 (RF-DETR) and 0.195 (flat-bug) under the comparison pipeline, and 0.820 against 0.096 under the framework-native evaluation. Tiling narrows the gap to 0.612 against 0.499, moving the two models in opposite directions. The confusion analysis (Figure 6.9, Table 6.23) settles what kind of failure this is: flat-bug boxes 95 % of Zoraptera and misnames 94 % of them, most often as Diptera. This is a naming failure, not a detection failure. Since the confusion is limited to one detector, its origin lies in that model's learned features rather than in the six source specimens.

A second, smaller disagreement involves Archaeognatha, a class that is neither specimen-poor (181 training specimens) nor small (95 px): on whole trays flat-bug scores 0.54 against 0.74 for RF-DETR (Table 6.5), and it is the class with which flat-bug confuses Zygentoma most often (Figure 6.9). The two disagreements may be a single one, an asymmetric confusion between Zygentoma and Archaeognatha in the convolutional detector.

6.6.7 The residual scale offset on real data

Section 4.11 showed three of the five annotated orders to be 1.75 to 2.58× smaller in the synthetic corpus than on real trays. Its prediction was that, should this matter, Coleoptera, Orthoptera and Diptera would lag on real data while Araneae and Hemiptera, whose scales agree, would hold up. On measurement (Table 6.24) the predicted ordering did not appear; what appeared instead was close to its reverse:

Table 6.24: Scale offset against real-data AP@50:95 (flat-bug tiled), per annotated order. The four rare orders are each six to eleven animals on one tray (Table 4.3).

Order	Scale offset	Real AP@50:95
Diptera	2.58× (largest offset)	**0.860** (best)
Orthoptera	1.75×	0.840
Coleoptera	1.88×	0.820
Hemiptera	1.01× (matched)	0.740
Araneae	0.99× (matched)	**0.625** (worst)

The result holds under the mixed regime. Every model was trained with the five percent real share, in which the offset orders appear at their real scale; Section 4.11 introduced that share as the corrective. What is shown, then, is that under the mixed regime the offset produced no visible penalty on these frames. Its counterpart, a synthetic-only model, is named in Section 7.3. Because four of the five per-order figures rest on six to eleven animals from a single tray each, the measurement can say that the predicted ordering was not observed on the one tray available per order, not that the offset costs nothing. The ordering actually observed (Araneae worst, Diptera best) runs roughly against the offset and does not track sample size; no interpretation is given to it.

6.6.8 Size of the domain gap

With the class set fixed (Table 6.17), the measured synthetic-to-real drop is 0.02 to 0.11 in mAP@50:95, and 0.084 for the best model. Published gaps for arthropod imagery under a change of imaging pipeline have the same order of magnitude. Holding out a sub-dataset costs the flat-bug detector about 7 % of its F1 [41]. YOLO detectors trained on citizen-science imagery reach a localisation F1 of 0.70 on field time-lapse against the 0.89 the same authors had reported in-distribution, a cross-study difference of roughly 0.19 that is indicative rather than controlled [56]. Fine-grained classification can lose up to 37 percentage points [24]. As each of these figures is a different metric (a change in F1, in top-1 accuracy, in mAP), the comparison is one of magnitude rather than value. A drop of this size is what the base model itself shows under a comparable change of protocol, and nothing more than that should be claimed.

Why the gap is small also needs stating plainly, because it bounds what the result means. The synthetic trays composite real specimen cutouts onto real photographs of the same trays, taken at the same imaging station, so the two domains share the tray, the station and, once resampled, the pixel density. The colour cast and the kind of debris differ (Section 4.11), and the placement of the specimens, the appearance of each pasted specimen and the soil model are synthetic. Given this narrow domain change, the finding is best read as evidence that a compositing generator built on the same station yields a small localisation gap, not that synthetic-to-real transfer is settled. In controlled-acquisition domains, of which a fixed tray under a fixed imaging station is one, synthetic-only training is known to match real training [60]. The same literature cautions that a synthetic validation score may be a poor predictor of real performance [61], so the synthetic 0.823 is reported as within-generator generalisation rather than as a deployment estimate. As five percent of the training signal is real, the figure describes the mixed regime the thesis proposes.

7. Conclusion and Future Work

7.1 Summary

The aim of this thesis was to remove the annotation bottleneck from automated insect monitoring. Detection training data would be generated instead of labelled, while a small quantity of real annotated imagery, mixed into the training set, would supply the domain realism that no generator can provide. A pipeline was built to this end. It segments individually identified specimens from the BIOSCAN collections, scales each by a size proxy derived from the crop metadata and calibrated once against real trays, and renders the cutout at the pixel density of the deployment camera under a class-balancing schedule. Each rendered cutout is then pasted onto photographs of real trap trays, with a soiling model added whose parameters were measured on genuinely soiled imagery. The result is more than a million exact COCO annotations across 22 arthropod orders. Ninety-one real annotated photographs of nineteen physical trays were split by tray. The forty-six training frames were oversampled (18 copies of each) so that they formed five percent of the training signal, then merged in; fourteen photographs of six trays were held back. Two detectors, a set-prediction transformer with a self-supervised backbone and an arthropod-specific YOLO derivative pretrained on real arthropod imagery, were trained on this corpus in both whole-tray and tiled regimes. A single comparison pipeline scored all five resulting models (the tiled transformer counted before and after its continued training) on the synthetic split and on the held-out real photographs.

Principal result. On the held-out real photographs, every one of the five models detects insects at mAP@50:95 between 0.86 and 0.89; class-agnostic detection on the clean trays of this station is therefore near saturation for all of them. Run zero-shot on the same frames, the released flat-bug checkpoint reaches 0.891 on that view, within the bootstrap interval of every model, so in detection alone the models trained here are on a par with an off-the-shelf arthropod detector. What the corpus contributes is the order-level label. When scored on order-level classification over the five annotated orders, the best model, the flat-bug derivative trained and run on native-resolution tiles, reaches 0.777. In the specimen-poor orders the remaining errors are insects that were found but given the wrong name; in the two smallest orders a considerable share is missed as well. Isolating the contribution of the five percent real share is identified as future work: a synthetic-only counterpart has since been run and is reported in Section 7.3, while a real-only counterpart at matched budget is the control that remains.

Principal findings. With respect to generation, the generator renders at the deployment pixel density, derives its soiling model from measurements on real trays, and balances classes according to a schedule. The per-class results show that, over the specimen counts present in the corpus, per-class accuracy is not predicted by the number of distinct source specimens, and that the weakest orders are limited by confusion between look-alike orders rather than by the supply of arrangements (Section 6.5.5). With respect to evaluation design, the corpus is partitioned by specimen and the real set by physical tray, and the annotation convention is regarded as one component of the domain gap, one that a synthetic benchmark cannot see on its own. With respect to the two detectors, which differ in pretraining domain and in architecture, the comparison is between the models as trained. On whole trays they are close; under tiling the convolutional model is more accurate on this corpus, shows no per-class collapse, and costs less per pass. With respect to the inference regime, native-scale tiling is the largest single lever at the resolution ratio used (0.44×), worth +0.05 to +0.15, and it applies to training and inference together. A whole-tray model run on tiles loses 0.20 to 0.44, and a tile-trained model run on whole trays loses 0.16 to 0.19; a model is therefore trained in the regime in which it will be deployed. The two directions fail differently: on tiles it was not trained for, a model misses animals (recall halves); on a whole tray it was not trained for, it emits spurious detections (precision falls while recall is unchanged). With respect to the synthetic-to-real gap, for a model trained with five percent real data and the class set fixed to the five annotated orders, the drop from synthetic to real is 0.02 to 0.11 in mAP@50:95 (0.084 for the best model). This is of the same order as the gaps published for changes of imaging system in this domain, and it is small because the domain change is narrow: the synthetic trays are composed on photographs of trays from the same station.

Methodological findings. The first is that a corpus must be partitioned by the entity from which the data derive, not by the artefact (Sections 4.7 and 4.9.3): the physical specimen for the synthetic corpus, the physical tray for the real set. A split at image level leaks the entity across the partition, negligibly for abundant entities and completely for rare ones, so that the per-class table reports its highest scores for precisely those classes on which it is least trustworthy. In a preliminary experiment, up to a third of the headline number under an image-level split was attributable to leakage (Section 4.7.3). The second is that an error in its own annotation convention is invisible to a synthetic benchmark (Section 4.8). Because a synthetic test split shares every convention of its training split, the recorded boxes must be checked against independently annotated data; that check is the box-scale sweep on the real photographs, which peaks at unity for every model (Section 6.5.6). In any account of the synthetic-to-real problem the label gap therefore stands beside the appearance and content gaps, as the one component to which a synthetic benchmark is structurally blind. The third is that only an assertion over the generated output catches a fault in code that runs without error (Section 4.13). A faulty generator can yield plausible images and plausible numbers, so the verification suite asserts every invariant over what was actually produced, and each assertion has been shown to fire when its fault is reintroduced. The figures behind these statements appear in Section 6.6 and are not repeated here.

7.2 Limitations

Size of the real test set. Fourteen photographs of six physical trays: 479 boxes around roughly 196 distinct animals belonging to five orders, of which four each appear on only one tray. The tray-level bootstrap intervals span 0.10 to 0.33 (Section 6.5.6), so the per-order figures on real data are indicative only; a larger real set that includes soiled trays is the obvious next step.

Specimen supply. Across the whole of BIOSCAN, several orders hold fewer than twenty specimens, and six taxa with fewer than ten were excluded; coverage would widen with a second source collection. Two rare orders (Scutigeromorpha, Zoraptera) are rendered at sizes that differ from the real animal, because only a handful of records underlie their size estimate (Section 4.11).

Label set. The groups typical of pitfall traps, Collembola, Isopoda, Acari and ants, lie outside the detector's coverage: they are either missing from the source pool at usable counts or not separable at order rank (Section 4.2.1).

Training length of the recommended model. Training of flat-bug tiled stopped after 15 of the 30 permitted epochs, with the validation metric still climbing at the final epoch (0.744 to 0.831 over the run); its figures therefore represent a floor for this recipe, not a ceiling.

Duplicate boxes at tile seams. The recommended model produces duplicate detections wherever an insect crosses a tile border (35 of its 46 false positives on the real frames are of this kind); Section 6.6.2 identifies a merge criterion that removes them.

7.3 Future Work

7.3.1 Immediate experiments

Removal of seam duplicates. Replace the cross-tile merge criterion with intersection over the smaller box, which in the worked example suppresses the clipped copy at 0.87 where IoU reaches only 0.37; alternatively, discard every detection touching an interior tile border. A third option, applied at training time, is to mask or inpaint the instances that fall below the visibility threshold of Section 5.2.3 instead of leaving them unlabelled, as the published flat-bug recipe does. The first two are inference-time changes that need no retraining, and either should raise the best-F1 of flat-bug tiled from 0.94 into the 0.96 to 0.97 band in which the other models sit.

Component baselines. The synthetic-only half of this control has since been run. The published flat-bug Large checkpoint was fine-tuned on the synthetic corpus alone, with the same warm start, the same recipe and the same 15 epochs as the mixed model. It was then measured in the detection-only view on the resampled copy of the fourteen held-out photographs (479 annotated insects). Zero-shot, the released checkpoint emits 493 detections; after fine-tuning on the synthetic corpus alone it emits 66; after fine-tuning on the synthetic corpus together with the five percent real share it emits 547. Counts rather than mAP are reported, since at so low a recall the trapezoidal variant used throughout inflates by roughly 0.13 to 0.19, and an mAP figure for the synthetic-only row would therefore not be comparable with the published rows. Fine-tuning a working arthropod detector on the synthetic corpus alone thus reduced it from 493 detections to 66, and five percent real data prevented that reduction. This shows that synthetic data alone does not transfer to the real frames; it does not show that the corpus contributed nothing. Whether the corpus does little, or builds the representation that lets forty-six photographs suffice, is settled only by the remaining control: a real-only model trained from random initialisation on the same 77 real photographs at a matched budget and scored through the comparison pipeline. Both readings are consistent with these counts. A third baseline, the two-stage RF-DETR of Section 6.4, separates architecture from pretraining domain.

The reverse cross-regime penalty. A tile-trained model run on whole trays loses 0.16 to 0.19, and it loses this by inventing detections while recall is almost unchanged (Section 6.5.7). Unlike the forward direction, there is no seam to blame here; the candidate explanation is the

0.44× change of scale by itself, which a downscaled-tile control (training tiles resized to whole-tray scale) would test for the cost of one inference pass per model.

The Zoraptera confusion. It is now established that this is a naming failure (flat-bug draws boxes around 95 % of them and gives 94 % of them the wrong label, most often Diptera). Because the confusion is confined to one model, its origin lies in the learned features of that detector, and a targeted study of that feature space is the natural continuation.

7.3.2 Corpus extensions

Additional specimens for the confused orders. The naming-failure finding (Section 6.6.4) converts an open-ended call for "more data" into a short list. The orders that are mistaken for one another (Zygentoma with Archaeognatha and Lepidoptera, Polyxenida with Lepidoptera, Zoraptera with Diptera) need additional distinct specimens, so that the model receives *contrast* between them. Further trays would not help, since composition is not what limits performance; nor would higher resolution, since the insects are already found. For these orders BIOSCAN is exhausted, and the specimens must therefore come from another source. The number that suffices depends on the order and on the architecture. Within this corpus, seven clean whole-animal specimens sufficed for Odonata and Phasmatodea, and six sufficed for Zoraptera under one architecture but not under the other, whereas eight for Zygentoma and eighteen for Polyxenida were too few for both.

Screening for fragment photographs. Because the BIOSCAN records do not state what each photograph shows, whether a whole adult, a larva, a detached limb or another body part, a fragment can be recognised only from the image itself. No automatic detector can tell a bad matte from a photograph of a fragment, but the source metadata may predict it: specimen-size fields that are low outliers within their own group are a plausible proxy. Cross-tabulating these against the pixel-based flags would show how widespread the problem is across the collection. One check comes first: whether several image records share a sample identifier. Should the same physical individual have been photographed more than once, the distinct-specimen counts reported throughout are optimistic.

The mixing ratio. An oversampling factor of eighteen fixed the real share of the training signal at five percent, and no sweep was run. The synthetic-only control above puts the floor above zero; the question that remains is how far below five percent that floor lies.

Posture modelling. The most conspicuous part of the content gap still left unmodelled is the difference between a specimen flattened under a microscope and one lying in a drained tray. The real-data result under the mixed regime was reached without modelling it, yet the synthetic-only control above shows that the content gap does bite once real anchoring is absent; a cheap ablation would establish how much of that gap is due to posture.

7.3.3 Evaluation

More annotated trays, clean and soiled. Fourteen clean photographs covering six trays and five orders suffice to show that detection works and to indicate the direction of the differences between models. Per-order figures on the rare orders, and any statement about the soiled trays that the synthetic corpus models, need a larger set. Among all the extensions, annotating further sets of real trays, including soiled ones, for training, validation and testing would contribute most to improving the models and to broadening what can be claimed for them. It would provide a larger real share in training, a validation set on which to choose operating points, and a test set spanning the soiled end of the range.

Operating points. MAP integrates over the whole confidence range, which makes it the right metric for comparing models and the wrong one for judging whether a system is usable. What an ecologist needs is a count, and a count requires a fixed operating point; at such a point, the trade-off between a model emitting few confident boxes and one emitting many uncertain ones looks quite different from what mAP suggests. The result becomes actionable once a recommended threshold is reported alongside the precision and recall it yields for each order. A first figure of this kind is the precision and recall of RF-DETR tiled, the best detection-only model, at its best-F1 point: both exceed 0.96 on the fourteen frames, while flat-bug tiled reaches 0.94 until its seam duplicates are removed.

7.3.4 Deployment

Choice of deployment model. The cost measurements settle this more sharply than accuracy alone would. Under a throughput budget, flat-bug whole-tray delivers 0.71 on the five real orders at 0.17 s per tray, six times cheaper than the tiled configuration of the transformer at the same accuracy. Under an accuracy constraint, flat-bug tiled delivers 0.78 at 0.57 s, with the seam duplicates of Section 6.6.2 still to be removed before a count. No configuration of the transformer lies on the frontier. In both cases the model is deployed in the regime in which it was trained: running the cheaper whole-tray model with tiled inference is not a third option (Section 6.5.7).

Model-assisted annotation. Because class-agnostic detection on the clean trays of this station is near saturation, the immediately useful product of the current model is its set of proposals. Pre-annotating real trays with flat-bug tiled, so that experts correct the *names* rather than drawing the boxes, lowers the marginal cost of exactly the real data that the preceding subsections ask for. In practice the predictions of the model are exported as pre-annotations in the Label Studio format; a new tray then opens in the annotation tool with every box already drawn, and the specialist's task is reduced to confirming or renaming each one. Published labour figures place manual identification at roughly 20 s of photography plus 80 s of expert review per specimen [68], and the labelling budget for a new site at 10 to 20 images per species [69]; this is the cost that such proposals would displace.

A smaller model for the edge. The imaging station itself, or a camera trap of the kind surveyed in Section 3.5, provides only a fraction of the compute used here. A compact student detector trained by knowledge distillation, with flat-bug tiled as the teacher providing boxes and class scores on unlabelled tray photographs, would retain most of the accuracy at a fraction of the cost per pass; the tiled regime carries over to the student unchanged.

Integration into the monitoring pipeline. The detector is a single stage of the system outlined in Section 3.4: detection and cropping at the imaging station, identification further downstream by a model better suited to fine-grained determination, and DNA evidence wherever it is available. The naming-failure finding makes the interface sharper. A detector that finds 97 % of the insects and misnames some of them is a good front end for an identification stage that only ever sees crops, and the taxonomically sensible confusions, silverfish for bristletail, are precisely those a barcode resolves.

Acronyms

AP Average Precision. 17, 19, 22, 37, 43, 57–60, 66–68, 77, 79, 80

BIN Barcode Index Number. 8

CNN Convolutional Neural Network. 11, 18
COCO Common Objects in Context. 5, 11, 17, 18, 20, 24, 25, 37, 38, 48, 51, 54, 57, 84
COI Cytochrome c Oxidase subunit I. 3, 8

DETR DEtection TRansformer. 12, 13, 17, 20, 51
DNA Deoxyribonucleic Acid. 3, 4, 8, 18, 20, 21, 88

GPU Graphics Processing Unit. 13, 32, 53, 54, 56

IoU Intersection over Union. 11–13, 17–19, 39, 52, 55–58, 60, 72, 74, 77, 78, 80, 86

mAP mean Average Precision. 37, 56–59, 67, 72, 83, 86, 88

NMS Non-Maximum Suppression. 11–13, 15, 17, 18, 51, 52, 58, 60, 68, 80

RGBA Red-Green-Blue-Alpha. 16, 24, 25, 27

SAHI Slicing Aided Hyper Inference. 15, 20, 22
SGD Stochastic Gradient Descent. 53
SSL Self-Supervised Learning. 11

ViT Vision Transformer. 11, 12, 15

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List of Figures

1.1 The monitoring workflow and the detector's place in it 2

2.1 A BIOSCAN specimen record and its derived cutout 8
2.2 Dense prediction versus set prediction 12

4.1 The generation pipeline 25
4.2 Fragment specimens in the Odonata source records 29
4.3 Two generated trays from the test split 33
4.4 Annotation of one generated tray 33
4.5 Class balance by construction 34
4.6 Tray soiling, real and generated 36
4.7 A real pitfall-trap tray from the imaging station 41
4.8 Ground-truth annotation on a real tray 42
4.9 Group structure of the real set 44
4.10 Rendered box size, real versus synthetic 46
4.11 Generated and real tray at matched scale 47

6.1 IoU between concentric boxes as a function of size ratio 57
6.2 Tiled inference 60
6.3 Per-class AP of the four principal models 64
6.4 Per-class change from whole-tray to tiled training 66
6.5 Per-class AP against distinct specimens 68
6.6 Box-scale sweep on the real photographs 71
6.7 Qualitative results of flat-bug tiled on real frames 73
6.8 Precision-recall curves on the held-out real frames 77
6.9 Confusion matrices for the two tiled models 78
6.10 Accuracy against inference cost 79

List of Tables

4.1 Composition of the real annotated set 40
4.2 Group-disjoint split of the real set 43
4.3 Composition of the real test split 44
4.4 Composition of the mixed corpus 45
4.5 Median box size of the five annotated orders, real versus synthetic 45
4.6 The corpus in summary 48
4.7 Per-class composition of the synthetic corpus 49

5.1 The two detectors compared 52

6.1 Training regime against inference regime 59
6.2 Best F1 and its confidence threshold on the held-out real frames 60
6.3 All five models on the synthetic test split 61
6.4 Whole-tray results on the mixed test split 62
6.5 Per-class whole-tray results on the mixed test split 63
6.6 RF-DETR, whole-tray versus tiled 63
6.7 Per-class change under tiling by specimen count 65
6.8 The three weakest classes under whole-tray and tiled RF-DETR 65
6.9 RF-DETR tiled before and after continued training 66
6.10 Rare and small classes under the two tiled models 66
6.11 Correlation of per-class AP with specimen count 67
6.12 Real held-out photographs, detection only 68
6.13 Real held-out photographs, five annotated orders 69
6.14 Tray-level bootstrap intervals on the real photographs 70
6.15 Per-order AP on the real held-out photographs 70
6.16 Box-scale sweep on the real photographs 70
6.17 The synthetic-to-real gap with the class set held fixed 72
6.18 False positives on the held-out real frames 72
6.19 Training regime against inference regime on the real photographs 74
6.20 Detections on tile seams 75
6.21 Inference cost against real five-order accuracy 76
6.22 The four principal models across the three evaluations 77
6.23 Confusion breakdown for the weakest classes 81
6.24 Scale offset against real-data AP per order 83